

3.2 Website

3.2.1 Authentication

3.2.1.1 Login

Step 1: Access the FDA System Login Page

The user (Admin, Super Admin or Monitor) navigates to the FDA System administration portal at the official URL: fda-website.vercel.app.

The authentication page is displayed with a split-screen layout:

- Left panel: A branded gradient panel (blue to cyan) displaying the FDA System logo (water-drop icon), the headline 'Flood Monitoring System', and two feature cards: 'Real-time' and 'Safe Route'.
- Right panel: A centered login form area with the title 'Welcome Back' and a subtitle notice explaining this portal is for Admin, Authority, and Super Admin roles only.

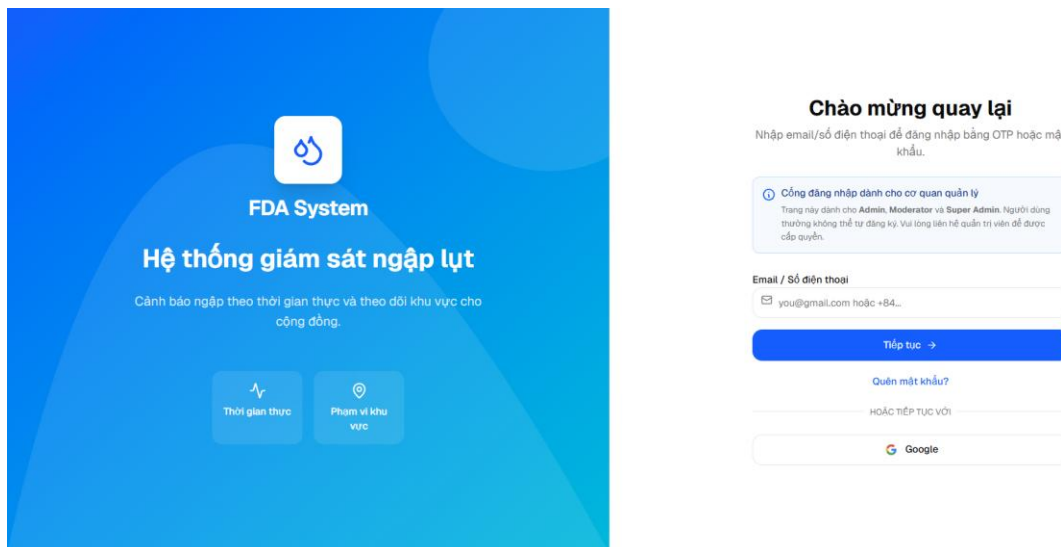


Figure 32: Login – Login Page Overview

Step 2: Enter Email or Phone Number

The user enters their registered email address or phone number into the 'Email / Phone number' input field.

The user then clicks 'Continue' to proceed.

The system calls the identifier-check API to determine the appropriate authentication method for the account.

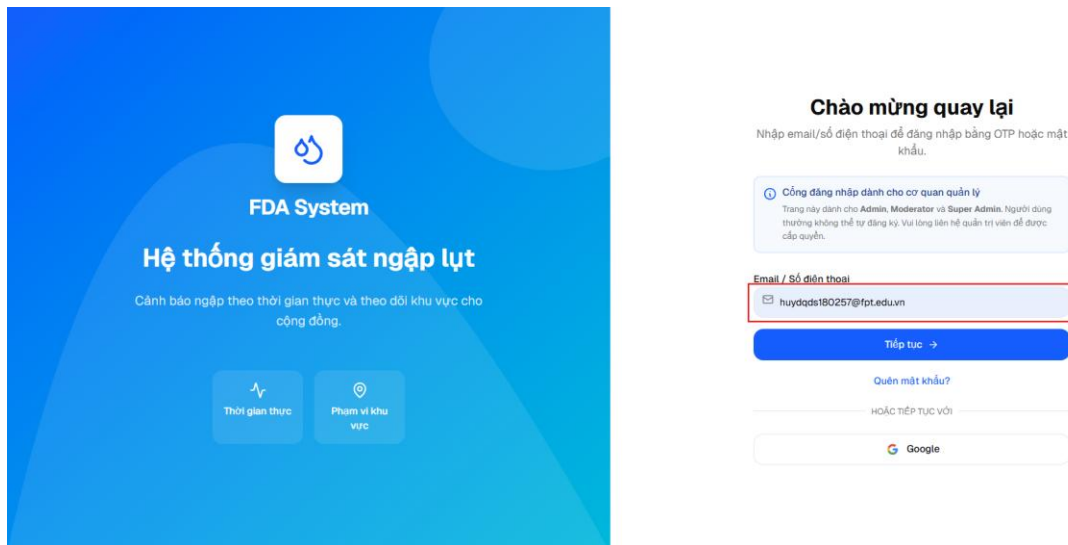


Figure 33: Login – Enter Identifier

Step 3.1: Login with Password

If the system determines the account uses password-based authentication, the form transitions to a password entry step:

- A text line displays: “**Account:** <user's email/phone>”
- A “**Password**” field appears with a lock icon on the left and a show/hide eye icon on the right.
- A “**Forgot password?**” link appears next to the password label.
- A “**Sign In**” → primary button and a “**Back**” secondary button are shown.

The user enters their password and clicks 'Sign In' to submit.

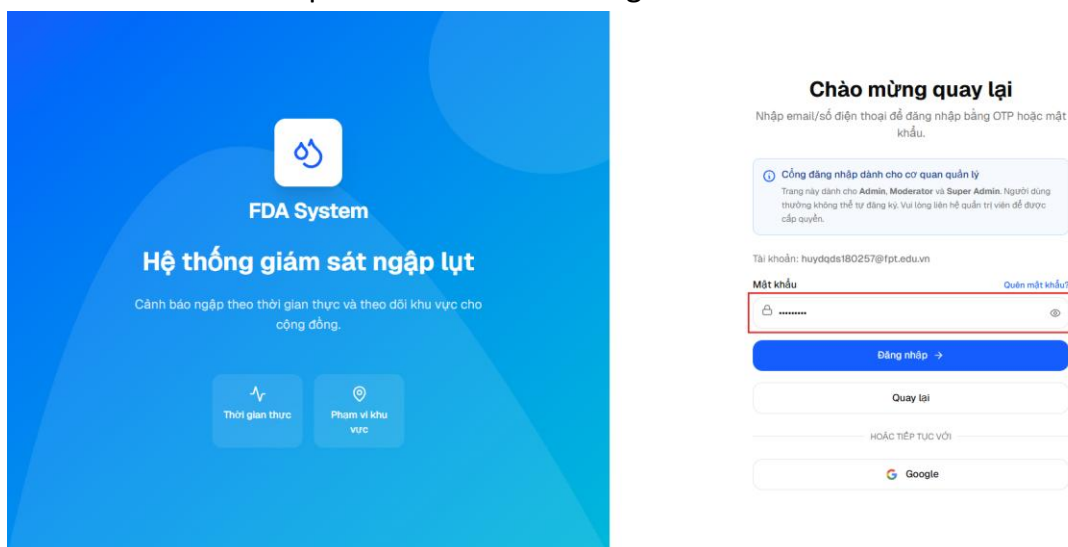


Figure 34: Login – Password Entry Form

Step 3.2: Login with OTP

If the system determines the account uses OTP-based authentication, the system automatically sends an OTP to the identifier and the form transitions to an OTP entry step:

- Header text shows: **“OTP sent to: <identifier>”**
- A 6-digit OTP input field is displayed.
- A countdown timer shows OTP expiry time (e.g., '4:59').
- “Sign In” primary button and two secondary buttons: **“Change email/phone”** and **“Resend OTP”**.

The user enters the 6-digit OTP received via email or phone and clicks **“Sign In”**.

The image shows two parts of the login interface. On the left is a blue sidebar for the 'FDA System' with the title 'Hệ thống giám sát ngập lụt' and a warning about real-time monitoring. On the right is the main login form titled 'Chào mừng quay lại'. It includes a login link for Admin, Moderator, and Super Admin, an OTP resend link for user huydqds180257@fpt.edu.vn, and a 6-digit OTP input field with a 4:48 countdown timer. Below the input field is a blue 'Xác thực OTP' button, and two links for 'Đổi email/số điện thoại' and 'Gửi lại OTP'. At the bottom, there is a 'HOẶC TIẾP TỤC VỚI' section with a Google login option.

Figure 35: Login – OTP Entry Form

Step 4: Successful Login – Redirect to Dashboard

Upon successful credential verification, the system:

- Validates the user's role (SUPER_ADMIN, ADMIN, or MODERATOR).
- Redirects Admin/Super Admin roles to the '/admin' dashboard.
- Redirects Authority roles to the '/authority' dashboard.

Displays a green toast notification: 'Signed in successfully!'

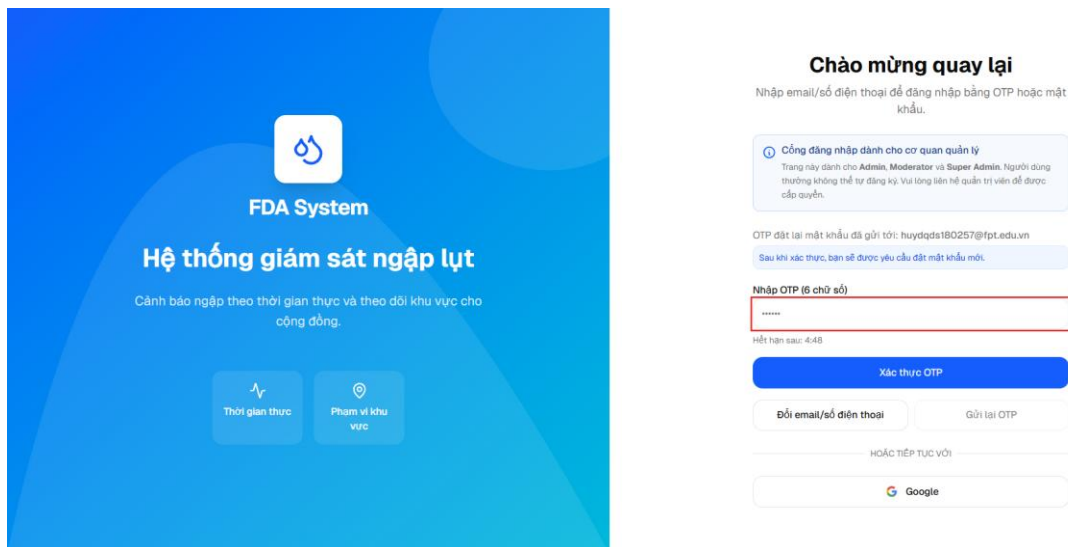


Figure 36: Login – Successful Login (Role-based Redirect)

Step 5: Login with Google Account

As an alternative, the user can click the 'Google' button below the form separator.

- The system calls the Google OAuth initialization API to generate the authorization URL.
- The browser is redirected to Google's sign-in page (accounts.google.com).
- The user selects their Google account and grants permission.
- Google redirects back to /auth/callback, completing authentication and redirecting to the role-based dashboard.

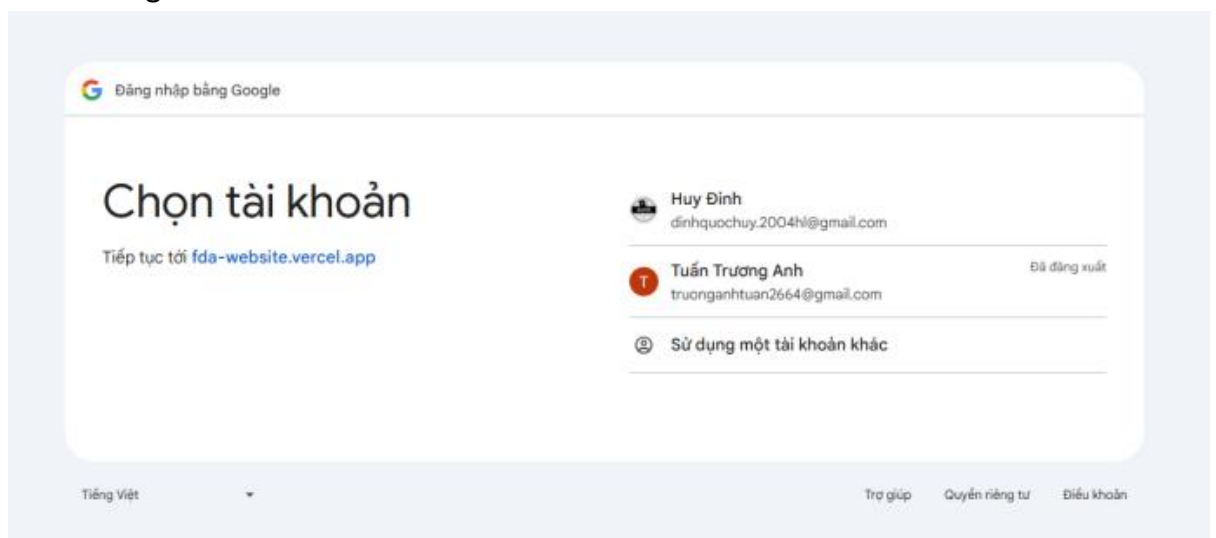


Figure 37: Login – Google OAuth Login Flow

3.2.1.2 Forgot and Reset Password

Step 1: Initiate Forgot Password from the Login Form

On the Login screen (identifier step or password step), the user clicks the “Quên mật khẩu?” link.

The system requires the identifier field to contain a value before initiating forgot password.

If the identifier field is empty, a warning is shown: “Vui lòng nhập email hoặc số điện thoại.”

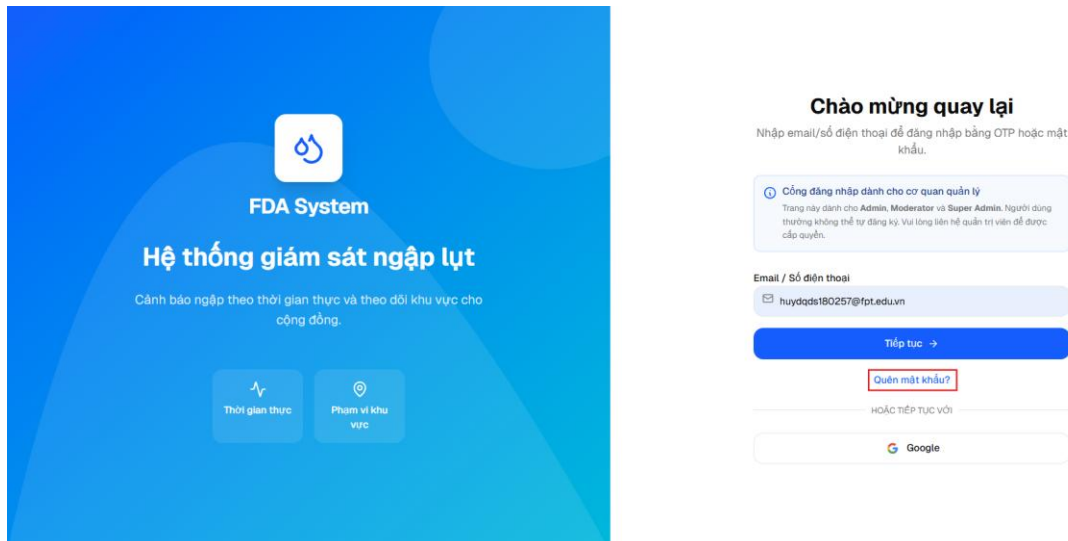


Figure 38: Forgot Password – Trigger Link Location

Step 2: OTP Sent for Password Reset

With a valid identifier entered, the system calls the OTP send API and transitions the form to OTP verification mode (FORGOT_PASSWORD mode).

A success toast is displayed: “Đã gửi OTP đặt lại mật khẩu!”

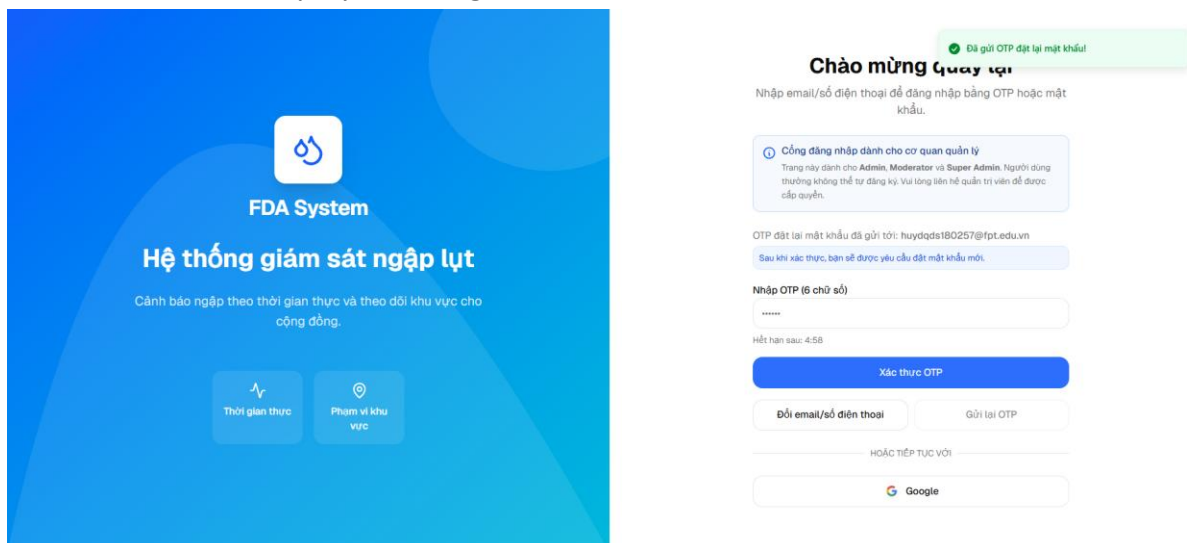


Figure 39: Forgot Password – OTP Sent Confirmation

Step 3: Enter and Verify OTP

The user retrieves the 6-digit OTP from their email or phone.

The user types the OTP into the input field and clicks 'Verify OTP'.

If the OTP is fewer than 6 digits: yellow toast “Xác thực OTP”.

If the OTP has expired, the “Gửi lại OTP” button becomes clickable to request a new code.

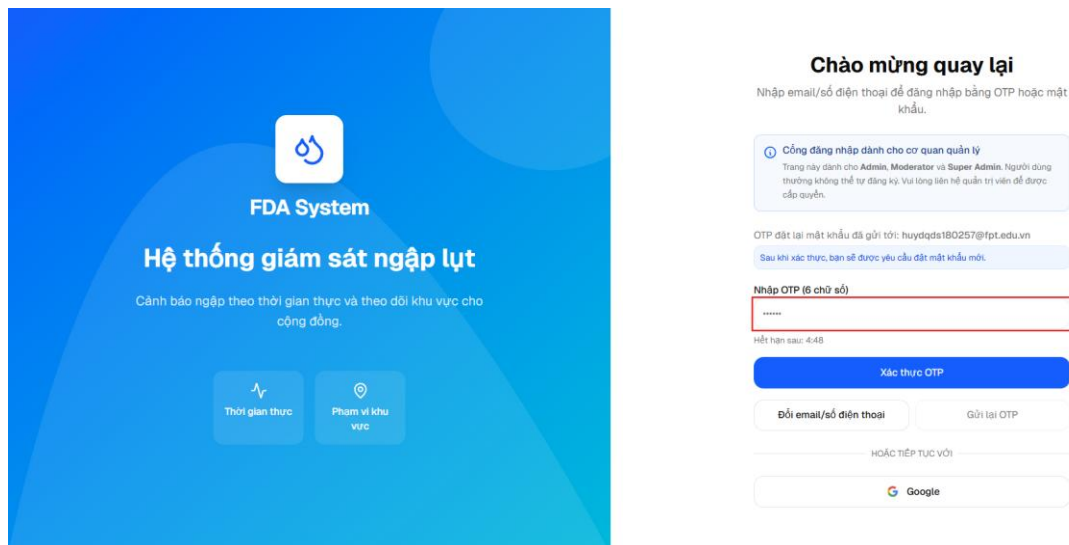


Figure 40: Forgot Password – OTP Input Interaction

Step 4: Set a New Password

Upon successful OTP verification in FORGOT_PASSWORD mode, the system logs the user in and automatically opens the 'Set a new password' modal over the dashboard. Inside the dialog, the user enters:

- New Password – must satisfy all five security requirements.
- Confirm Password – must match the new password exactly.

Requirements are shown as a real-time checklist:

- At least 8 characters
- Contains uppercase (A-Z)
- Contains lowercase (a-z)
- Contains a number (0-9)
- Contains a special character (!@#\$...)

A password match indicator also appears when the user types in the “Xác nhận mật khẩu” field.

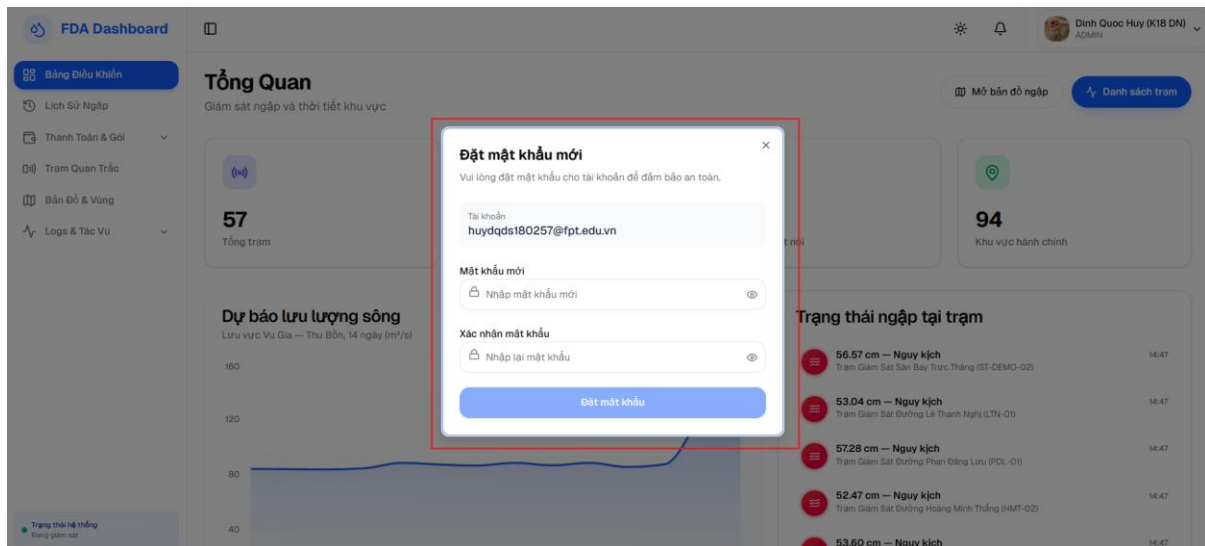


Figure 41: Forgot Password – Set New Password Modal

Step 5: Confirm Reset and Log In Normally

After the user clicks “Đặt mật khẩu” and the API confirms success:

- Green toast: “Đặt lại mật khẩu thành công!”
- The modal closes automatically.
- The user is fully logged in and can use the system normally.

For future logins, the user signs in with their email and the newly set password (password-based flow).

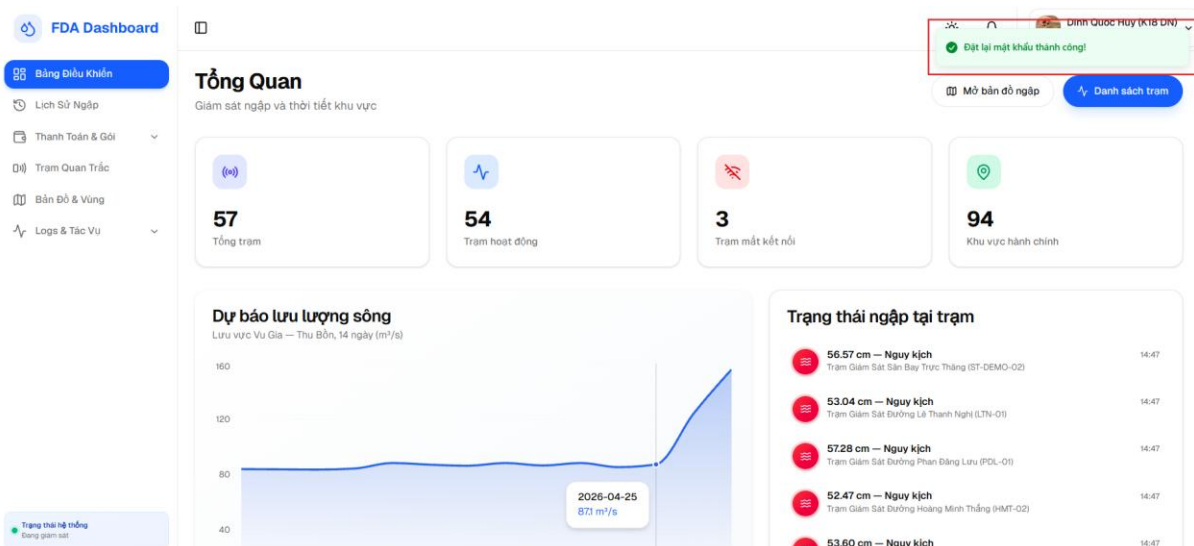


Figure 42: Forgot Password – Password Reset Successful

3.2.2 Profile Management

3.2.1.1 Information Tab

Step 1: Open the Profile Panel

- Click on your avatar or display name at the top-right corner of the dashboard.
- From the dropdown menu, select “**Hồ sơ**” to open the profile modal.
- The panel opens on the Information tab by default.

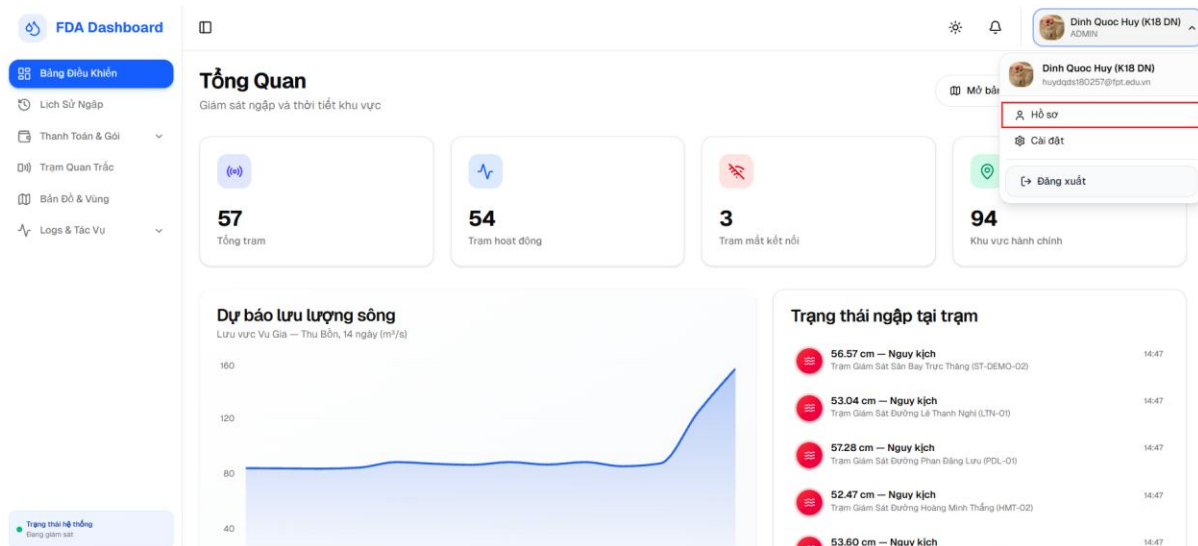


Figure 43: View Profile

Step 2: View Account Information

The Information tab displays the following details:

- **Họ và tên** - Your display name. This field can be edited.
- **Email** - The email address associated with your account. This field cannot be changed.
- **Số điện thoại** - Your phone number registered to the account (view only on this tab; updates are done in the Phone Number tab).
- **Nhà cung cấp / Trạng thái** - The login method used (e.g., local or Google) and the current account status (e.g., Active).
- **Trạng thái xác nhận** - Shows whether your Email and Phone Number have been verified.
- **Vai trò** - The role(s) assigned to your account (e.g., User, Admin, Authority, Super Admin).

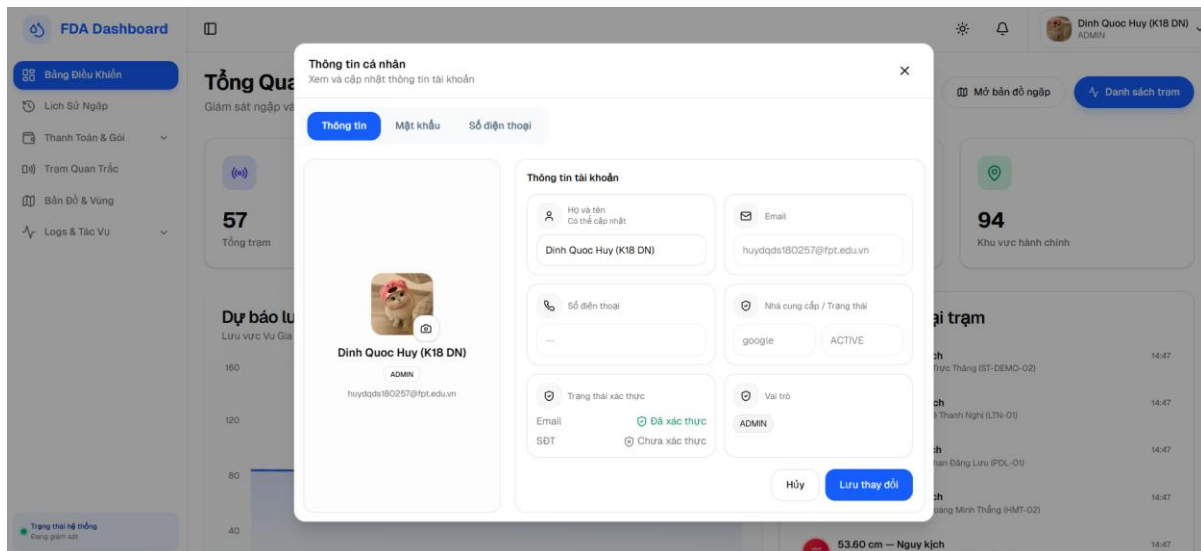


Figure 44: View account profile information

Step 3: Update Your Full Name

- Click inside the “Họ và tên” field and type your new name.
- The name entered here will be displayed throughout the system.

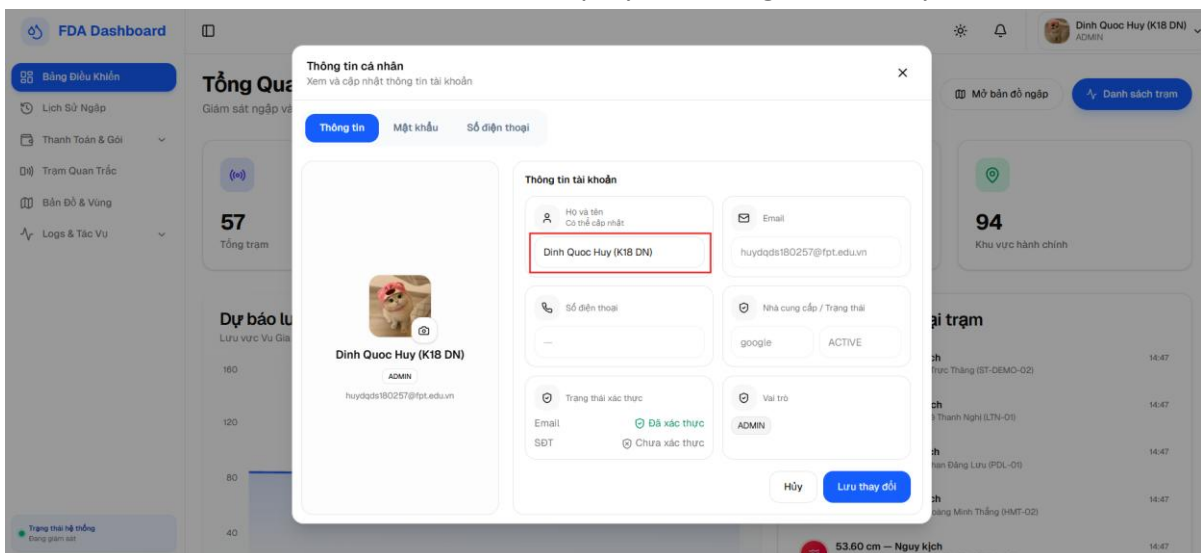


Figure 45: View account profile information

Step 4: Update Your Profile Picture

- On the left side of the panel, click the small camera icon located at the bottom-right corner of your avatar.
- A file picker will open. Select an image file from your device (JPG, PNG, etc.).
- The new image will appear as a preview immediately. It will only be saved after you click Save.

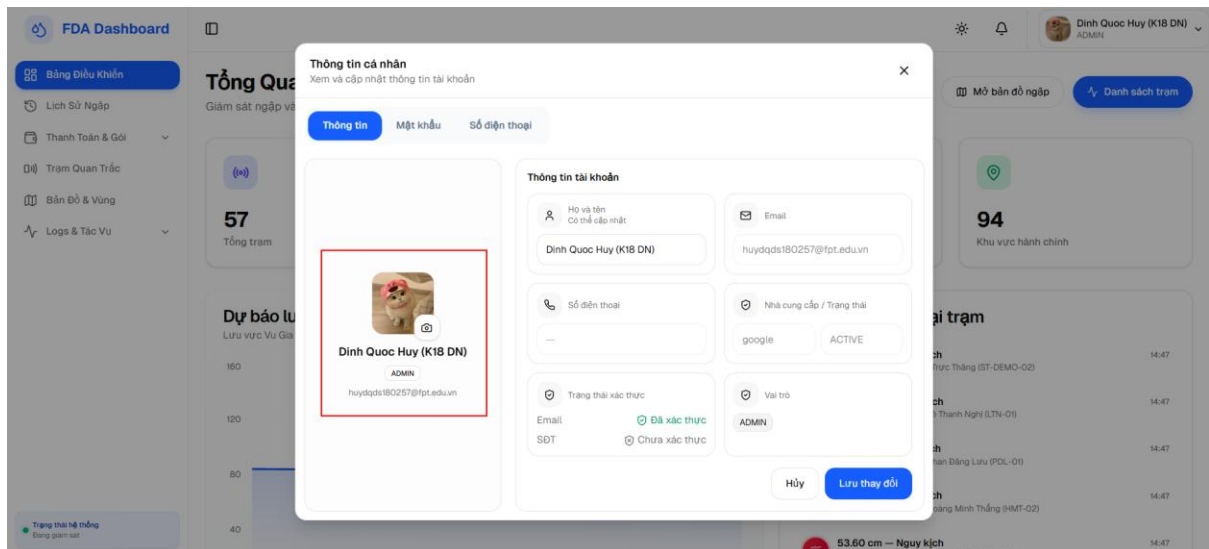


Figure 46: View account profile information

Step 5: Save or Cancel Changes

After making your changes, click the "Lưu thay đổi" button at the bottom-right of the form.

- **Success** - The profile is updated and the panel closes automatically.
- **Cancel** - Click "Hủy" to discard all unsaved changes and restore the original values.

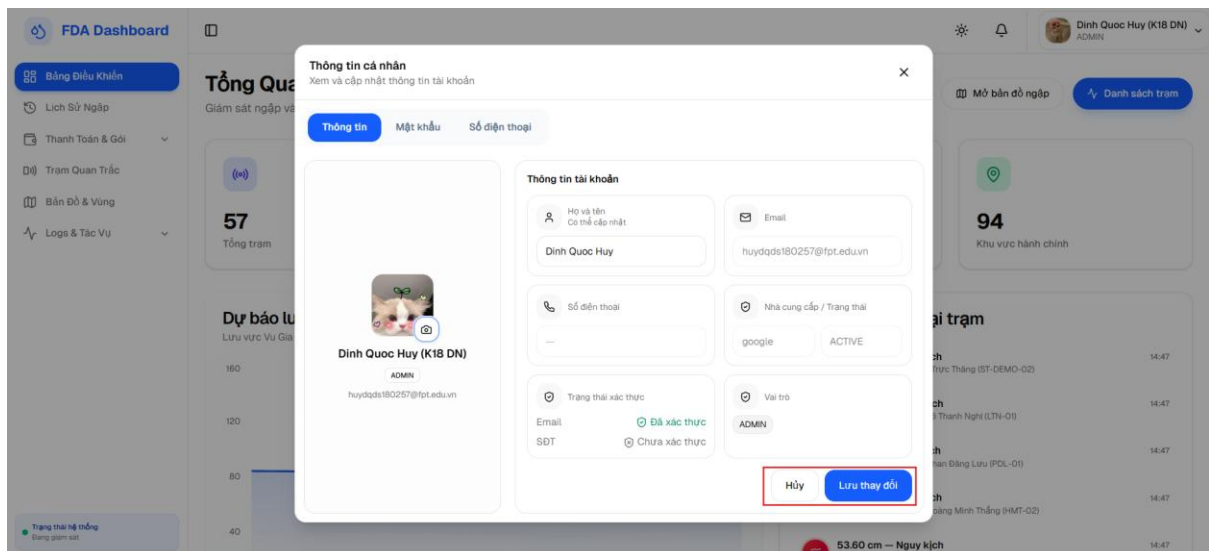


Figure 47: View account profile information

3.2.1.2 Password Tab

Step 1: Switch to the Password Tab

- Inside the profile panel, click the “**Mật khẩu**” tab at the top of the right section.

- The Change Password form will appear.

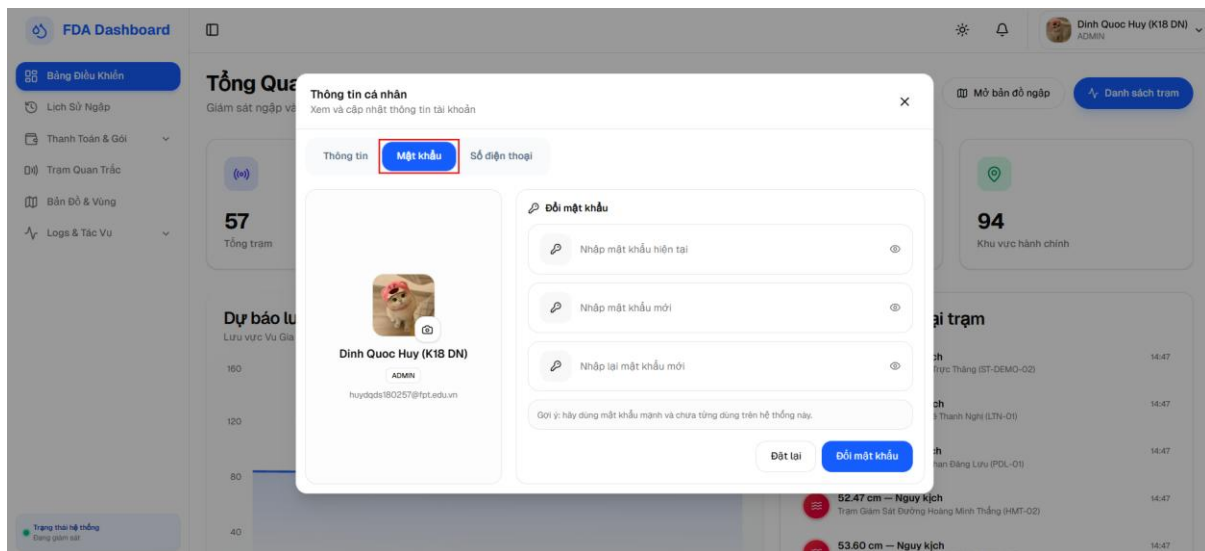


Figure 48: View account password information

Step 2: Fill in the Password Fields

The form contains three required fields:

- **Nhập mật khẩu hiện tại** - Enter the password you are currently using to log in. Use the eye icon on the right of the field to show or hide the characters.
- **Nhập mật khẩu mới** - Enter the new password you want to set. Use the eye icon to show or hide the characters.
- **Nhập lại mật khẩu mới** - Re-enter the new password to confirm it matches. Use the eye icon to show or hide the characters.

Note: The new password and the Confirm New Password must be exactly the same. If they do not match, the system will display an error and prevent the

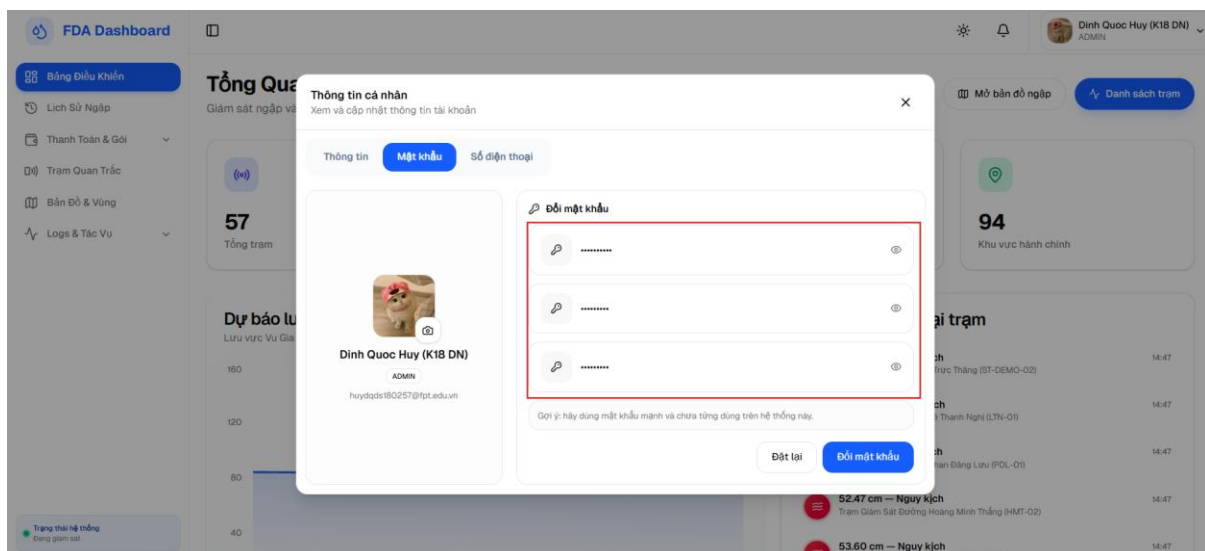


Figure 49: View account password information

Step 3: Submit the Password Change

After filling in all three fields, click the “Đổi mật khẩu” button at the bottom-right.

- **Success** - The password is updated and the panel closes automatically. You will need to use the new password the next time you log in.
- **Error (fields empty)** - If any field is left blank, a message will appear: "Vui lòng điền đầy đủ thông tin."
- **Error (passwords do not match)** - If the New Password and Confirm New Password do not match, a message will appear: "Mật khẩu xác thực không khớp."
- **Error (wrong current password)** - If the current password entered is incorrect, the system will show an error from the server.

Click “Reset” to clear all three fields and start over without closing the panel

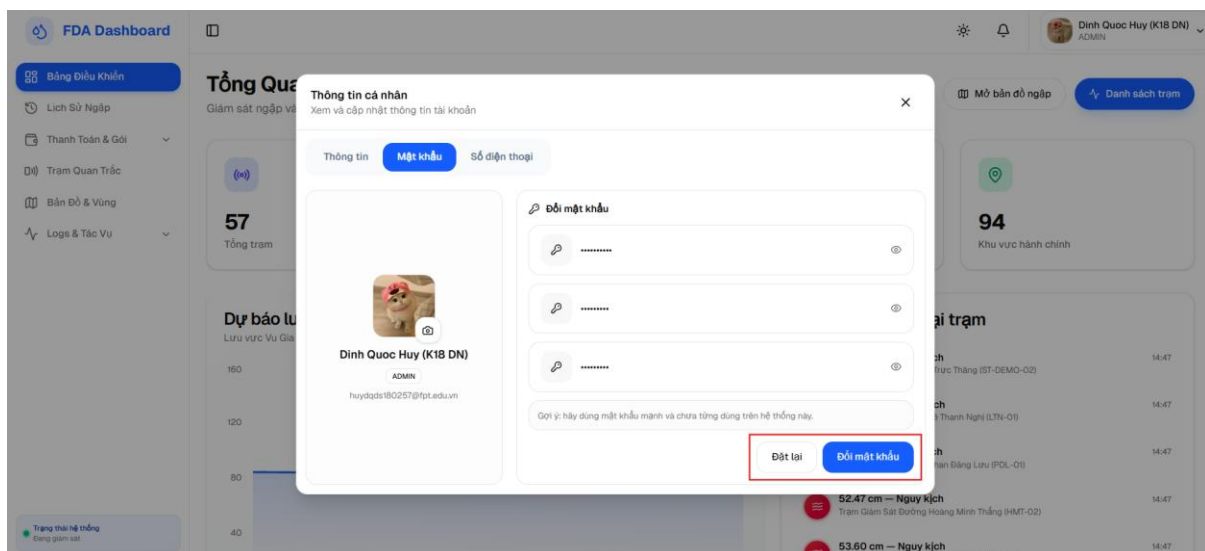


Figure 50: Submit the Password Change

3.2.1.3 Phone Number Tab

Step 1: Switch to the Phone Number Tab

- Inside the profile panel, click the “Số điện thoại” tab at the top of the right section.

- The Update Phone Number form will appear.

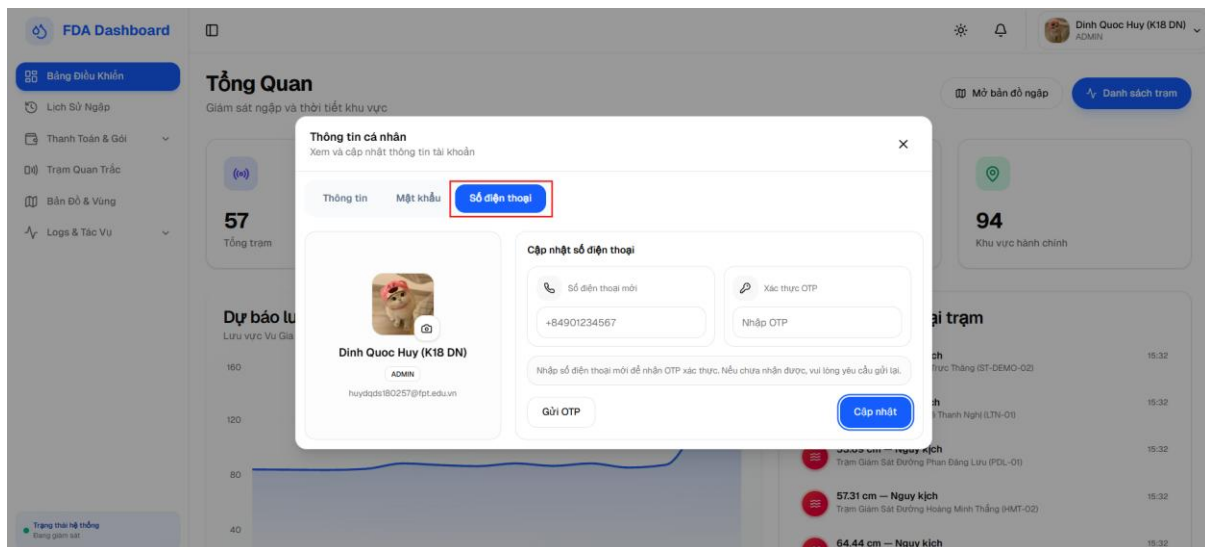


Figure 51: Phone Number Tab

Step 2: Enter the New Phone Number

- In the "Số điện thoại mới" field, type the phone number you want to link to your account.
- Use the international format if applicable (e.g., +84901234567 for a Vietnamese number).
- Make sure the number is correct, as the OTP verification code will be sent to this number.

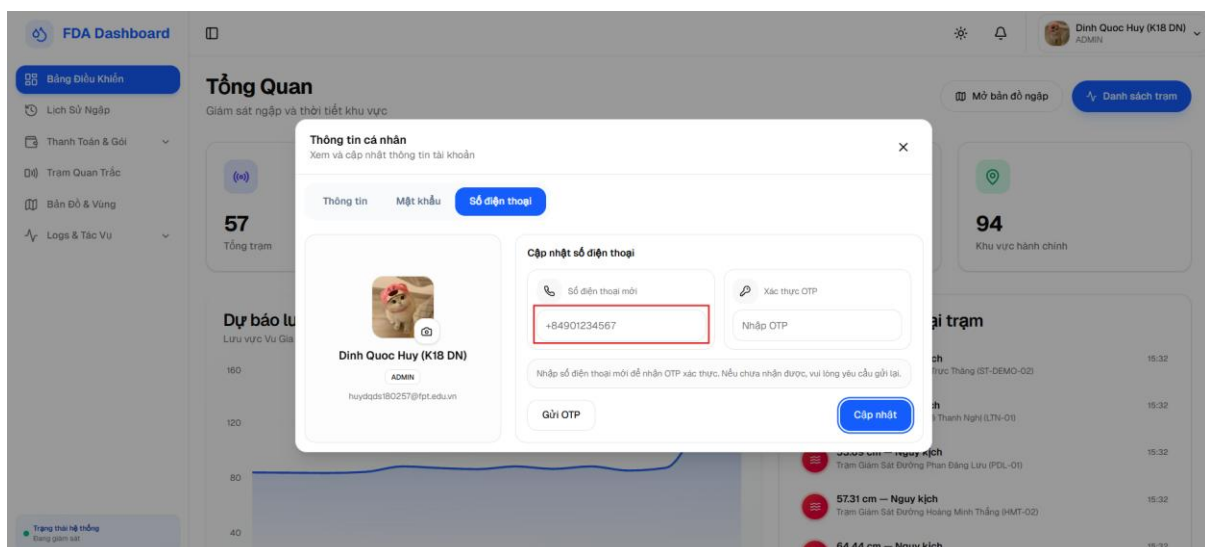


Figure 52: Enter the New Phone Number

Step 3: Request an OTP Code

Click the “Gửi OTP” button on the left side of the action bar.

- **Success** - An OTP code will be sent as an SMS to the phone number you entered. A countdown timer will appear, showing how many seconds remain before the OTP expires.
- **Error (empty field)** - If the phone number field is empty, the system will show a message: "Vui lòng điền đầy đủ thông tin."
- **Error (invalid number)** - If the number is not valid or cannot receive SMS, the system will display an error.

Note: The OTP code is valid for a limited time (shown by the countdown timer). If it expires before you enter it, click “Gửi OTP” again to receive a new code.

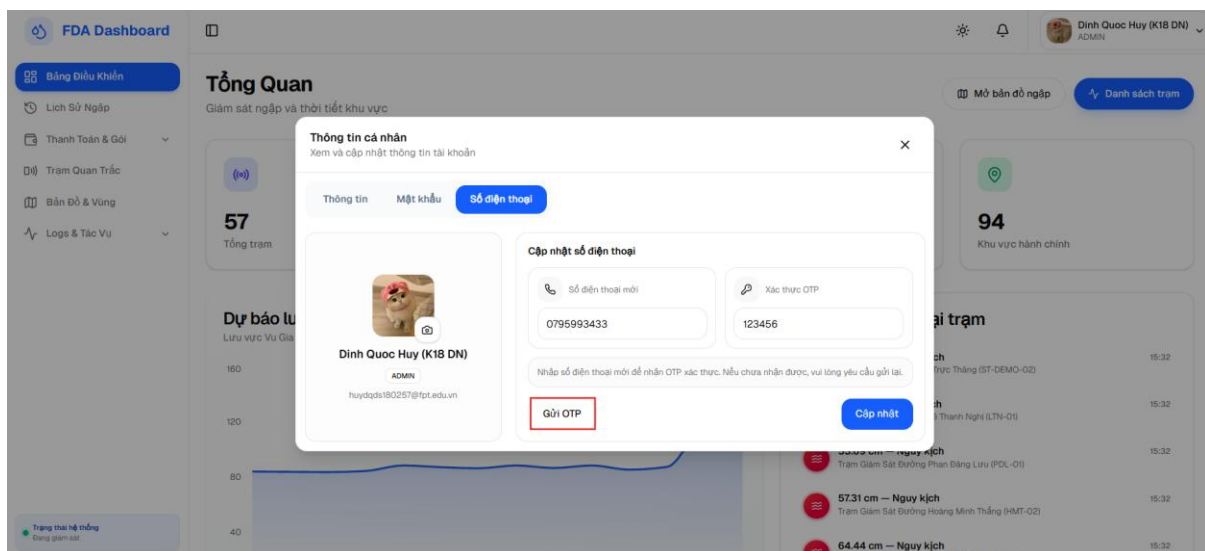


Figure 53: “Send OTP” button

Step 4: Enter the OTP Code

- Check your phone for the SMS message containing the OTP code.
- Type the 6-digit OTP code into the “Xác thực OTP” field.

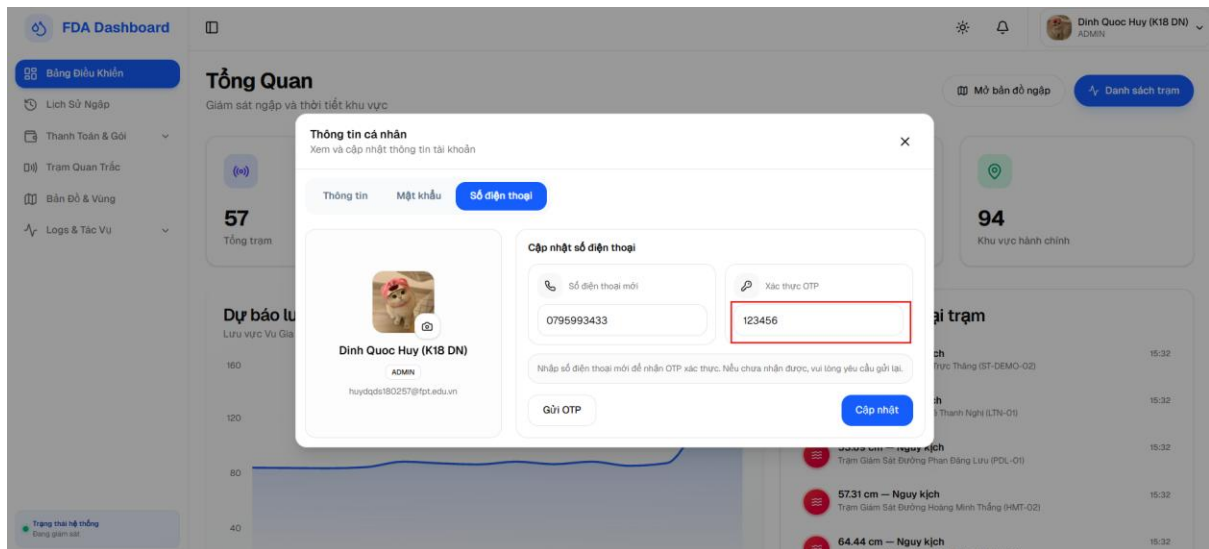


Figure 54: Enter the OTP Code

Step 5: Confirm the Update

Click the “Cập nhật” button on the right side of the action bar.

- **Success** - The phone number is updated and the panel closes automatically. Your new phone number now appears on the Information tab.
- **Error (OTP expired or wrong)** - If the code is incorrect or has expired, an error message will appear. Click “Gửi OTP” again to request a new code.
- **Error (fields empty)** - If either field is empty, the system will show: “Vui lòng điền đầy đủ thông tin.”

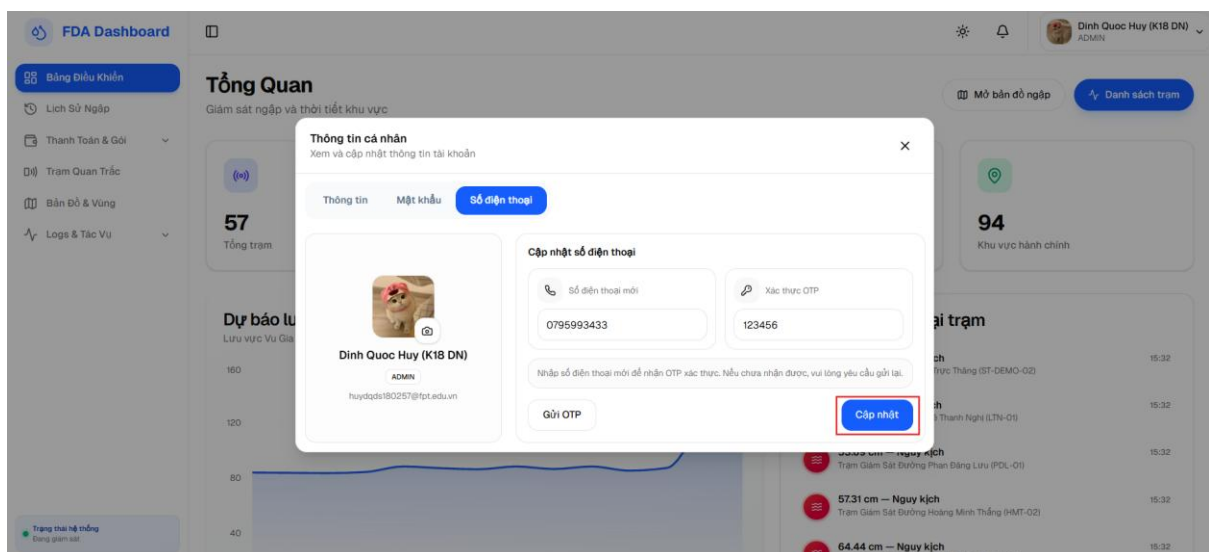


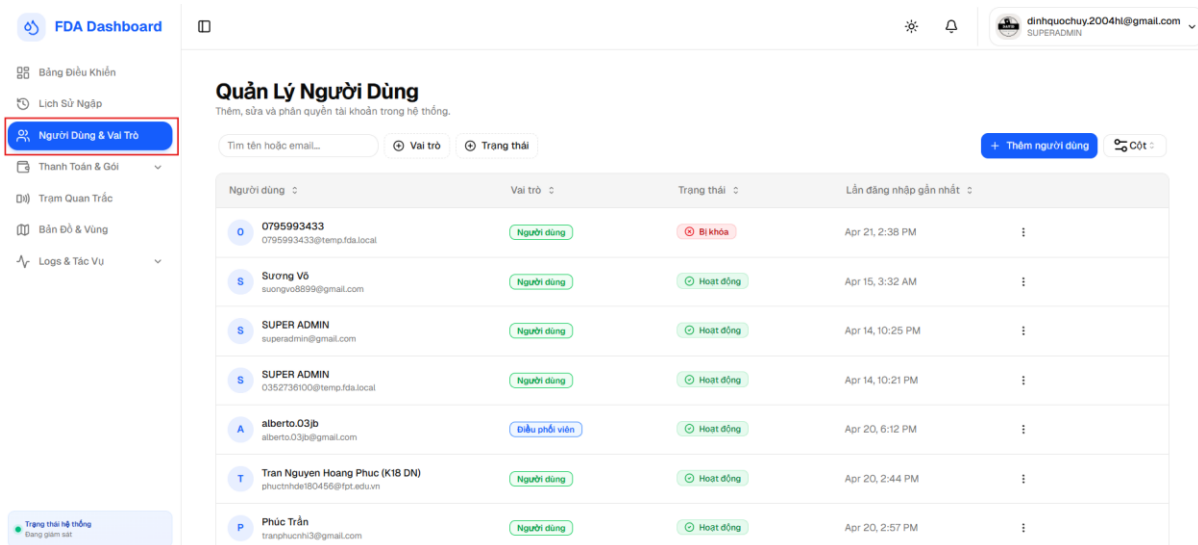
Figure 55: Confirm the Update

3.2.3 User Management

3.2.3.1 View and Search Users

Step 1: Navigate to the Users & Roles Tab

- After logging in with the Super Admin account, the system redirects to the Admin Dashboard.
- On the left-hand sidebar, click on "Người Dùng và Vai Trò" to open the user management section
- The main area now displays a data table listing all registered users in the system.



Quản Lý Người Dùng
Thêm, sửa và phân quyền tài khoản trong hệ thống.

Tim tên hoặc email... Vai trò: Trạng thái: + Thêm người dùng Cột

Người dùng	Vai trò	Trạng thái	Lần đăng nhập gần nhất
0795993433 0795993433@temp.fda.local	Người dùng	Bi khóa	Apr 21, 2:38 PM
Sương Võ suongvo889@gmail.com	Người dùng	Hoạt động	Apr 15, 3:32 AM
SUPER ADMIN superadmin@gmail.com	Người dùng	Hoạt động	Apr 14, 10:25 PM
SUPER ADMIN 0352736100@temp.fda.local	Người dùng	Hoạt động	Apr 14, 10:21 PM
alberto.03b alberto.03b@gmail.com	Điều phối viên	Hoạt động	Apr 20, 6:12 PM
Tran Nguyen Hoang Phuc (K18 DN) phucthndn80456@fpt.edu.vn	Người dùng	Hoạt động	Apr 20, 2:44 PM
Phúc Trần tranphuch3@gmail.com	Người dùng	Hoạt động	Apr 20, 2:57 PM

Figure 56: View User List

Step 2: Understand the User Table

The table presents the following columns for each user account:

- **Người dùng** – Displays the user's full name and email address, along with an avatar initial for quick recognition.
- **Vai trò** – Shows the assigned role(s) of the user, displayed as a colored badge:
 - **Super Admin** – Highest-level administrator
 - **Admin** – System administrator
 - **Moderator** – Government/authority-level account
 - **User** – Standard end-user account
- **Trạng thái** – Shows the current account status:
 - **Active** – The account is in normal use
 - **Banned** – The account has been blocked from logging in
 - **Inactive** – The account exists but is not currently active
- **Lần đăng nhập gần nhất** – The most recent date and time the user logged into the system.

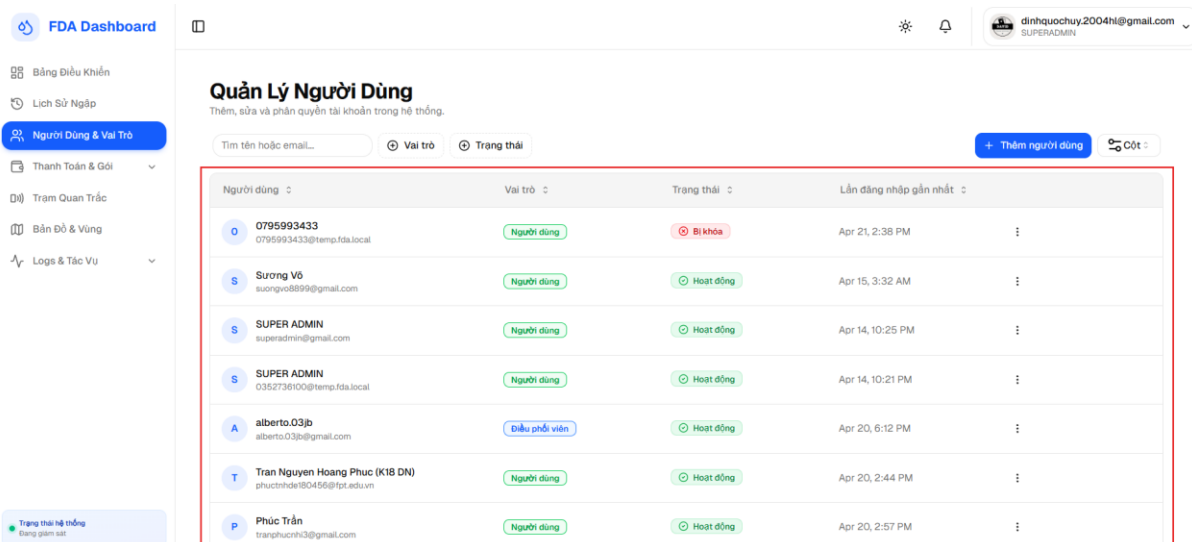


Figure 57: View User List

Step 3: Search and Filter the User List

The toolbar above the table provides multiple ways to find specific users quickly:

- **Search box** – Type a name or email address to filter results as you type. Results update automatically.
- **Role filter** – Click the “**Vai trò**” dropdown to show only users who have a specific role (e.g., Admin, Authority, User).
- **Status filter** – Click the “**Trạng thái**” dropdown to display only users with a chosen account status (e.g., Active, Banned).
- **Multiple filter** – Click the “**Cột**” dropdown to display optional columns for filter.

Multiple filters can be combined at the same time for a more refined search.

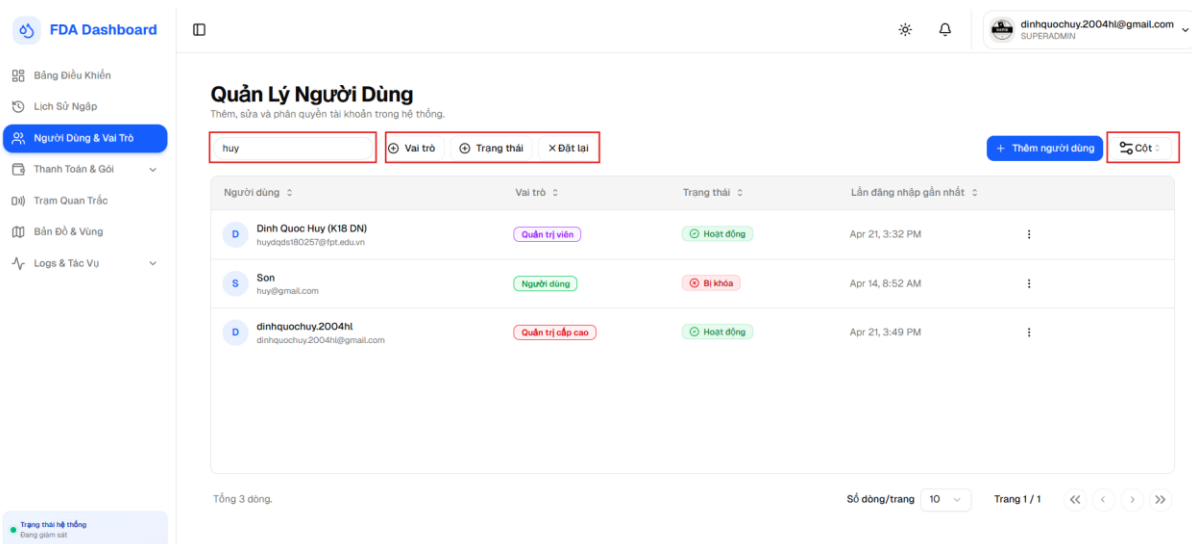


Figure 58: Search User

3.2.3.2 Create a New User

Step 1: Open the Create User Form

- In the User Management page, click the “+ Thêm người dùng” button located at the top-right of the user table toolbar.
- A dialog box titled “Tạo người dùng mới ” will appear on the screen.

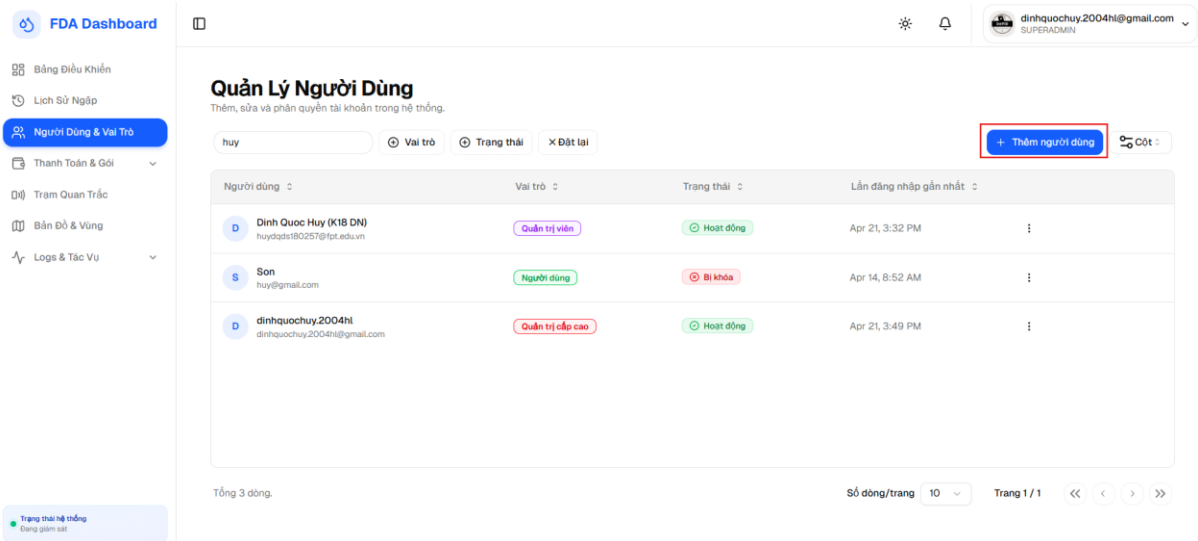


Figure 59: Add User

Step 2: Fill in User Information

The form contains the following fields (fields marked with * are required):

- **Email *** – Enter a valid email address for the new user (e.g., user@example.com). This email will be used for login and system notifications.
- **Mật khẩu *** – Set an initial password for the account. The password must be at least **6 characters** long.
- **Họ và tên *** – Enter the user's full name as it will be displayed in the system.
- **Số điện thoại** (optional) – Enter a 10–11 digit Vietnamese phone number. This field is not required but can be provided for contact purposes.
- **Trạng thái** – Select the initial account status from the dropdown:
 - **Active** – The user can log in immediately after the account is created.
 - **Inactive** – The account is created but the user cannot log in yet.
 - **Banned** – The account is created in a blocked state.
- **Vai trò *** – Choose the role assigned to this user. Available roles are:
 - **User** – Standard citizen/end-user (default)
 - **Admin** – System administrator
 - **Super Admin** – Highest-level administrator

- **Moderator** – Government or regulatory authority

The screenshot displays the 'FDA Dashboard' interface. The main section is titled 'Quản Lý Người Dùng' (User Management) with a subtitle 'Thêm, sửa và phân quyền tài khoản trong hệ thống' (Add, edit and assign permissions to accounts in the system). A modal window titled 'Tạo người dùng mới' (Create new user) is open, showing the following fields:

- Email ***: Input field with 'nguyenvana@example.com'.
- Mật khẩu ***: Input field with placeholder 'Nhập mật khẩu (tối thiểu 6 ký tự)' (Enter password (minimum 6 characters)).
- Họ và tên ***: Input field with 'Nguyễn Văn A'.
- Số điện thoại**: Input field with '0901234567'.
- Trạng thái**: Dropdown menu with 'Hoạt động' (Active) selected.
- Vai trò ***: Dropdown menu with 'Người dùng' (User) selected.

At the bottom of the modal are buttons for 'Hủy' (Cancel) and 'Tạo người dùng' (Create user). The background shows a table of existing users with columns for name, email, and status. The sidebar on the left contains navigation links like 'Bảng Điều Khiển', 'Lịch Sử Ngập', 'Người Dùng & Vai Trò', 'Thanh Toán & Gói', 'Tram Quan Trắc', 'Bản Đồ & Vùng', and 'Loge & Tác Vụ'.

Figure 60: View Create User Form

Step 3: Submit the Form

After filling in all required fields, click the “Tạo người dùng” button at the bottom of the dialog.

- If all information is valid, the system will create the account and display a success notification: “Tạo người dùng thành công!” The new user will immediately appear in the user table.
- If any required field is missing or invalid (e.g., incorrect email format, password too short, invalid phone number), the system will highlight the problematic field in red and display a clear error message asking you to correct it.

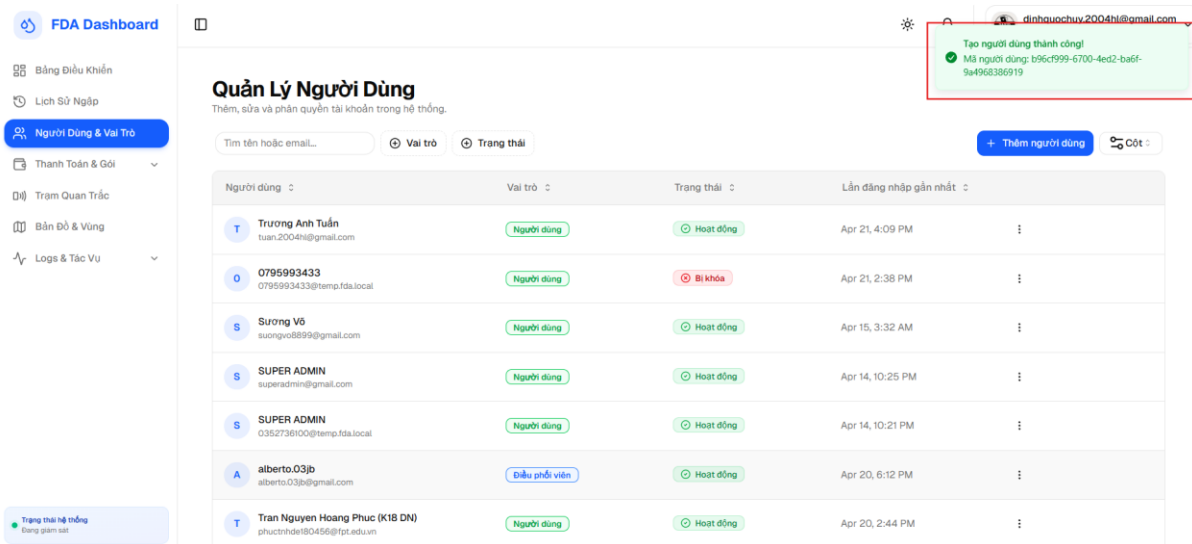


Figure 61: Submit the Form

3.2.3.3 Edit an Existing User

Note: Only user accounts that were **created by super admin** can be edited. Accounts registered directly by users through the app cannot be edited from this interface.

Step 1: Open User Management Module

- In the user table, locate the user account you want to update.
- Click the : (three-dot menu) icon located at the far-right of that user's row.
- A small dropdown menu will appear showing the available actions.

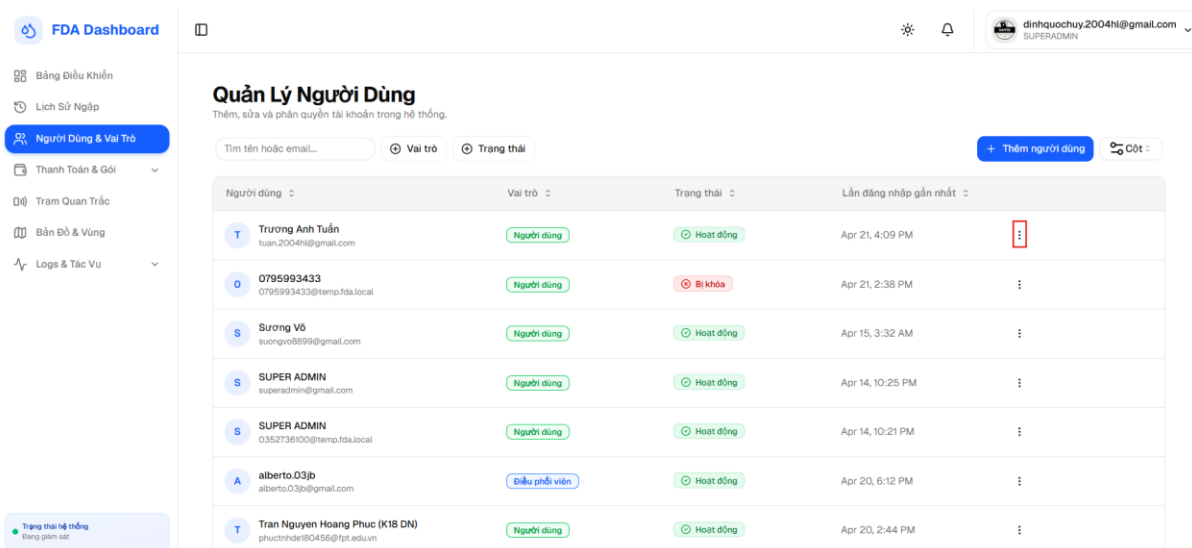


Figure 62: Edit User

Step 2: Select “Sửa”

- Click ‘Sửa’ from the dropdown menu.

- An “**Sửa người dùng**” dialog box will appear, pre-filled with the user's current information.
- If the selected user was not created by an admin, the **Edit** option will be disabled (grayed out), and an error toast will appear indicating that only admin-created accounts can be edited.

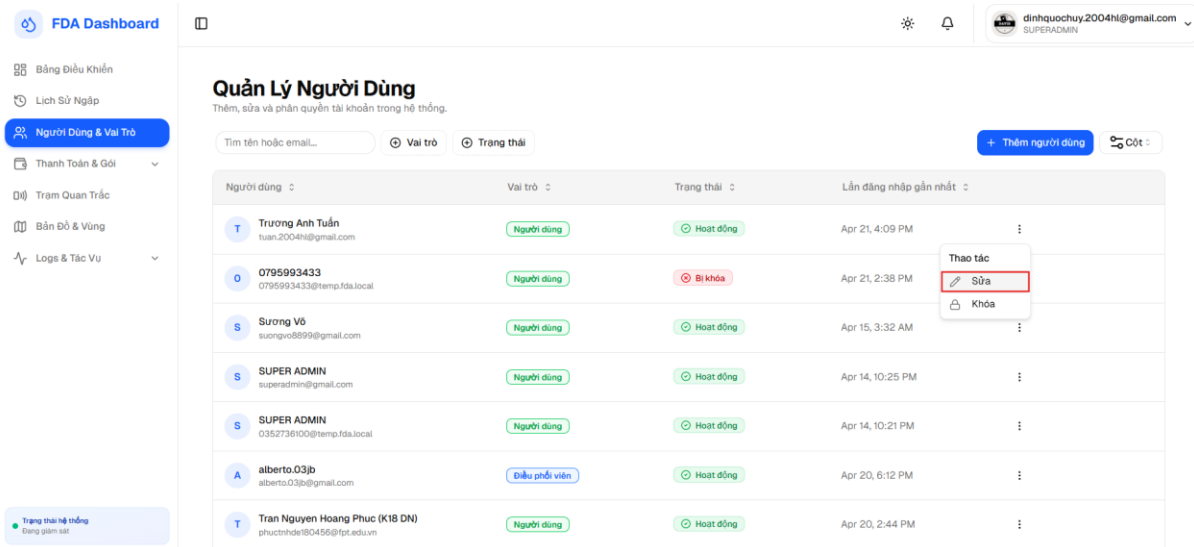


Figure 63: Edit User

Step 3: Update User information

This edit form contains the following fields:

- **Email** – This field is shown for reference only and **cannot be changed**.
- **Họ và tên** – Update the user's display name.
- **Số điện thoại** (optional) – Update or add a phone number (10–11 digits).
- **Trạng thái** – Change the account's current status (Active / Inactive / Banned).
- **Vai trò** – Change the user's assigned role. Use the radio buttons to select one of the available roles:
 - User
 - Admin
 - Super Admin
 - Moderator

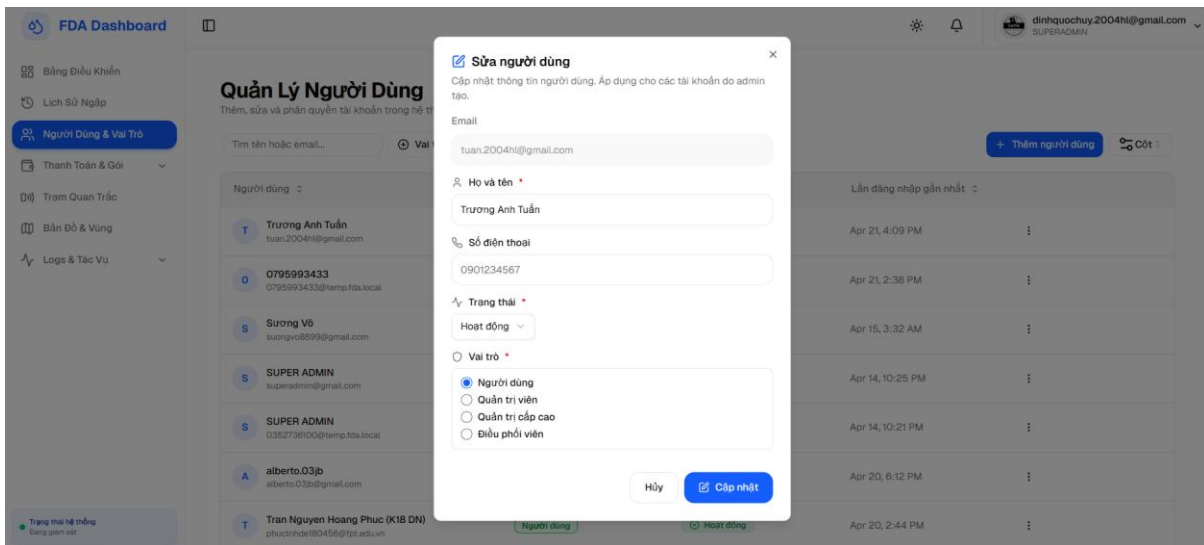


Figure 64: Update User information

Step 4: Save the changes

Click the "Cập nhật" button to save all changes.

- If the update is successful, a notification will appear: "Cập nhật người dùng thành công!" The table will automatically refresh to reflect the changes.
- If an error occurs (e.g., a required field is empty or the phone number is in the wrong format), the system will display an error message next to the relevant field.

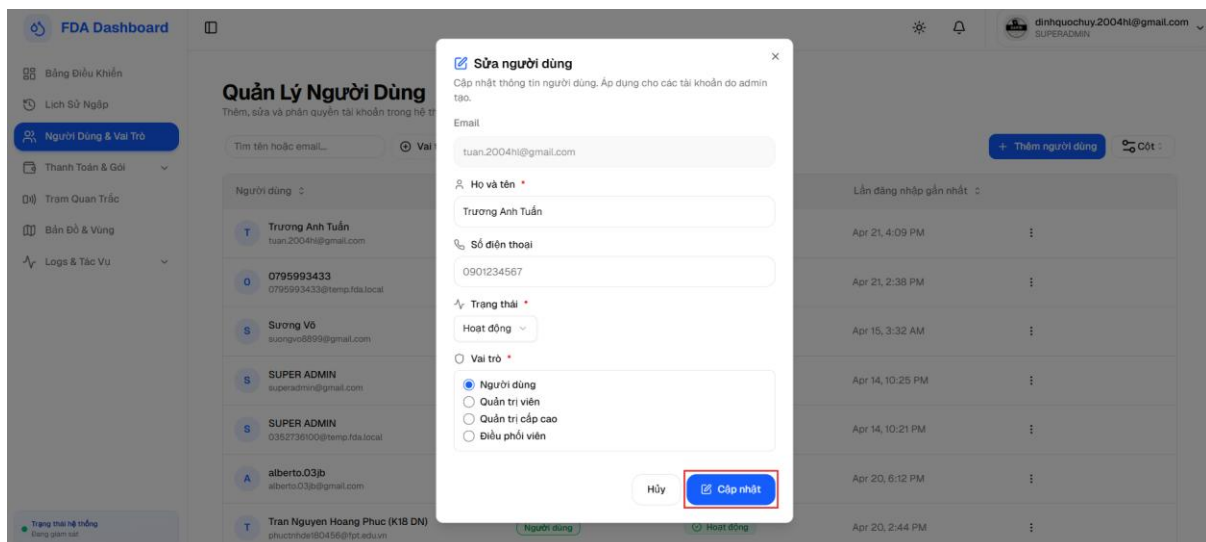


Figure 65: Save the changes

3.2.3.4 Ban/ Unban a User

This action lets the Super Admin immediately block a user from logging into the system, or restore access to a previously banned account.

Step 1: Open the Actions Menu

- In the user table, locate the user account you want to ban or unban.
- Click the : (three-dot menu) icon at the right end of that user's row.

The screenshot shows the 'Quản Lý Người Dùng' (User Management) page in the FDA Dashboard. The page has a sidebar with navigation links and a main content area with a table of users. The table has columns: 'Người dùng' (User), 'Vai trò' (Role), 'Trạng thái' (Status), and 'Lần đăng nhập gần nhất' (Last login). The user 'Trương Anh Tuấn' is highlighted, and the three-dot menu icon is visible at the end of his row.

Người dùng	Vai trò	Trạng thái	Lần đăng nhập gần nhất
Trương Anh Tuấn tuan.2004hi@gmail.com	Người dùng	Hoạt động	Apr 21, 4:09 PM
0795993433 0795993433@temp.fda.local	Người dùng	Bị khóa	Apr 21, 2:38 PM
Sương Võ suongvo8899@gmail.com	Người dùng	Hoạt động	Apr 15, 3:32 AM
SUPER ADMIN superadmin@gmail.com	Người dùng	Hoạt động	Apr 14, 10:25 PM
SUPER ADMIN 0352736100@temp.fda.local	Người dùng	Hoạt động	Apr 14, 10:21 PM
alberto.03jb alberto.03jb@gmail.com	Điều phối viên	Hoạt động	Apr 20, 6:12 PM
Tran Nguyen Hoang Phuc (K18 DN) phucthhe180456@fpt.edu.vn	Người dùng	Hoạt động	Apr 20, 2:44 PM

Figure 66: Ban User

Step 2: Select "Ban" or "Unban"

- If the user's current status is Active or Inactive, the menu will show “Khóa”.
- If the user's current status is already Banned, the menu will show “Mở khóa”.

Click the appropriate option.

The screenshot shows the 'Quản Lý Người Dùng' (User Management) page in the FDA Dashboard. The user 'Trương Anh Tuấn' is highlighted, and the three-dot menu icon is visible at the end of his row. The menu is open, showing options: 'Sửa' (Edit) and 'Khóa' (Ban).

Người dùng	Vai trò	Trạng thái	Lần đăng nhập gần nhất
Trương Anh Tuấn tuan.2004hi@gmail.com	Người dùng	Hoạt động	Apr 21, 4:09 PM
0795993433 0795993433@temp.fda.local	Người dùng	Bị khóa	Apr 21, 2:38 PM
Sương Võ suongvo8899@gmail.com	Người dùng	Hoạt động	Apr 15, 3:32 AM
SUPER ADMIN superadmin@gmail.com	Người dùng	Hoạt động	Apr 14, 10:25 PM
SUPER ADMIN 0352736100@temp.fda.local	Người dùng	Hoạt động	Apr 14, 10:21 PM
alberto.03jb alberto.03jb@gmail.com	Điều phối viên	Hoạt động	Apr 20, 6:12 PM
Tran Nguyen Hoang Phuc (K18 DN) phucthhe180456@fpt.edu.vn	Người dùng	Hoạt động	Apr 20, 2:44 PM

Figure 67: Select "Ban" or "Unban"

Step 3: Confirm the Action

- A confirmation dialog will appear, showing:
 - The email address of the user about to be banned or unbanned.
 - A short explanation of what the action does:
 - “Người dùng sẽ không thể đăng nhập sau khi bị khóa.”

- “Người dùng có thể đăng nhập lại sau khi được mở khóa.”
- Click “**Khóa người dùng**” (or “**Mở khóa người dùng**”) to proceed, or click “**Hủy**” to go back without making any changes.

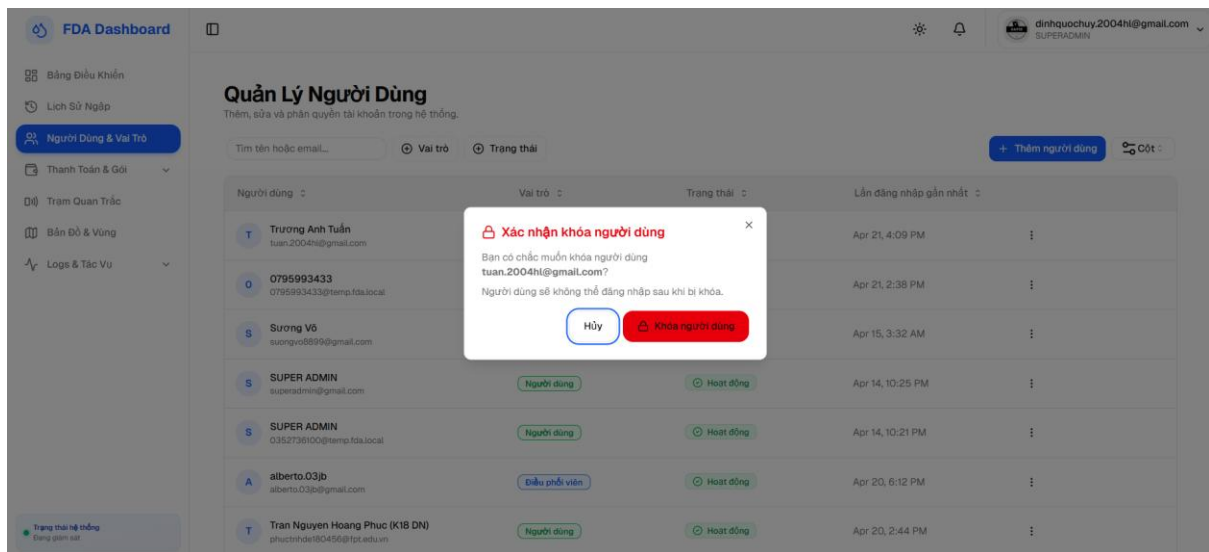


Figure 68: Confirm the Action

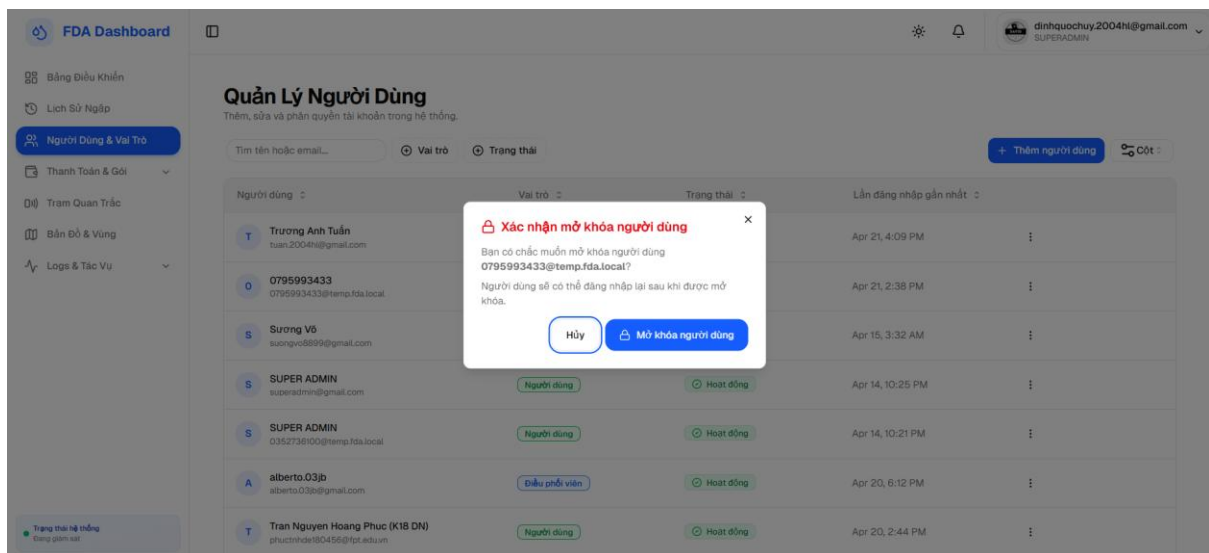


Figure 68: Confirm the Action

Step 4: Outcome after Ban/ Unban

- If the action is successful, a confirmation toast message will appear:
 - “Tài khoản [email] đã bị khóa.”
 - “Tài khoản [email] đã được mở khóa.”
- The user's **Trạng thái** badge in the table will immediately update to reflect the new state.
- A banned user will be immediately prevented from logging into the system until the Super Admin unbans their account.

FDA Dashboard

Quản Lý Người Dùng
Thêm, sửa và phân quyền tài khoản trong hệ thống.

Tim tên hoặc email... Vai trò Trạng thái + Thêm người dùng Cột

Người dùng	Vai trò	Trạng thái	Lần đăng nhập gần nhất
T Trương Anh Tuấn tuan.2004hi@gmail.com	Người dùng	Bi khóa	Apr 21, 4:09 PM
O 0795993433 0795993433@temp.fda.local	Người dùng	Bi khóa	Apr 21, 2:38 PM
S Sương Vẽ suongve889@gmail.com	Người dùng	Hoạt động	Apr 15, 3:32 AM
S SUPER ADMIN superadmin@gmail.com	Người dùng	Hoạt động	Apr 14, 10:25 PM
S SUPER ADMIN 0352736100@temp.fda.local	Người dùng	Hoạt động	Apr 14, 10:21 PM
A alberto.03jb alberto.03jb@gmail.com	Điều phối viên	Hoạt động	Apr 20, 6:12 PM
T Tran Nguyen Hoang Phuc (K18 DN) phucthhoai80456@tp.edu.vn	Người dùng	Hoạt động	Apr 20, 2:44 PM

Trạng thái hệ thống đang giám sát

Đã khóa người dùng
Tài khoản tuan.2004hi@gmail.com đã bị khóa.

Figure 69: Outcome after Ban

FDA Dashboard

Quản Lý Người Dùng
Thêm, sửa và phân quyền tài khoản trong hệ thống.

Tim tên hoặc email... Vai trò Trạng thái + Thêm người dùng Cột

Người dùng	Vai trò	Trạng thái	Lần đăng nhập gần nhất
T Trương Anh Tuấn tuan.2004hi@gmail.com	Người dùng	Hoạt động	Apr 21, 4:09 PM
O 0795993433 0795993433@temp.fda.local	Người dùng	Bi khóa	Apr 21, 2:38 PM
S Sương Vẽ suongve889@gmail.com	Người dùng	Hoạt động	Apr 15, 3:32 AM
S SUPER ADMIN superadmin@gmail.com	Người dùng	Hoạt động	Apr 14, 10:25 PM
S SUPER ADMIN 0352736100@temp.fda.local	Người dùng	Hoạt động	Apr 14, 10:21 PM
A alberto.03jb alberto.03jb@gmail.com	Điều phối viên	Hoạt động	Apr 20, 6:12 PM
T Tran Nguyen Hoang Phuc (K18 DN) phucthhoai80456@tp.edu.vn	Người dùng	Hoạt động	Apr 20, 2:44 PM

Trạng thái hệ thống đang giám sát

Đã mở khóa người dùng
Tài khoản tuan.2004hi@gmail.com đã được mở khóa.

Figure 69: Outcome after Unban

3.2.4 Station Management

3.2.4.1 View and Search Stations

Step 1: Navigate to the Station Management Page

- After logging in, click on “Trạm Quan Trắc” in the left-hand sidebar of the FDA Dashboard.
- The main area displays a table listing all monitoring stations registered in the system.

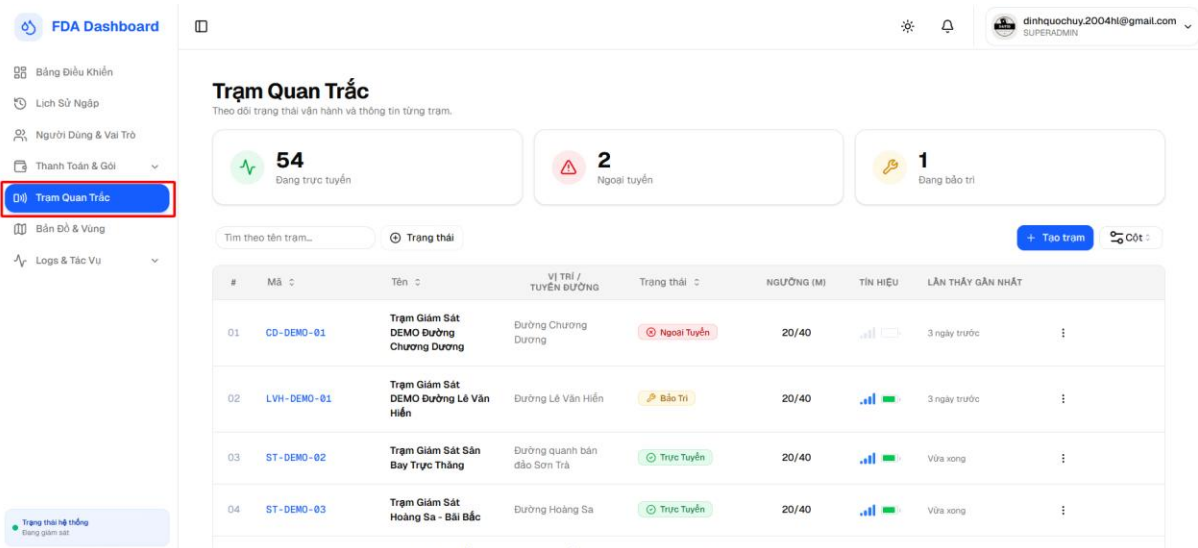
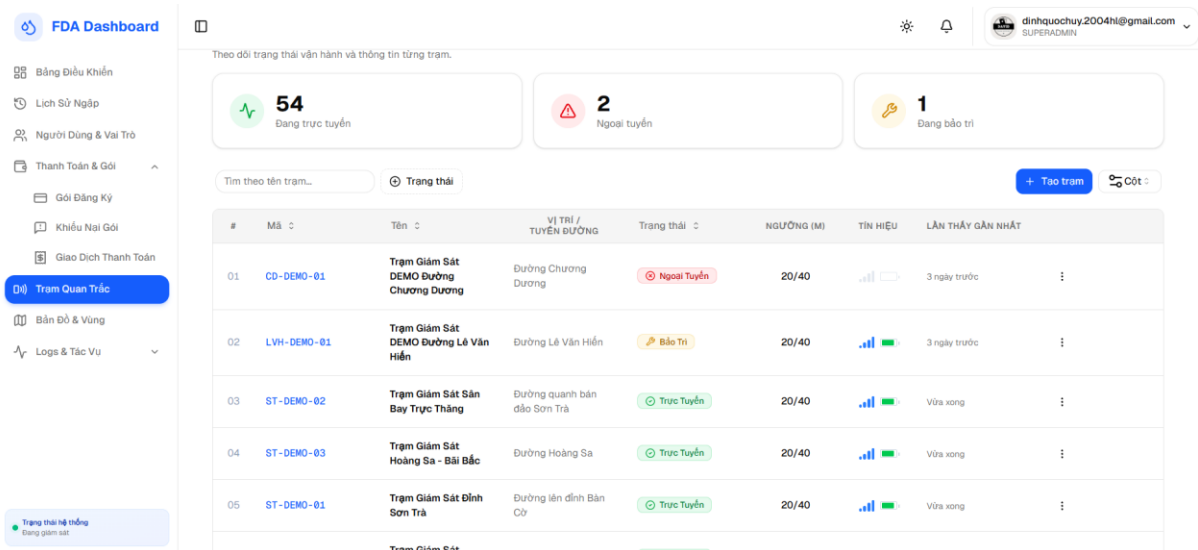


Figure 70: Station Management

Step 2: View Station Table

Each row in the table represents one monitoring station with the following information:

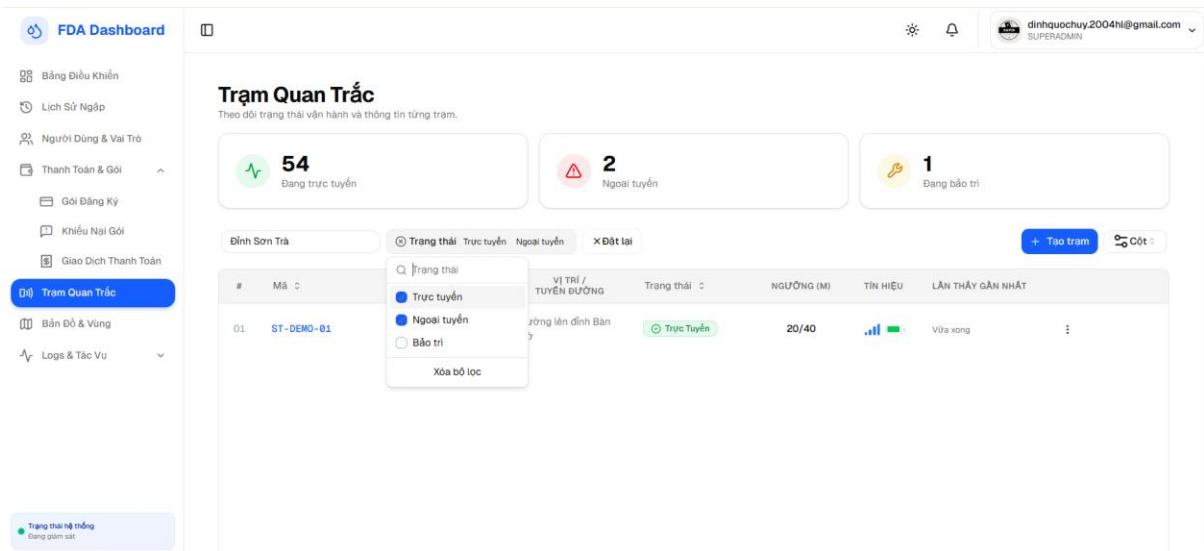
- **Mã** - The unique identifier code of the station (e.g., ST_DN_DRAGON_01). Used to quickly reference a station in the system.
- **Tên** - The full display name of the station (e.g., Dragon Bridge Station).
- **Tuyến đường** - The name of the road or area where the station is located.
- **Trạng thái** - The current operating status of the station, shown as a colored badge:
 - **Trực tuyến (Green)** - The station is running normally and sending data.
 - **Bảo Trì (Yellow)** - The station is temporarily offline for maintenance work.
 - **Ngoại Tuyến (Red)** - The station is not responding or has stopped sending data.
- **Ngưỡng (cm)** - Displays the configured threshold levels for the station (e.g., 20/40).
- **Tín** - Indicates the signal strength and connection quality of the station.
- **Lần thấy gần nhất** - Shows the last time the system received data from the station (e.g., “3 ngày trước”, “Vừa xong”).



Step 3: Search and Filter Stations

The toolbar above the table provides tools to quickly find a specific station:

- **Search by Name** - Type part of the station name in the search box. The table filters results automatically as you type.
- **Status filter** - Click the "Trạng thái" dropdown to show only stations with a specific status. You can select multiple statuses at the same time.



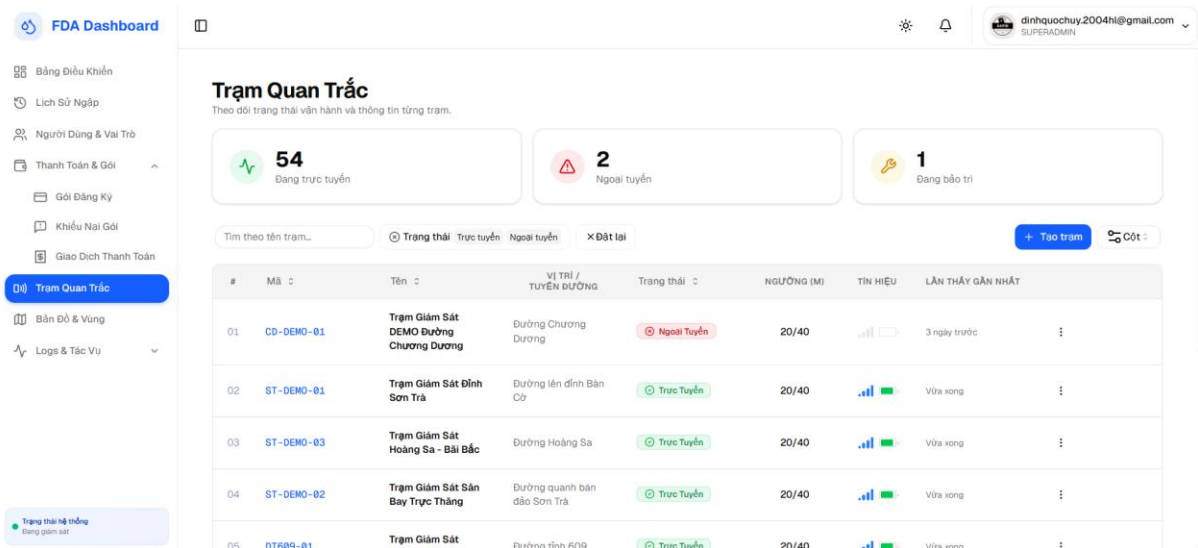


Figure 73: Filter Station

3.2.4.2 Add a New Station

Step 1: Open the Add Station Form

- On the Station Management page, click the “Tạo Trạm” button at the top-right of the toolbar.
- A dialog box titled “Tạo trạm mới” will appear on the screen.

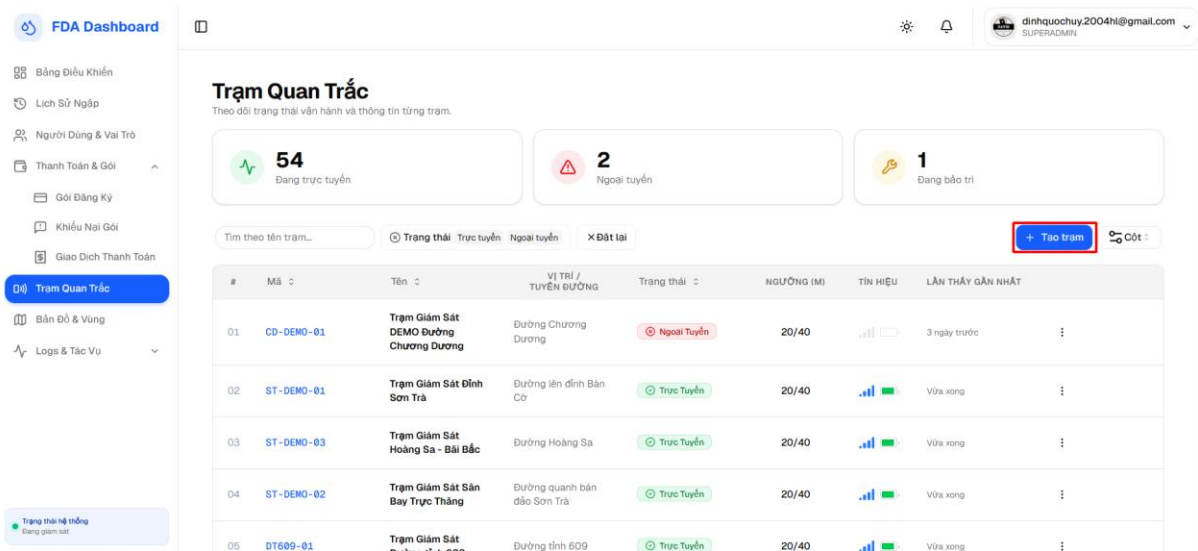


Figure 74: Add Station Button

Tạo trạm mới

Điền thông tin để tạo trạm quan trắc mới. Các trường có dấu * là bắt buộc.

Mã trạm *

Tên trạm *

Trạng thái *

Tên đường

Mô tả vị trí

Hướng

Vĩ độ

Kinh độ

Ngưỡng cảnh báo

Ngưỡng khẩn cấp

Figure 75: Add Station Form

Step 2: Fill in Basic Information

The form is divided into several sections. Start with the basic details (fields marked with * are required):

- **Mã trạm** - Enter a unique code to identify this station. Use a consistent naming pattern (e.g., ST_DN_DRAGON_01 for Station - Da Nang - Dragon Bridge - Unit 01). Must be at least 2 characters.
- **Tên** - Enter a descriptive name for the station (e.g., Dragon Bridge Station). Must be at least 2 characters.
- **Trạng thái** - Select the station's current operating state from the dropdown:
 - **Đang hoạt động** - The station is installed and operational.
 - **Ngoại tuyến** - The station is not currently sending data.
 - **Bảo trì** - The station is undergoing maintenance.
- **Tuyến đường** - Enter the name of the road or street where the station is placed (e.g., 2 Tháng 9 Street).

+ Tạo trạm mới

Điền thông tin để tạo trạm quan trắc mới. Các trường có dấu * là bắt buộc.

Mã trạm * Tên trạm *

Trạng thái * Tên đường

Mô tả vị trí 0 / 500

Hướng

Vĩ độ Kinh độ

Ngưỡng cảnh báo Ngưỡng khẩn cấp

Figure 76: Fill in Basic Information

Step 3: Add Location Details

Provide more information about where the station is physically located:

- **Mô tả vị trí** - Write a short description of the installation spot (e.g., "Located on the west side of Dragon Bridge, near the riverbank"). Maximum 500 characters.
- **Hướng** - Describe the water flow direction at this station (e.g., upstream, downstream, or road section).
- **Vĩ độ** - Enter the geographic latitude coordinate of the station (e.g., 16.061153). Must be between -90 and 90.
- **Kinh độ** - Enter the geographic longitude coordinate of the station (e.g., 108.221589). Must be between -180 and 180.
- **Offset hiệu chuẩn ($\pm$ cm)** – Adjusts the measured water level by a fixed value (in centimeters) to improve accuracy.

Note: Latitude and Longitude are used to show the station on the map. If you do not know the exact coordinates, you can look them up using Google Maps.

+ Tạo trạm mới

Điền thông tin để tạo trạm quan trắc mới. Các trường có dấu * là bắt buộc.

Mã trạm * Tên trạm *

Trạng thái * Tên đường

Mô tả vị trí

Hướng

Vi độ Kinh độ

Ngưỡng cảnh báo Ngưỡng khẩn cấp

Offset hiệu chuẩn (cm)

Figure 77: Add Location Details

Step 4: Set Alert Thresholds

Alert thresholds define the water levels that trigger flood warnings for this station:

- **Ngưỡng cảnh báo** - Set the water level (in meters) at which a Warning alert is sent (e.g., 0.5). When the measured water level reaches this value, the system will notify relevant users.
- **Ngưỡng khẩn cấp** - Set the water level (in meters) at which a Critical alert is sent (e.g., 1.2). This represents a more serious flooding condition that requires urgent attention.

Note: The Critical threshold should always be set higher than the Warning threshold to ensure alerts are triggered in the correct order.

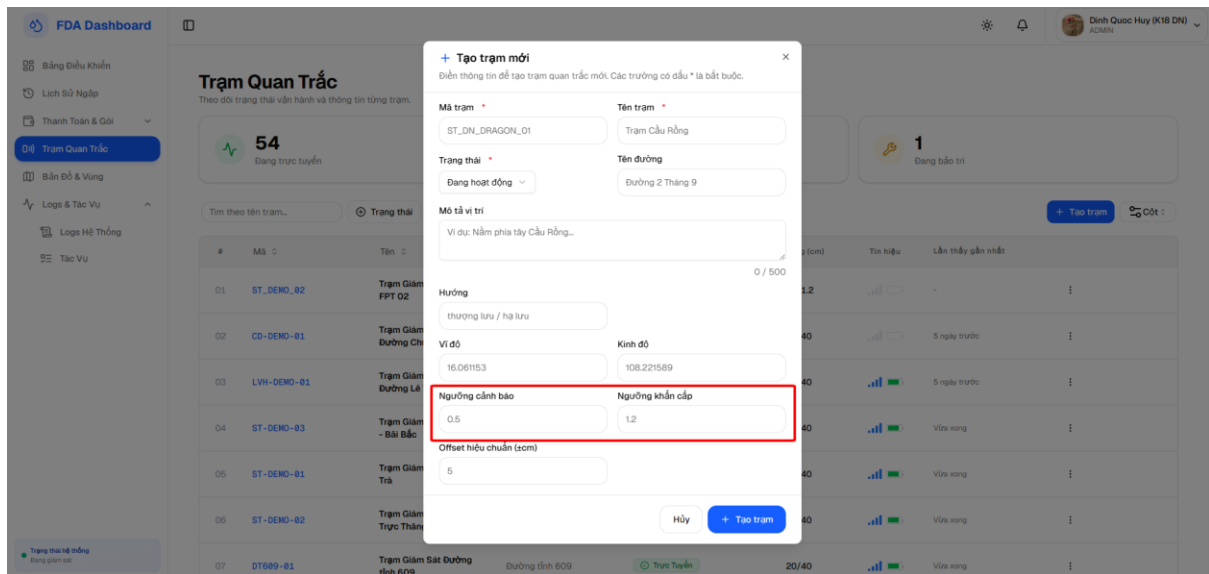


Figure 78: View Technical Chart

Step 5: Save the New Station

After filling in all required fields, click the “Tạo trạm” button at the bottom of the form.

- **Success** - A notification appears: "Station created successfully!" The new station will immediately appear in the station table.
- **Validation error** - If a required field is missing or a value is out of range (e.g., invalid coordinates), the field will be highlighted red with an error message. Correct the error and try

Click “Hủy” to close the form without saving anything.

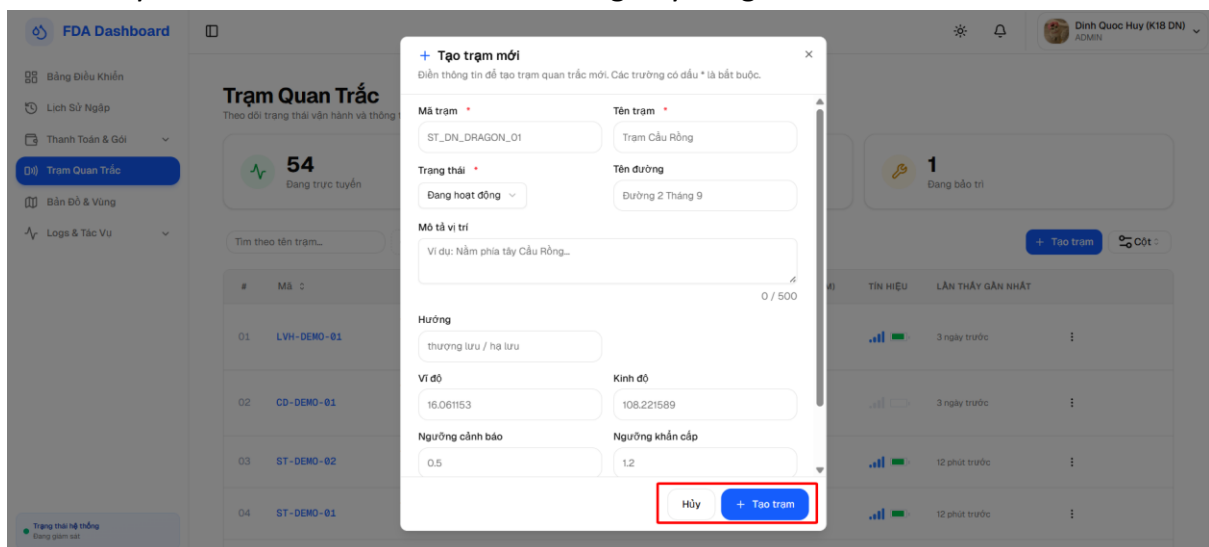


Figure 79: Save the New Station

3.2.4.3 Update a Station

Step 1: Open the Actions Menu

- In the station table, locate the station you want to update.

- Click the three-dot menu icon (kebab menu) at the far-right of that station's row.
- A dropdown menu will appear with two options: Update and Delete.

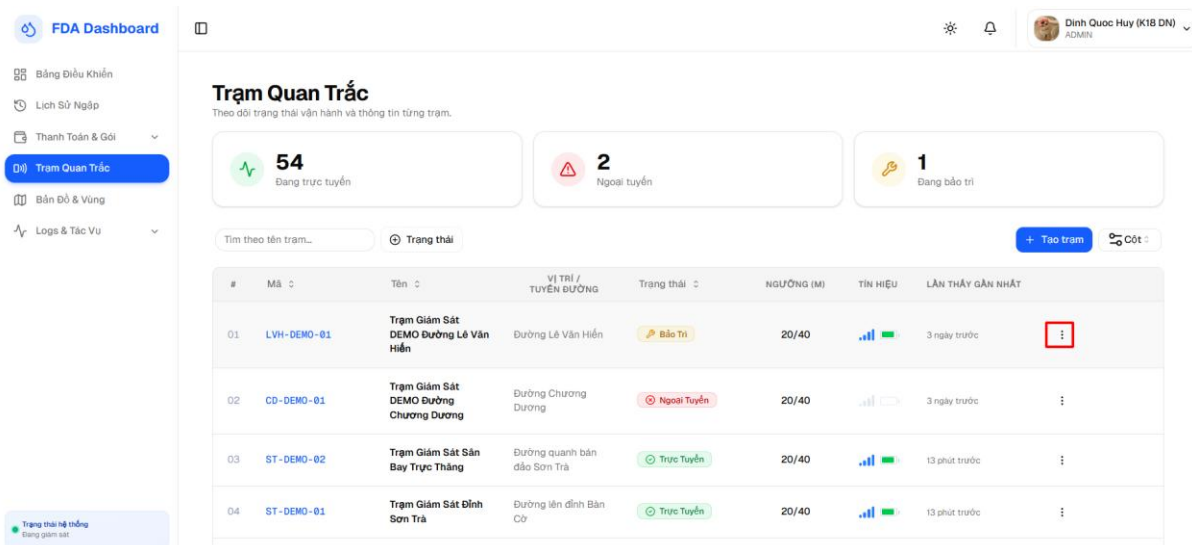


Figure 80: Open the Actions Menu

Step 2: Select “Update”

- Click “Cập nhật” from the dropdown menu.
- An “Sửa trạm” dialog box will appear, pre-filled with the station's existing information.

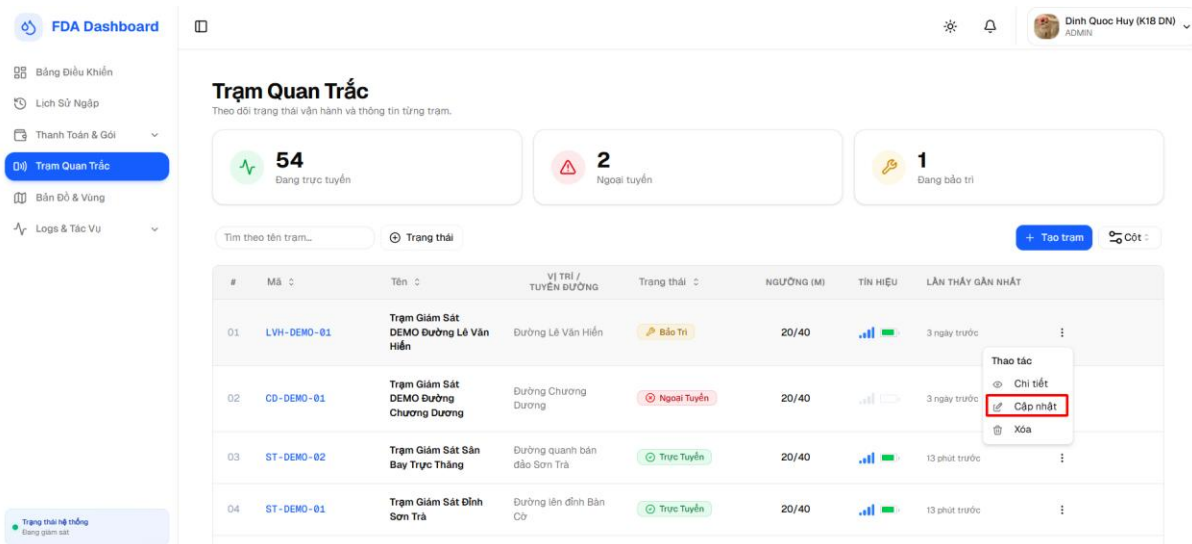


Figure 81: Click "Update"

Step 3: Modify the Station Information

The edit form contains the same fields as the create form. You can update any of the following:

- **Mã trạm** - Enter a unique code to identify this station. Use a consistent naming pattern (e.g., ST_DN_DRAGON_01 for Station - Da Nang - Dragon Bridge - Unit 01). Must be at least 2 characters.

- **Tên** - Enter a descriptive name for the station (e.g., Dragon Bridge Station). Must be at least 2 characters.
- **Trạng thái** - Select the station's current operating state from the dropdown:
 - **Đang hoạt động** - The station is installed and operational.
 - **Ngoại tuyến** - The station is not currently sending data.
 - **Bảo trì** - The station is undergoing maintenance.
- **Tuyến đường** - Enter the name of the road or street where the station is placed (e.g., 2 Thang 9 Street).

For more information about where the station is physically located:

- **Mô tả vị trí** - Write a short description of the installation spot (e.g., "Located on the west side of Dragon Bridge, near the riverbank"). Maximum 500 characters.
- **Hướng** - Describe the water flow direction at this station (e.g., upstream, downstream, or road section).
- **Vĩ độ** - Enter the geographic latitude coordinate of the station (e.g., 16.061153). Must be between -90 and 90.
- **Kinh độ** - Enter the geographic longitude coordinate of the station (e.g., 108.221589). Must be between -180 and 180.
- **Offset hiệu chuẩn ($\pm$ cm)** – Adjusts the measured water level by a fixed value (in centimeters) to improve accuracy.

The screenshot displays the 'Sửa trạm' (Edit Station) form in the FDA Dashboard. The form is a modal window with the following fields:

- Mã trạm**: ST-DEMO-03
- Tên trạm**: Trạm Giám Sát Hoàng Sa - Bãi Bắc
- Trạng thái**: Đang hoạt động
- Tên đường**: Đường Hoàng Sa
- Mô tả vị trí**: Tuyến đường ven biển phía Đông bán đảo, gần khu vực InterContinental. (68 / 500 characters)
- Hướng**: Đông
- Vĩ độ**: 16.11684
- Kinh độ**: 108.309818
- Ngưỡng cảnh báo**: 20
- Ngưỡng khẩn cấp**: 40

At the bottom of the form, there are buttons for 'Hủy' (Cancel) and 'Cập nhật trạm' (Update Station). The background shows the 'Trạm Quan Trắc' (Monitoring Station) overview with a table of stations and a sidebar with navigation options.

Figure 82: Modify the Station

Step 4: Save the Changes

After making all desired changes, click the “Cập nhật trạm” button.

- **Success** - A notification appears: "Station updated successfully!" The table will automatically show the updated information.

- **Validation error** - If a required field is empty or a value is invalid, the field is highlighted in red. Correct it and click “Cập nhật trạm” again.
- Click “Hủy” to close the dialog without saving any changes.

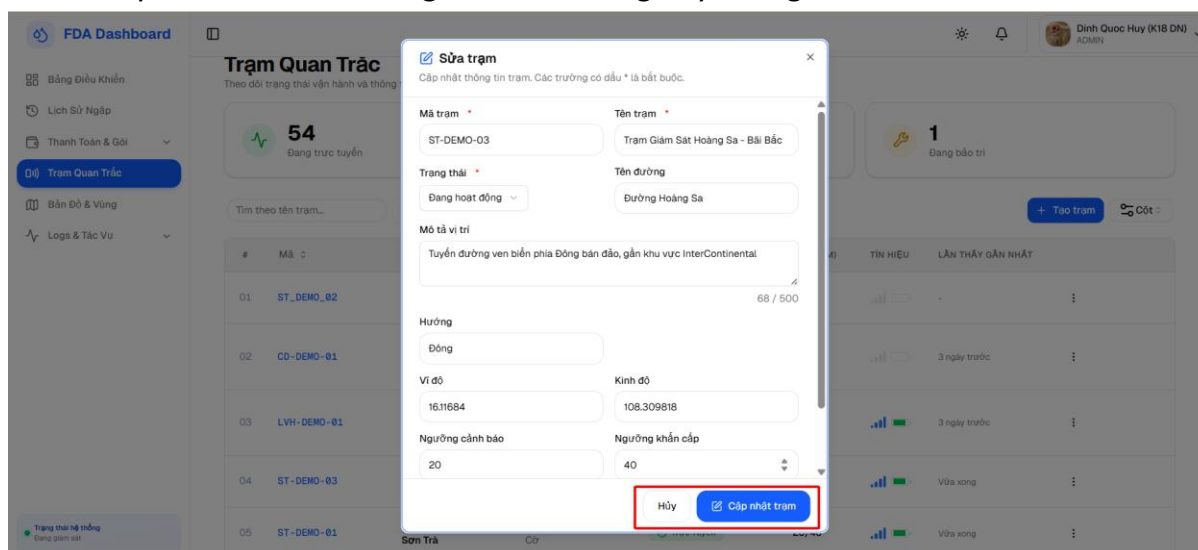


Figure 83: Save the Changes

3.2.4.4 Delete a Station

Step 1: Open the Actions Menu

- In the station table, locate the station you want to remove.
- Click the three-dot menu icon at the far-right of that station's row.

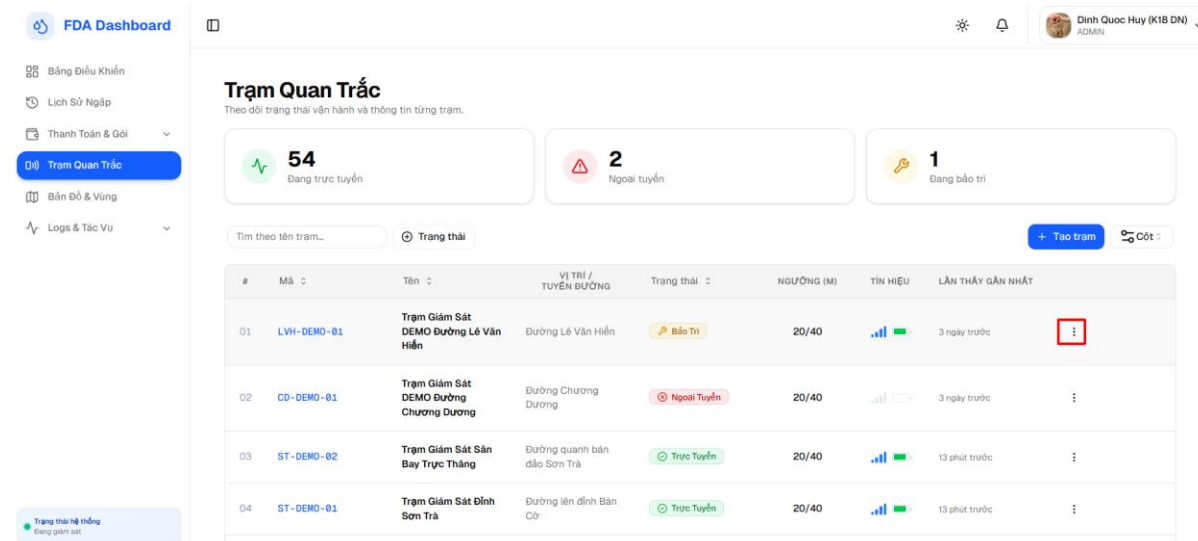


Figure 84: Delete a Station

Step 2: Select Delete

- Click “Xóa” from the dropdown menu.
- A confirmation dialog will appear asking you to confirm the action.

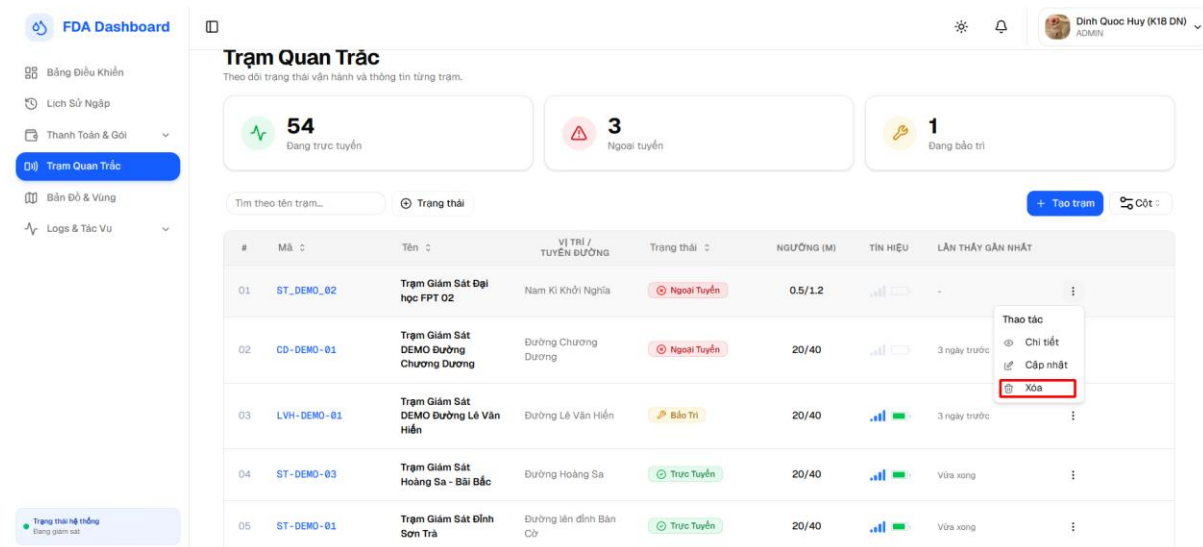


Figure 85: Select Delete

Step 3: Confirm the Deletion

Read the confirmation message carefully to ensure you have selected the correct station.

- **Confirm** - Click button “Tiếp tục” to permanently delete the station. The station will be removed from the table immediately.
- **Cancel** - Click “Hủy” to go back without deleting anything.

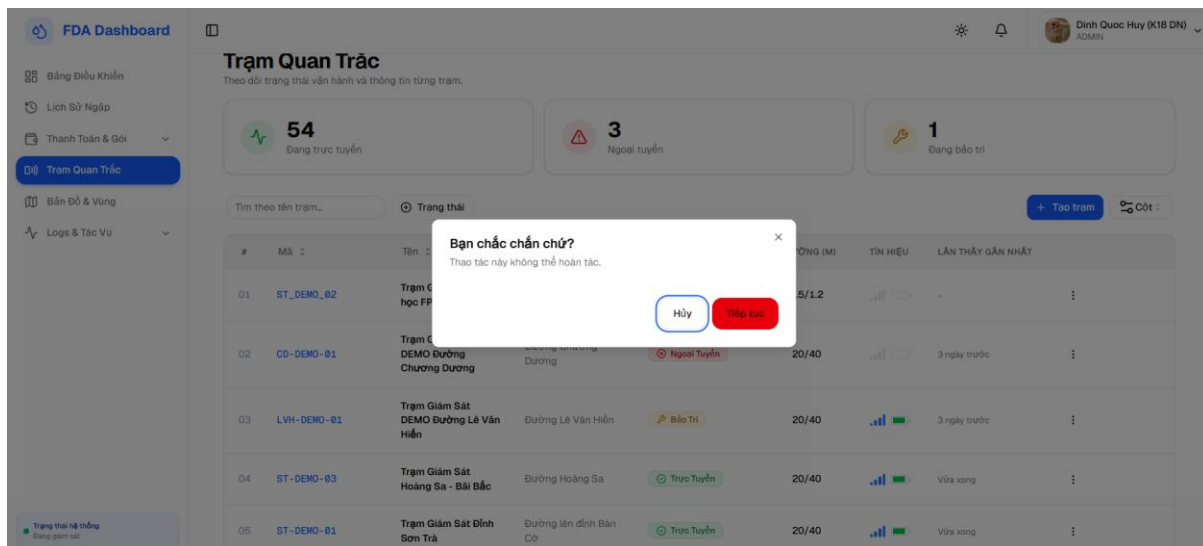


Figure 86: Deletion

3.2.5 Alert and Notification Management

This module allows moderators to configure automated alert messages and manage manual notifications sent to users regarding upcoming floods, weather changes, and system updates.

3.2.5.1 Alert Message Template Management

Step 1: Open the Alert Template Page

From the dashboard sidebar, expand “Cảnh Báo & Tin” and click “Mẫu Cảnh Báo”.

The page “**Mẫu Cảnh Báo**” displays a table of existing templates and an action to add a new template.

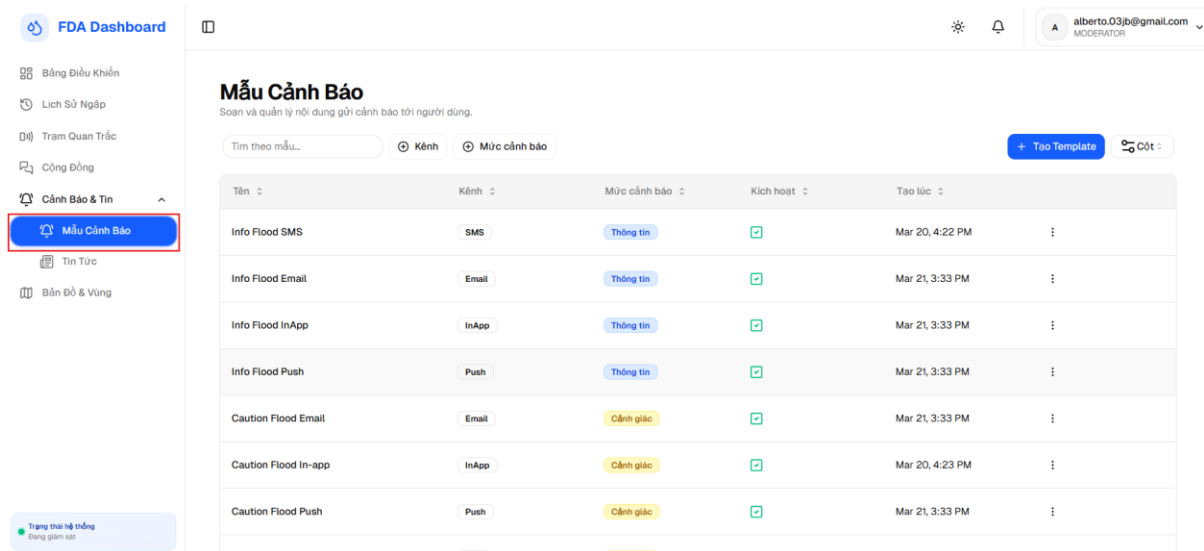


Figure 85: Alert Template Management - Templates Table

Step 2: Create a New Template

Click the “**Tạo Template**” button on the page .

A form dialog opens to enter the template information (name, condition fields if available, and message body).

Click “**Tạo**” button to create the template.

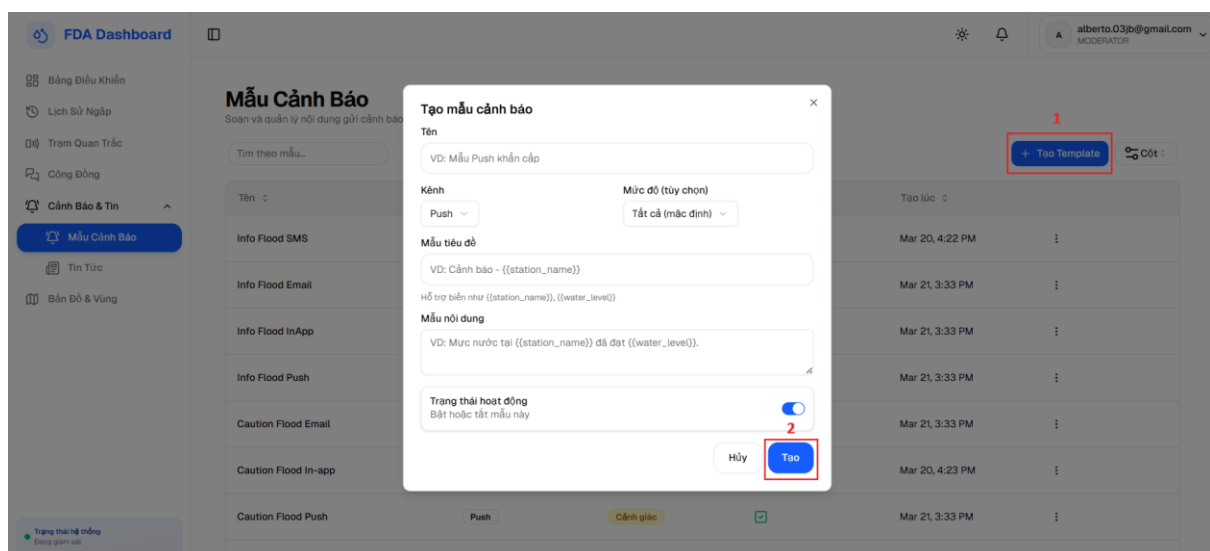


Figure 85: Alert Template Management - Create Template Dialog

Step 3: Preview, Edit, or Delete a Template

In the templates table, open the row actions menu (action button per template).

Available actions typically include:

- **“Xem trước”**: open the preview dialog to inspect the rendered template content.
- **“Sửa”**: update template fields, then save changes.
- **“Xóa”**: remove the template (confirmation may be required).

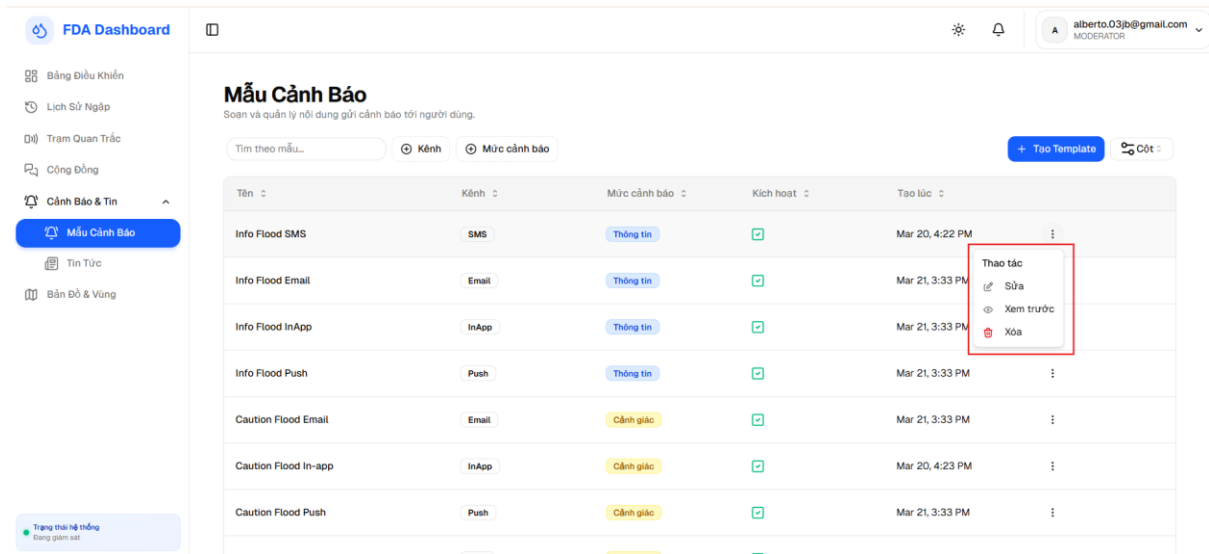


Figure 85: Alert Template Management - Row Actions

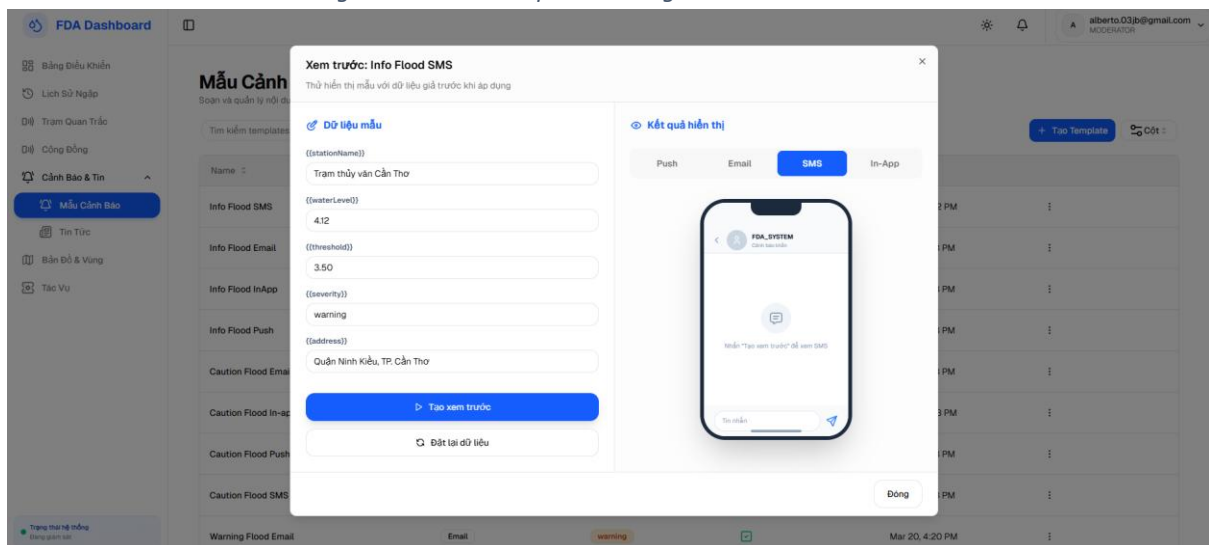


Figure 85: Alert Template Management - Preview

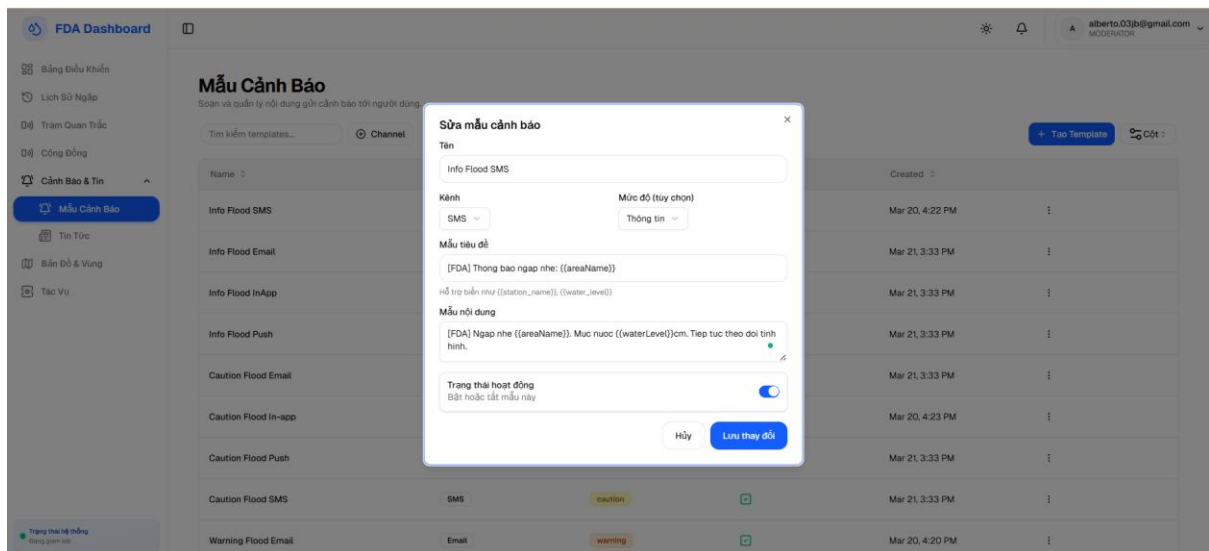


Figure 85: Alert Template Management - Edit

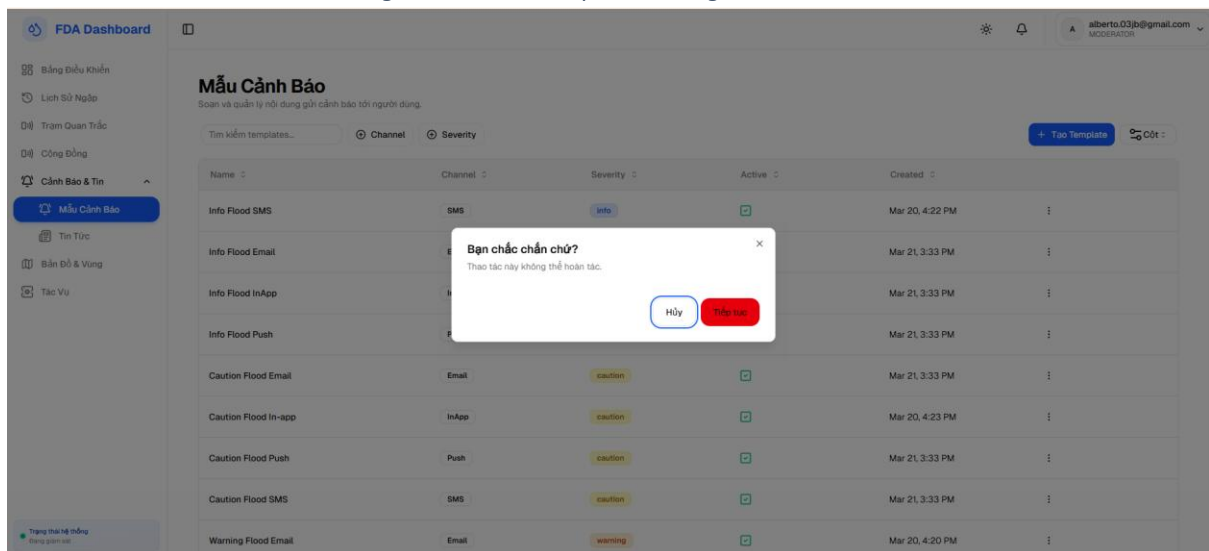


Figure 85: Alert Template Management - Delete

3.2.5.2 Send and Management Notification

This feature allows Moderator users to create and publish notifications/news to users. Notifications are managed in the **"Tin Tức"** page, where items can be drafted, scheduled, published, edited, and deleted.

Step 1: Open the News / Notification Page

From the dashboard sidebar, expand **"Cảnh Báo & Tin"** and click **"Tin Tức"**. The page **"Tin Tức"** displays a listing table of announcements/notifications.

Title	Status	Priority	Target	Scheduled At	Published At	Author	Views
Thông báo theo dõi điểm ngập tại... Khu công nghiệp Hòa...	Published	Normal	Region: Liên Chiêu, Thanh Khê	—	08:00 13/04/2026	Dinh Quoc Huy (K18 DN)	0
Cảnh báo toàn thành phố: Mưa...	Published	High	All	—	07:58 13/04/2026	Dinh Quoc Huy (K18 DN)	0
Cảnh báo học sinh và phụ huynh tại... Thanh Khê có nguy c...	Published	Normal	Region: Thanh Khê	—	07:57 13/04/2026	Dinh Quoc Huy (K18 DN)	0
Khuyến cáo sơ tán tạm thời tại vùng... Hòa Xuân có nguy cơ...	Published	Urgent	Region: Hòa Xuân	—	07:52 13/04/2026	Dinh Quoc Huy (K18 DN)	0
Cảnh báo giao thông ngập tại... Ngõ Hành Sơn ngập...	Published	High	Region: Ngõ Hành Sơn	—	07:51 13/04/2026	Dinh Quoc Huy (K18 DN)	0
Thông báo ngập cục bộ khu vực...	Published	Normal	Region: Sơn Trà	—	07:49 13/04/2026	Dinh Quoc Huy (K18 DN)	0

Figure 85: News - Listing Table

Step 2: Create a New Notification

Click "**Tạo Bản Tin**" (top-right).

A dialog title "**Tạo thông báo**" opens. Fill in the required fields:

- **Tiêu đề** (required)
- **Nội dung** (required)
- Optional: **Tóm tắt**, **Hình ảnh đại diện** (URL or Upload), **Đối tượng nhận** (Tất cả / Theo khu vực / Theo vai trò), **Độ ưu tiên** (Thấp/Bình thường/Cao/Khẩn cấp)
- Optional scheduling: enable "**Đặt lịch đăng**"* and select a future date/time

Click "**Tạo thông báo**" to save.

Figure 85: News - Create Notification Dialog

Step 3: Publish (Send) a Notification

In the listing table, open the row action menu "**Thao tác**".

If the item is in **draft/pending** status, click **"Đăng ngay"** to publish.

After publishing, a toast message appears: **"Đã đăng thông báo"**.

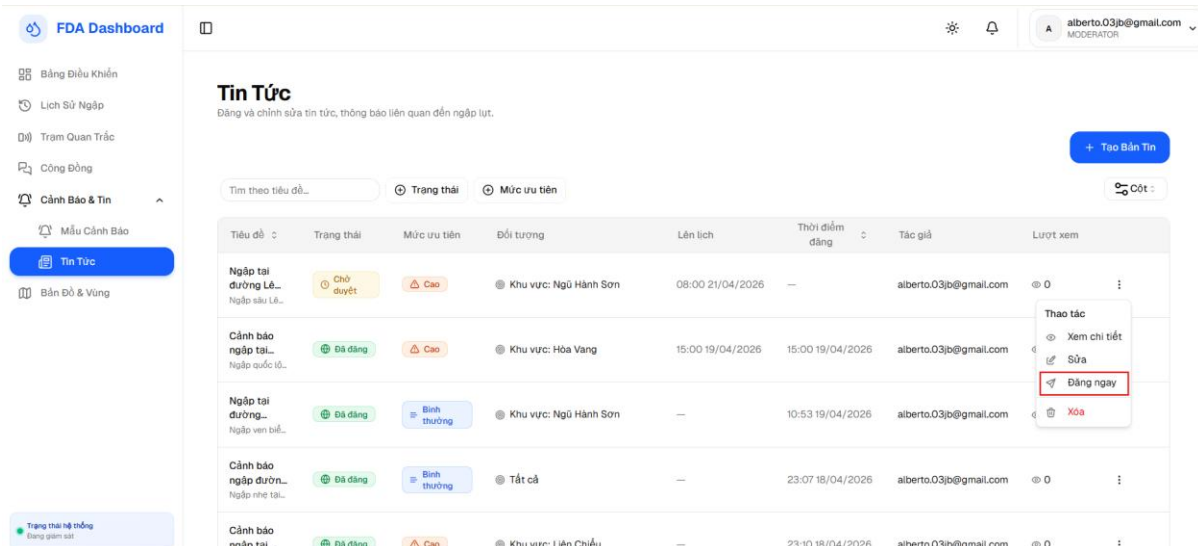


Figure 85: News - Push Action

Step 4: View, Edit, or Delete a Notification

In the same row action menu **"Thao tác"**:

- Click **"Xem chi tiết"** to open the details dialog.
- Click **"Sửa"** to edit (only available for draft/pending items).
- Click **"Xóa"** to remove the notification (a confirmation modal appears).

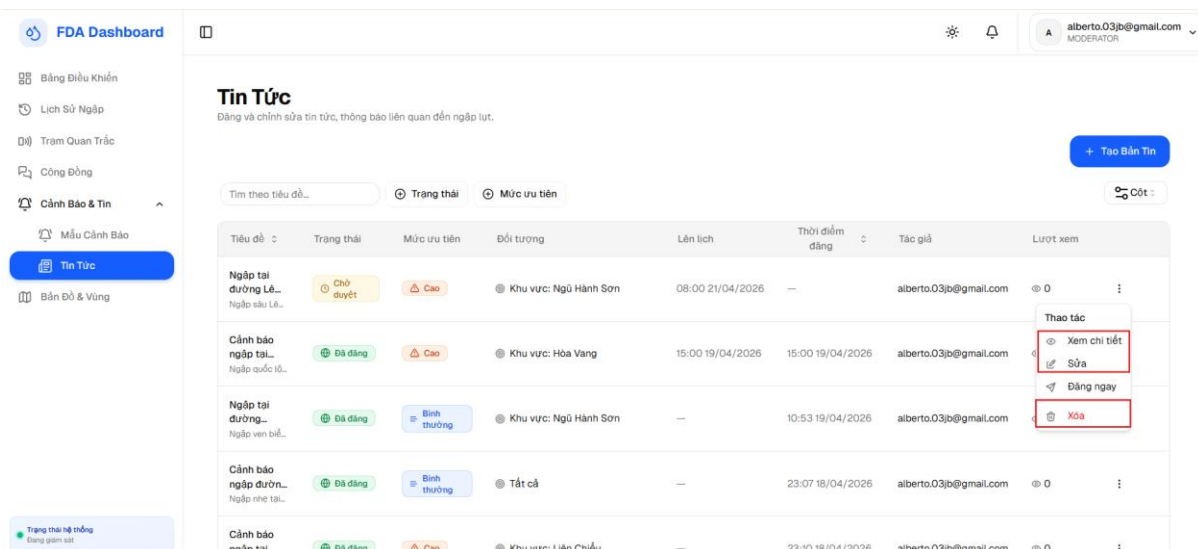


Figure 85: News - View/ Edit/ Delete Actions

3.2.6 Subscription and Payment Management

This module is used to handle premium features, manage different subscription tiers, and monitor customer payments in the system.

3.2.6.1 Management Subscription Plans

Step 1: Access Subscription Plans

From the Admin sidebar, open **"Thanh Toán & Gói"** and click **"Gói Đăng Ký"**.

The **"Gói Đăng Ký"** page shows the plans list/table and top actions (e.g., **"Ẩn gói không hoạt động"**, **"Tạo Gói"**).

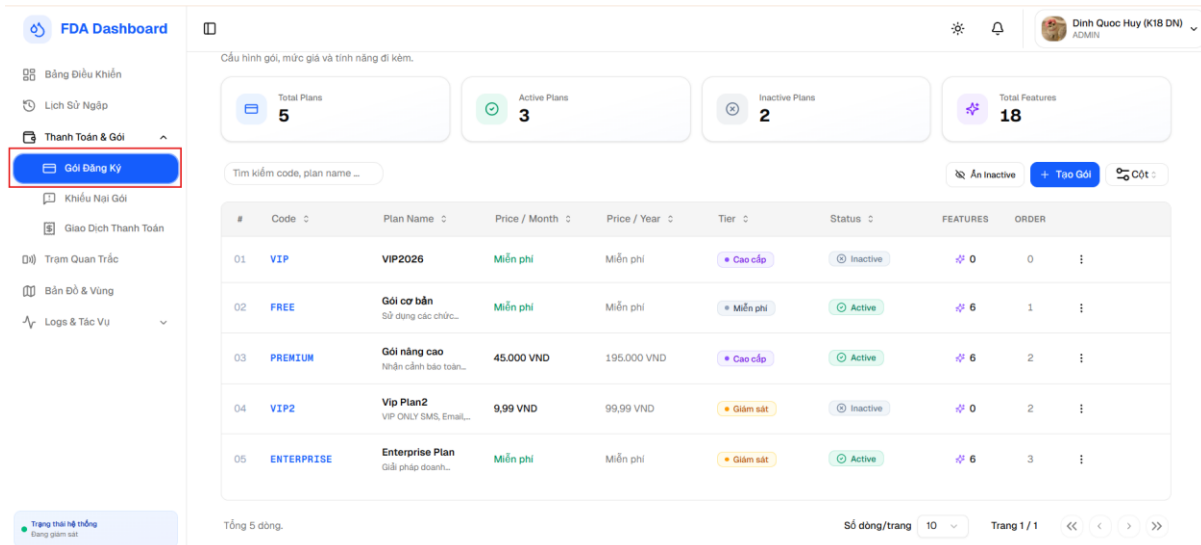


Figure 92: Subscription Plans Overview

Step 2: Add or Edit a Plan

Click **"Tạo Gói"** to create a plan, or use the edit action on an existing plan.

When creating a plan, a modal dialog **"Tạo gói mới"** opens.

Key fields in the modal include **"Mã gói"**, **"Tên gói"**, pricing (month/year), tier, sort order, and **Features** (button: **"Thêm tính năng"**).

Click **"Tạo Gói"** to submit, or **"Hủy"** to close.

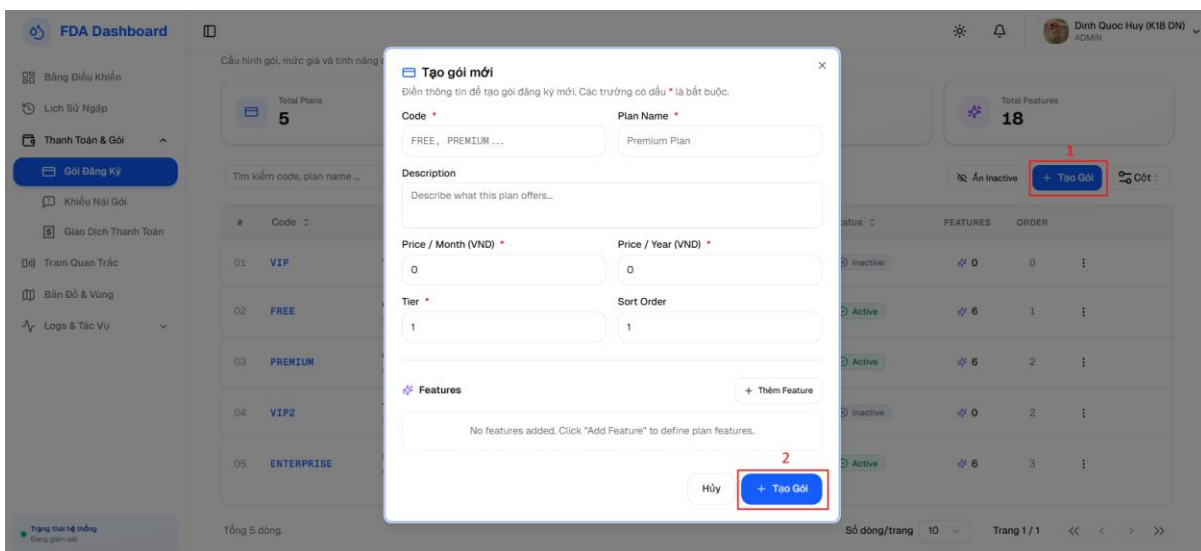


Figure 87: Add Subscription Plans

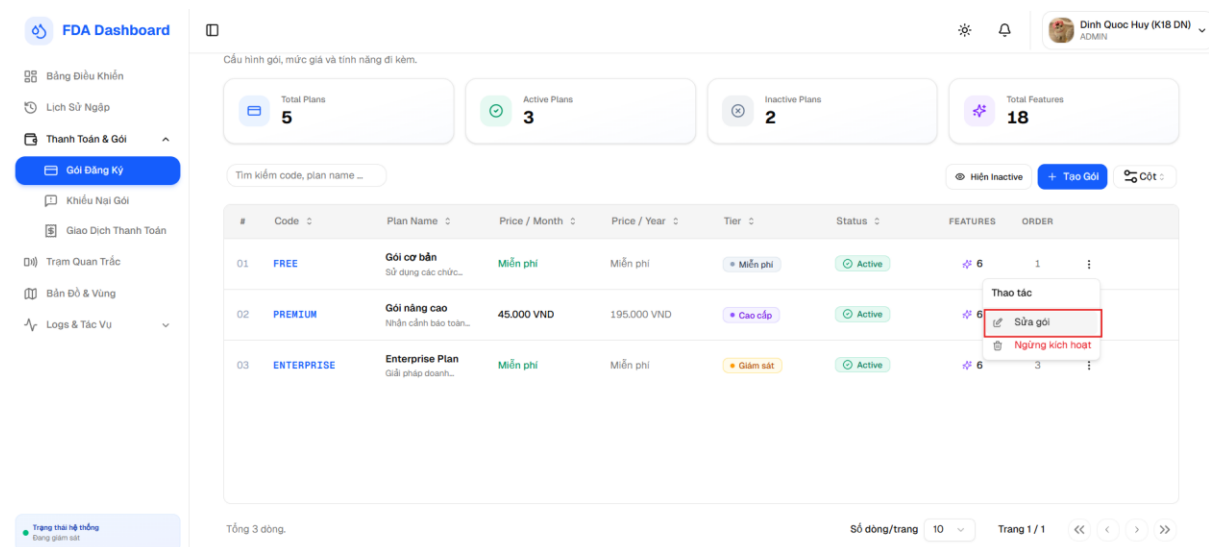


Figure 87: Edit Subscription Plans Action

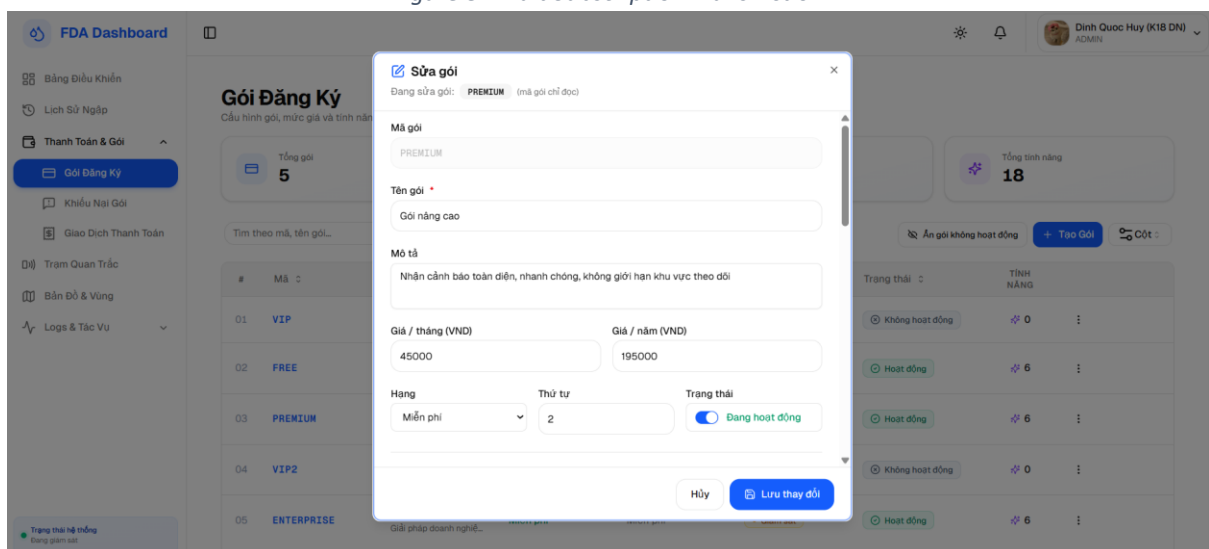


Figure 87: Edit Subscription Plans

3.2.6.2 Monitor Payment Transactions (Giao Dịch Thanh Toán)

Step 1: Open Payment Transactions

From the Admin sidebar, open "Thanh Toán & Gói" and click "Giao Dịch Thanh Toán".

The page shows a transactions table (user, plan, amount, created date, status).

Giao Dịch Thanh Toán
Xem lịch sử thanh toán và chi tiết từng đơn hàng.

Total Transactions: 36
Total Revenue: 162.009 VND
Pending: 1
Cancelled: 14

Tim kiếm order code, user, plan | Status

#	Order Code	User	Plan	Amount	Duration	Status	Payment Date
01	1776488589410	Hoàng Phúc Trần Nguyễn tphuc9398@gmail.com	Gói nâng cao PREMIUM	45.000 VND	1 month	Paid	12:03:23 18/4/2026
02	1776483828765	Hoàng Phúc Trần Nguyễn tphuc9398@gmail.com	Gói nâng cao PREMIUM	45.000 VND	1 month	Cancelled	10:43:40 18/4/2026
03	1776276391499	Hoàng Phúc Trần Nguyễn tphuc9398@gmail.com	Gói nâng cao PREMIUM	2.000 VND	1 month	Paid	01:07:45 16/4/2026
04	1776276325631	Hoàng Phúc Trần Nguyễn tphuc9398@gmail.com	Gói nâng cao PREMIUM	2.000 VND	1 month	Cancelled	01:05:25 16/4/2026
05	1776048105260	Phúc Trần tranphucnh3@gmail.com	Gói nâng cao PREMIUM	2.000 VND	1 month	Paid	09:42:02 13/4/2026

Figure 87: Payment Transactions - Table View

Step 2: Search and Filter Records

Use the toolbar (if available) to filter by:

- User identity (email/name)
- Payment status (e.g., “Đã thanh toán” / “Đã hủy” / “Chờ xử lý”)
- Date range (if provided)

Giao Dịch Thanh Toán
Xem lịch sử thanh toán và chi tiết từng đơn hàng.

Total Transactions: 36
Total Revenue: 162.009 VND
Pending: 1
Cancelled: 14

Tim kiếm order code, user, plan | Status: Paid x Bật lại

#	Order Code	User	Plan	Amount	Duration	Status	Payment Date
01	1776488589410	Hoàng Phúc Trần Nguyễn tphuc9398@gmail.com	Gói nâng cao PREMIUM	45.000 VND	1 month	Paid	12:03:23 18/4/2026
02	1776276391499	Hoàng Phúc Trần Nguyễn tphuc9398@gmail.com	Gói nâng cao PREMIUM	2.000 VND	1 month	Paid	01:07:45 16/4/2026
03	1776048105260	Phúc Trần tranphucnh3@gmail.com	Gói nâng cao PREMIUM	2.000 VND	1 month	Paid	09:42:02 13/4/2026
04	1776021041896	Phúc Trần tranphucnh3@gmail.com	Gói nâng cao PREMIUM	2.000 VND	1 month	Paid	02:12:57 13/4/2026
05	1776020482772	Hoàng Phúc Trần Nguyễn tphuc9398@gmail.com	Gói nâng cao PREMIUM	2.000 VND	1 month	Paid	02:02:18 13/4/2026

Figure 87: Payment Transactions - Filters

3.2.6.3 Handle Customers Disputes (Khiếu Nại Gói)

Step 1: Open the Subscription Disputes Page

From the Admin sidebar, open "Thanh Toán & Gói" and click "Khiếu Nại Gói".

The page title is "Khiếu Nại Gói".

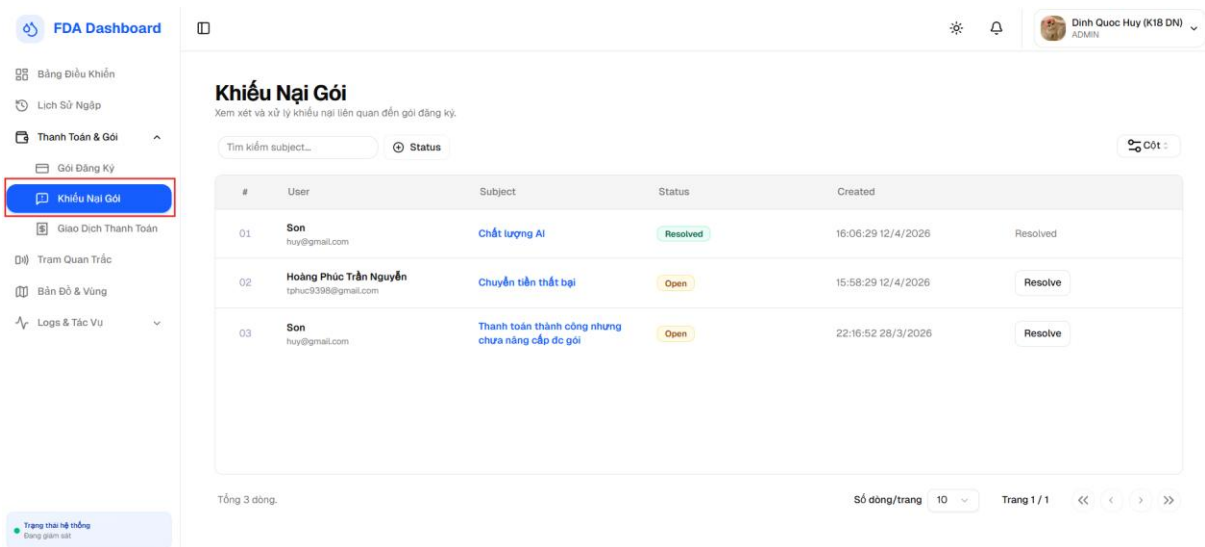


Figure 87: Subscription Disputes - Page Overview

Step 2: Filter Complaints (Subject / Status)

Use **Search** (placeholder: "Tìm kiếm subject...") and **Status** (Đang mở/ Đã xử lý/ Từ chối) to filter.

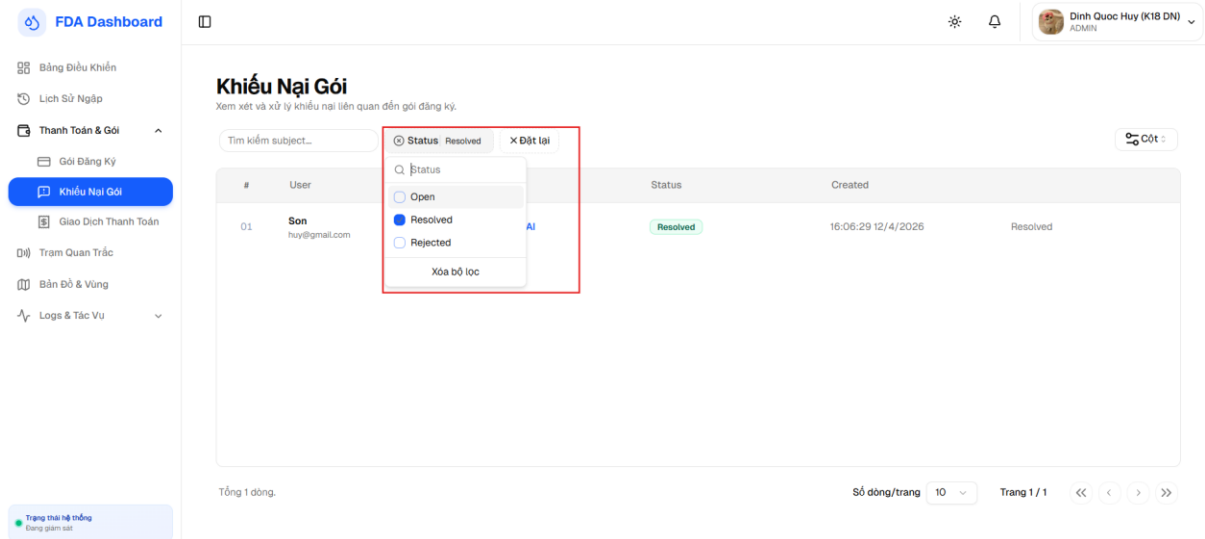


Figure 87: Subscription Disputes - Toolbar Filters

Step 3: View Complaint Details

Click a "Tiêu đề" value to open the "Chi tiết khiếu nại" dialog.

The dialog shows "Tiêu đề", "Người dùng", "Nội dung", "Trạng thái", "Tạo lúc", "Phản hồi của Admin" and "Xử lý lúc".

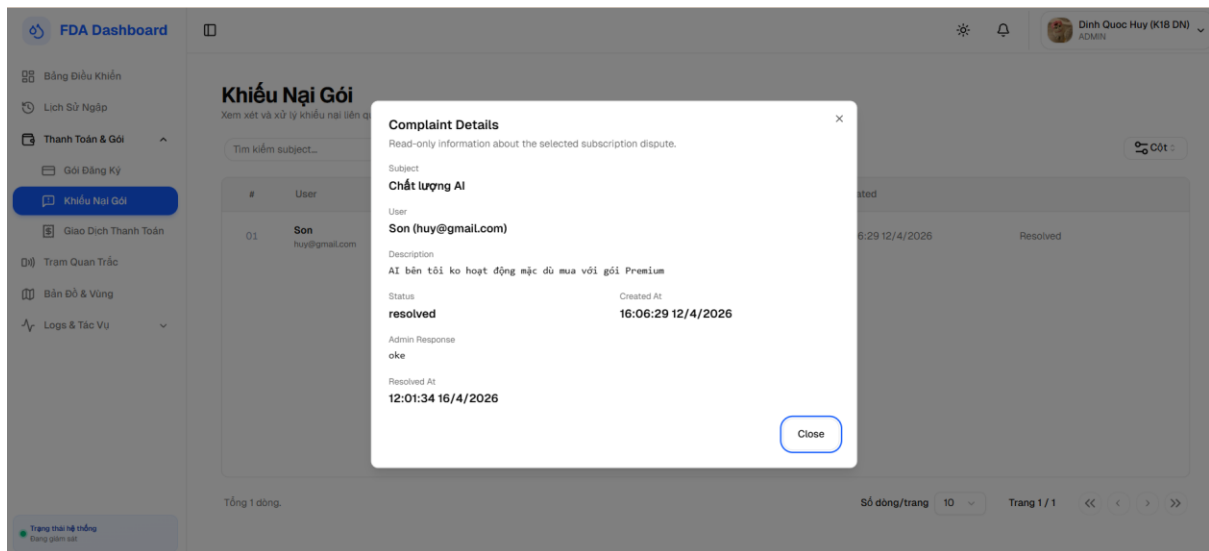


Figure 87: Subscription Disputes - Complaint DetailsDialog

Step 4: Resolve or Reject a Complaint

In the right-most action column (header may be blank), click **Resolve** for **Open** complaints.

In the “**Xử lý khiếu nại gói**” dialog, enter “**Phản hồi của Admin**”, choose “**Đã xử lý**” or “**Từ chối**”, then click “**Gửi**”.

After success, a toast appears and the table refreshes.

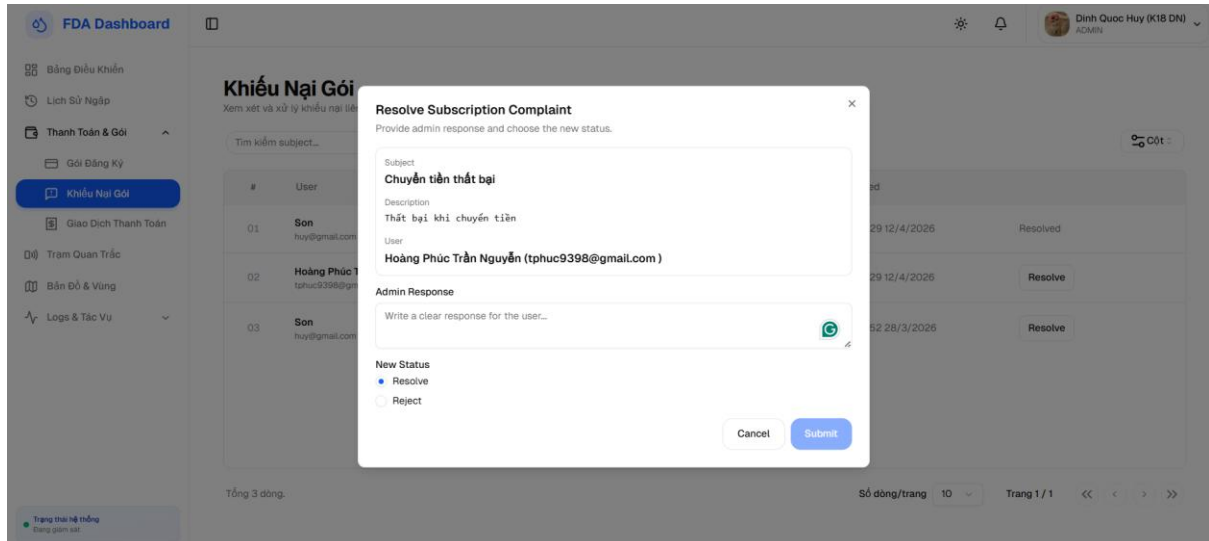


Figure 87: Subscription Disputes - Resolve/ Reject Dialog

3.2.7 Flood Map Realtime

The map provides a live, geographic overview of all monitored zones, sensor statuses, and active flood warnings.

3.2.7.1 View Real-time Flood Map

Step 1: Open the Map Interface

From the dashboard side menu, select "**Bản Đồ & Vùng**" tab.

The system loads a full-screen interactive map centered on the monitored city/region.

On the map, monitoring stations appear as colored circular markers. These markers represent the latest observed condition (real-time snapshot) for each station.

Users can visually scan the map to identify areas of higher concern and hover on specific markers for real-time details.

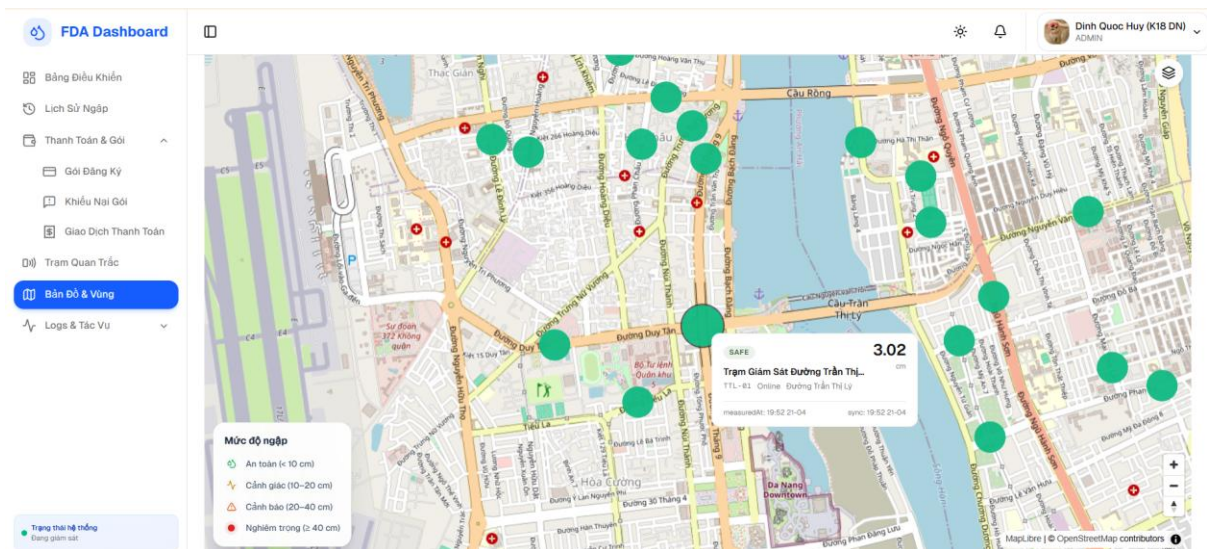


Figure 87: Real-time Flood Map - Overview

Step 2: View the Flood Severity Legend

At the bottom-left corner of the map, locate the legend panel labeled "**Mức độ ngập**".

This panel indicates flood severity levels and the corresponding visual colors shown on the map.

Use the legend to interpret marker colors at a glance:

- **Green(safe): An toàn**– Safe / normal water level
- **Yellow(caution): Cảnh giác** – Elevated water level (early risk)
- **Orange(warning): Cảnh báo** – High water level (warning range)
- **Red (critical): Nghiêm trọng** – Critical flooding level (highest risk)

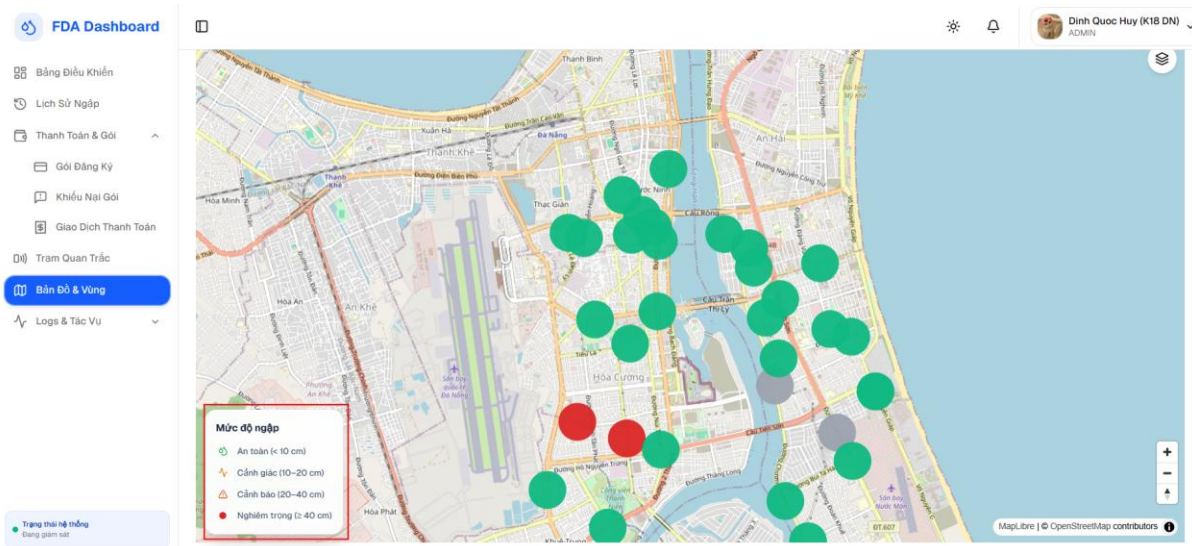


Figure 87: Flood Severity Legend

3.2.7.2 Monitor Station Sensor Data

Step 1: Select a specific Station On the map

On the map, locate the colored station markers representing monitoring stations. Click on any station marker to open a station detail card.

Step 2: Read Live Metrics

In the station detail card, review the station's real-time information, typically including:

- Current water level (e.g., meters and/or centimeters, depending on the display)
- Additional station metrics shown on the card (e.g., thresholds, distance, sensor height if provided)
- The last data update time (timestamp shown at the bottom area of the card)

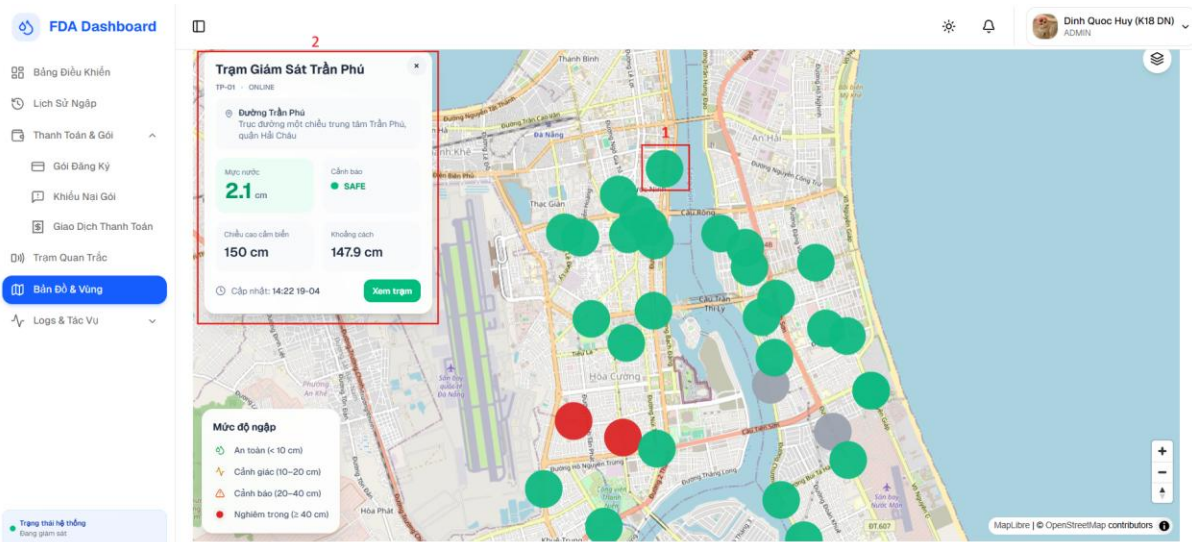


Figure 87: Live Metrics of Station Card

3.2.7.3 Filter Map Layers and Data

This feature allows users to control which datasets are visible on the map through the layer panel ""**Lớp Bản Đồ**". Users can also switch the base map type.

Step 1: Access the Map Filter

On the map view, locate the layer panel button at the top-right corner (map control area).

Click the button to open the panel titled "**Lớp Bản Đồ**".

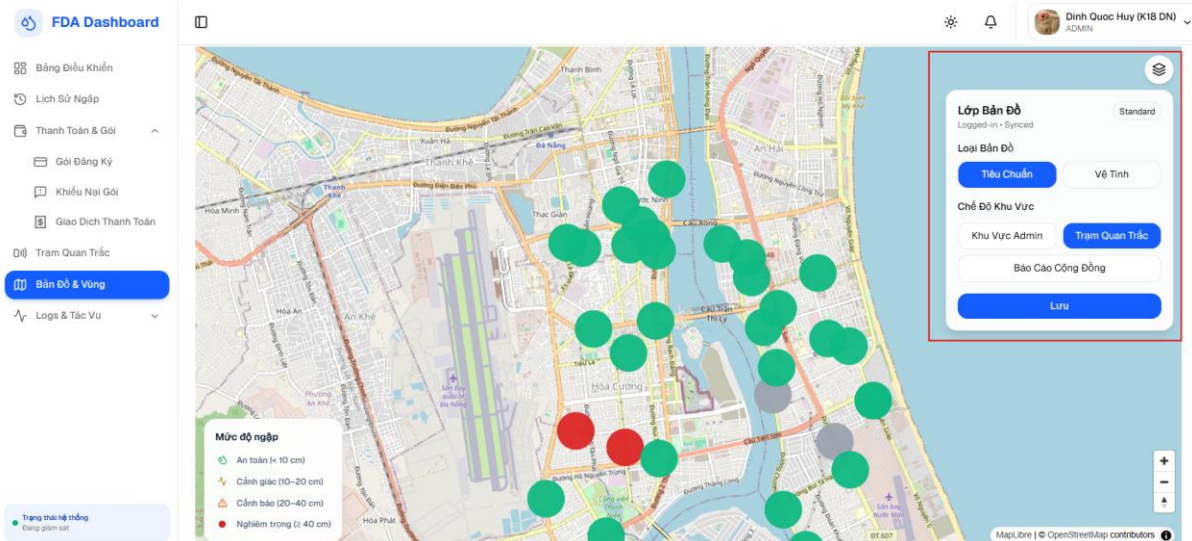


Figure 87: Map Layer Panel

Step 2: Switch Base Map Type

Inside "**Lớp Bản Đồ**", use the "**Loại nền**" controls to switch between base map styles:

- "**Tiêu chuẩn**" (standard road map)
- "**Vệ tinh**" (satellite view), if available

After selecting, click "**Lưu**" to apply changes.

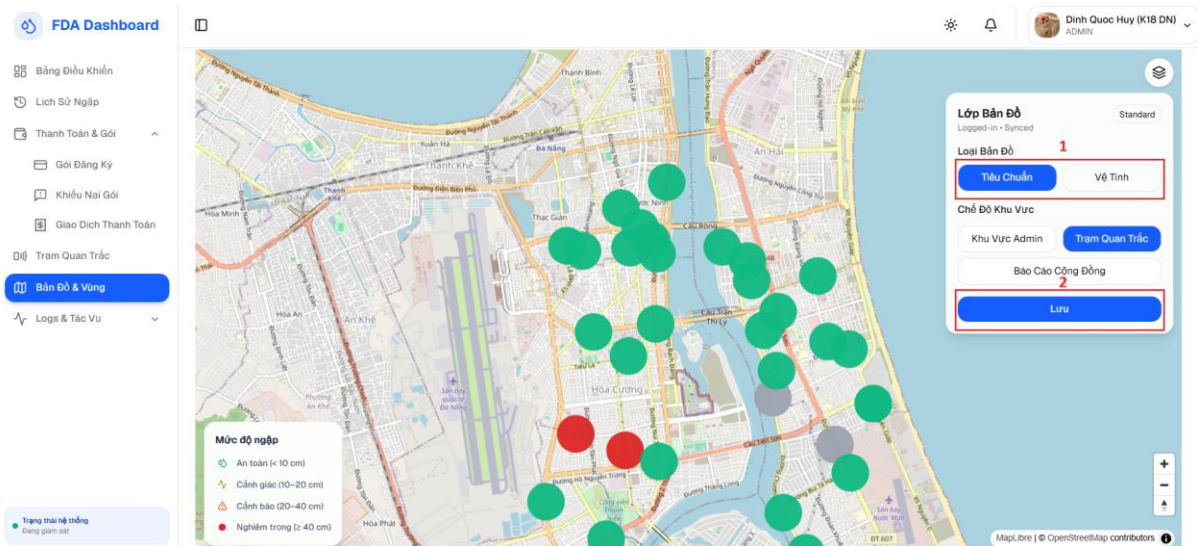


Figure 87: Base Map Switch

Step 3: Toggle Data Layers

In the same panel, locate the "**Chế độ Khu Vực**" and other layer toggles. These controls allow you to show/hide different datasets, such as:

- "**Khu Vực Admin**": Administrative areas
- "**Trạm Quan Trắc**": Monitoring stations
- "**Báo Cáo Cộng Đồng**": Community reports - for real-time citizen-submitted reports

After making changes, click "**Lưu**".

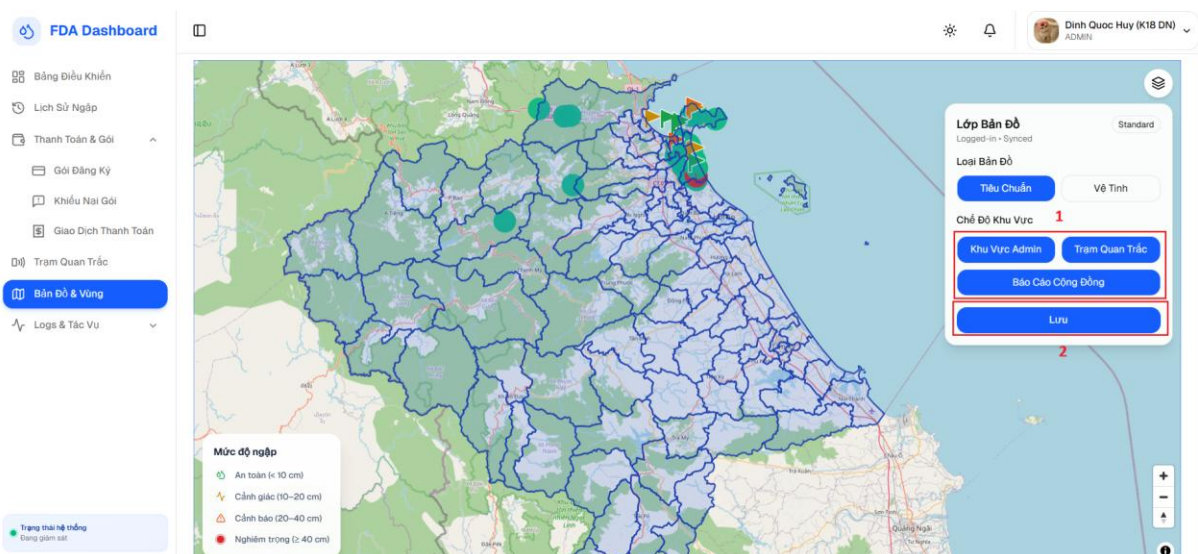


Figure 87: Layer Toggles

Step 4: View Community Reports (On Map)

When the "**Báo Cáo Cộng Đồng**" layer is enabled:

- Community report markers become visible on the map alongside station markers.
- Click a community report marker to open the report detail popup card.

The popup card (left side) typically includes:

- Title: "**Phản ánh cộng đồng**"
- Report content short description under the title
- **Key fields** displayed as short label/value rows (depending on report data), such as:
 - "**Mức độ**": severity
 - "**Tin cậy**": trust / confidence indicator
 - "**Trạng thái**": status
 - "**Gửi lúc**": a timestamp displayed on the card as a date/time line.

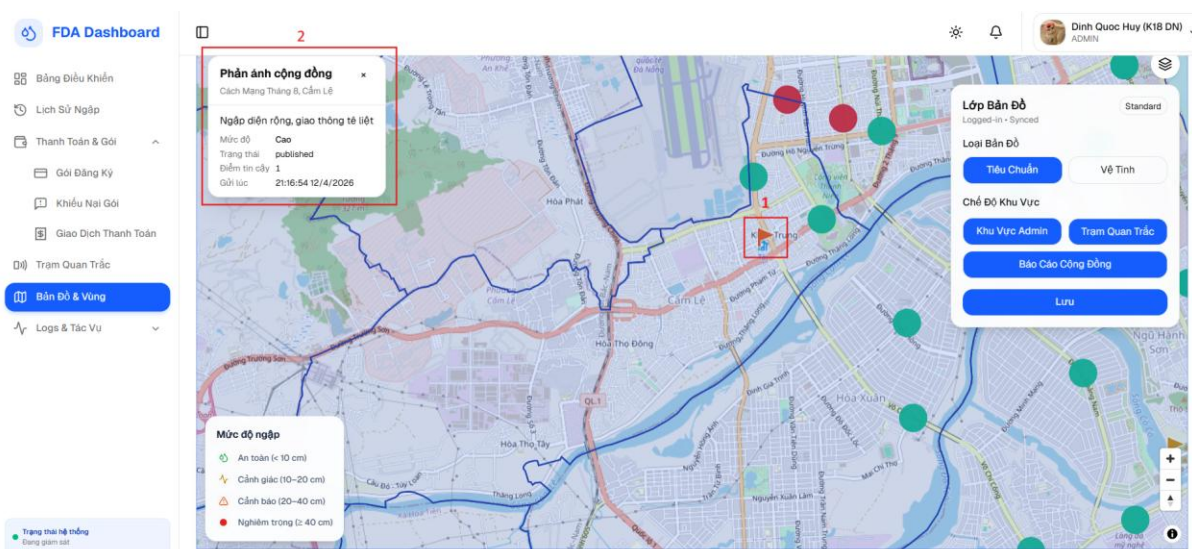


Figure 87: Community Report - Popup Card on Map

3.2.8 AI-based Flood Prediction

This section describes how to use the AI-powered flood prediction feature in the FDA System. The feature consists of three main capabilities:

3.2.8.1 View AI-based Flood Prediction

Step 1: Navigate to the Flood Map

- In the left-hand sidebar, click tab “Bản Đồ và Vùng”

- The map view loads, displaying administrative area polygons over a base map layer.

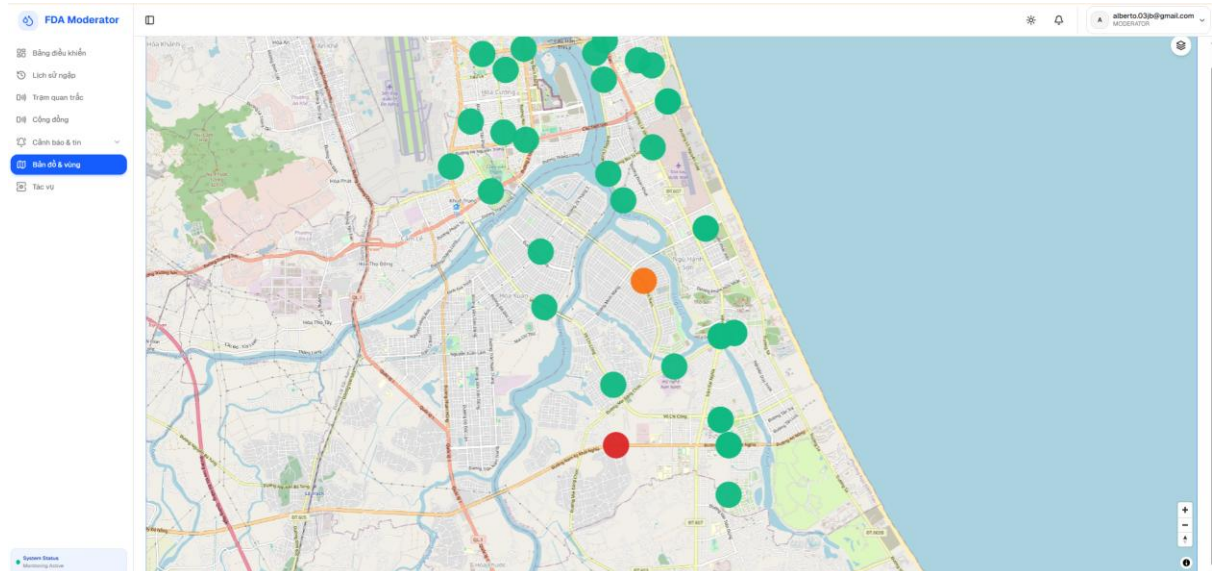


Figure 87: Navigate to the Flood Map

Step 2: Confirm Required Map Layers are Active

Before running AI prediction, ensure the following layers are enabled via the “**Lớp Bản Đồ**”

- “**Khu Vực Admin**”: toggle ON to show administrative area boundaries. Clicking an area triggers the prediction card.
- “**Trạm Quan Trắc**”: toggle ON to show sensor station markers. Station data contributes to the AI prediction.

Open the Layer Panel by clicking the Layers button on the map interface. Adjust toggles as needed, then click “**Lưu**” to apply changes

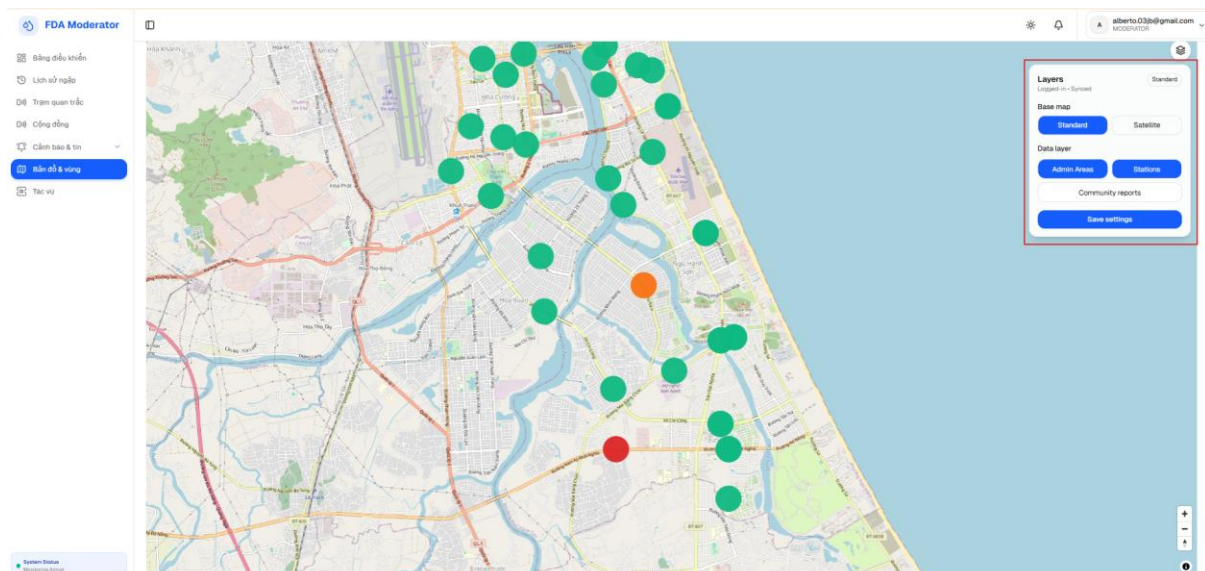


Figure 87: Confirm Required Map Layers are Active

Step 3: Select an Administrative Area on the Map

Click directly on any administrative area polygon (ward or district boundary) displayed on the map.

The system highlights the selected area by zooming the map camera to fit the area's boundary.

The **“Area Detail Card”** slides in from the left side of the screen, showing:

- Area name and administrative code
- Two action buttons: **“Chụp ảnh vệ tinh”** and **“Dự đoán AI”**

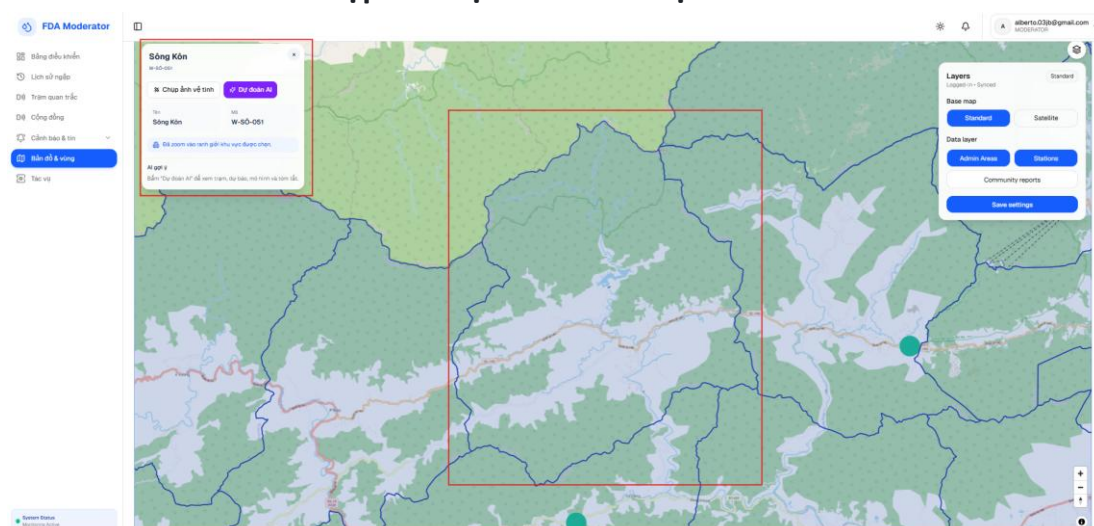


Figure 87: Filter Payment Records

Step 4: Trigger AI Flood Prediction

Click the **“Dự đoán AI”** button (purple/violet button with a sparkles icon).

The button label changes to **“Đang gọi AI”** and becomes disabled while the request is in progress.

The system calls the backend API to assemble data from multiple sources (weather data, satellite imagery, sensor stations, and terrain analysis) and compute the ensemble prediction.

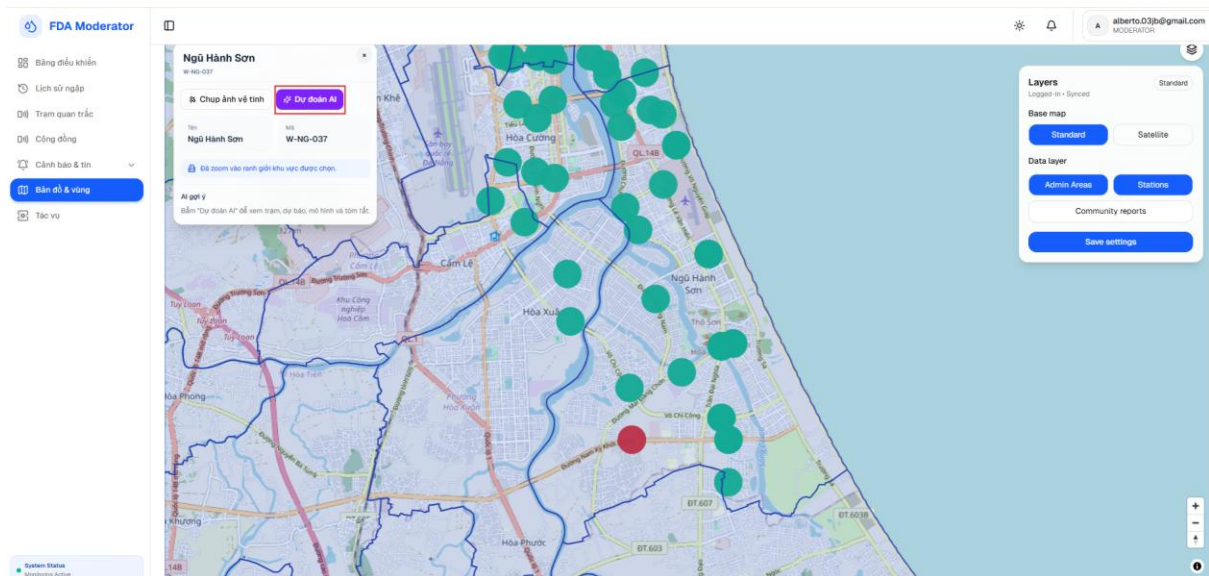


Figure 87: Filter Payment Records

Step 5: Review the AI Prediction Result

Upon successful completion, the “**PredictFloodAiPanel**” section expands within the Area Detail Card, displaying:

The button label changes to “**Đang gọi AI**” and becomes disabled while the request is in progress.

The system calls the backend API to assemble data from multiple sources (weather data, satellite imagery, sensor stations, and terrain analysis) and compute the ensemble prediction.

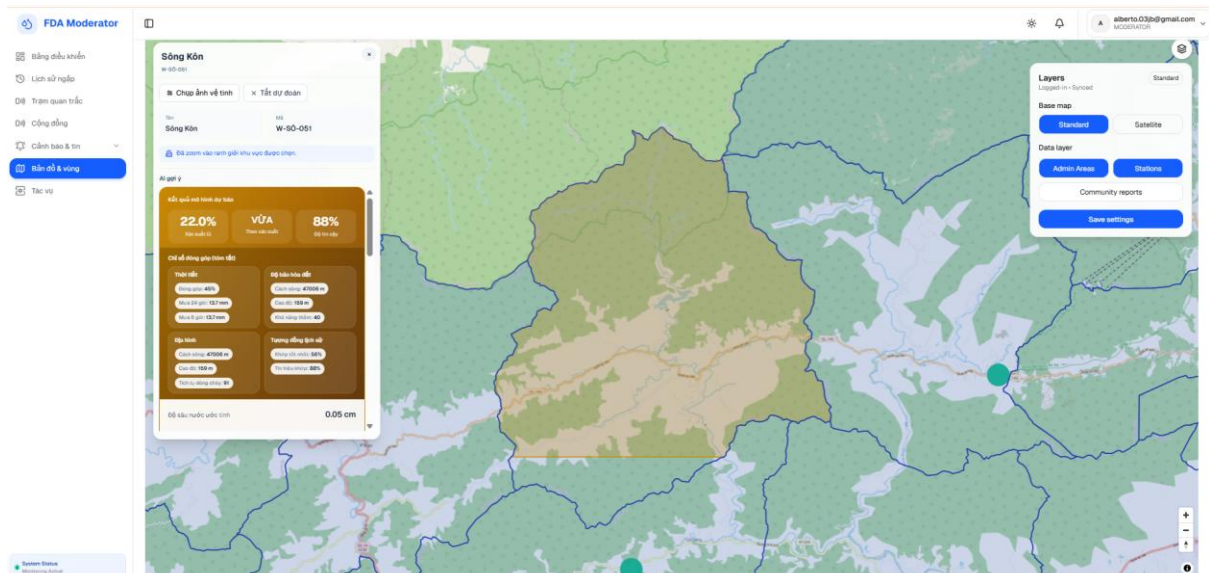


Figure 87: Review the AI Prediction Result

3.2.8.2 Evaluate Administrative Area-based Flood Updates

This feature enables authorized users to evaluate the accuracy of AI flood predictions by comparing the predicted forecast windows against real-time water level data from sensor stations. The comparison is facilitated through the “**Flood History**” module.

Step 1: Trigger AI Flood Prediction

After navigate to “**Area Detail Card**”, click the “**Dự đoán AI**” button (purple/violet button with a sparkles icon).

The button label changes to “**Đang gọi AI**” and becomes disabled while the request is in progress.

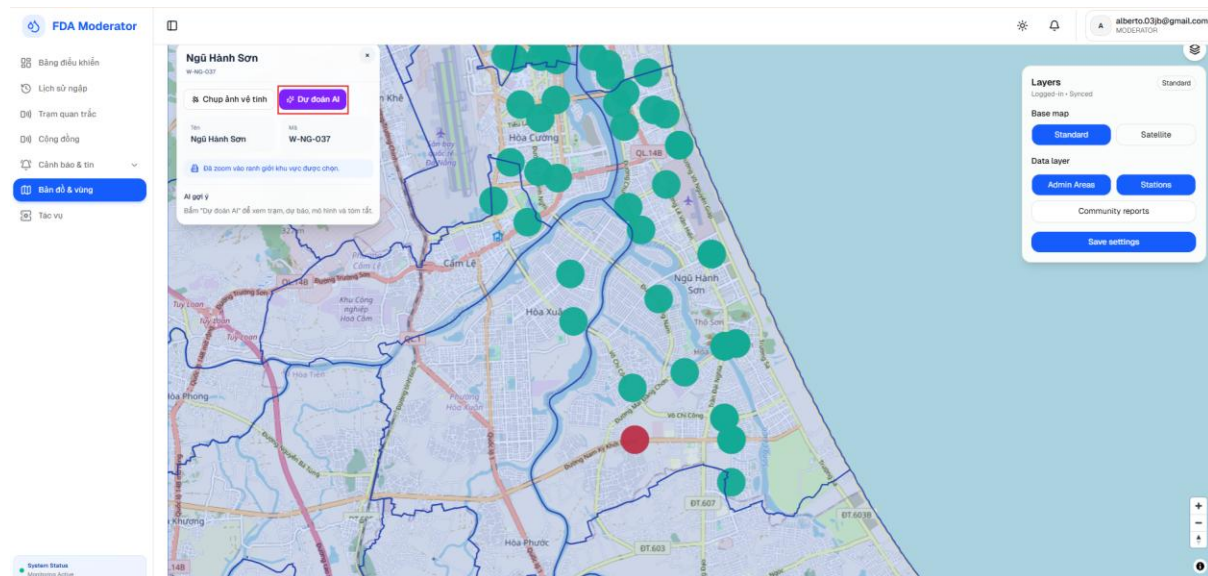


Figure 87: Trigger AI Flood Prediction

Step 2: Review result compare AI Precision with Actual Data

The comparison results are displayed in two sections: “**Xu hướng theo khung giờ**” and “**Trạm đo gần khu vực**”.

“**Xu hướng theo khung giờ**”: displays AI prediction values and corresponding severity levels for different time horizons (e.g., +3h, +5h, +7h).

“**Trạm đo gần khu vực**” : provides an overview of the number of monitoring stations within the selected area. When expanded, it displays a list of stations along with their detailed water level data.

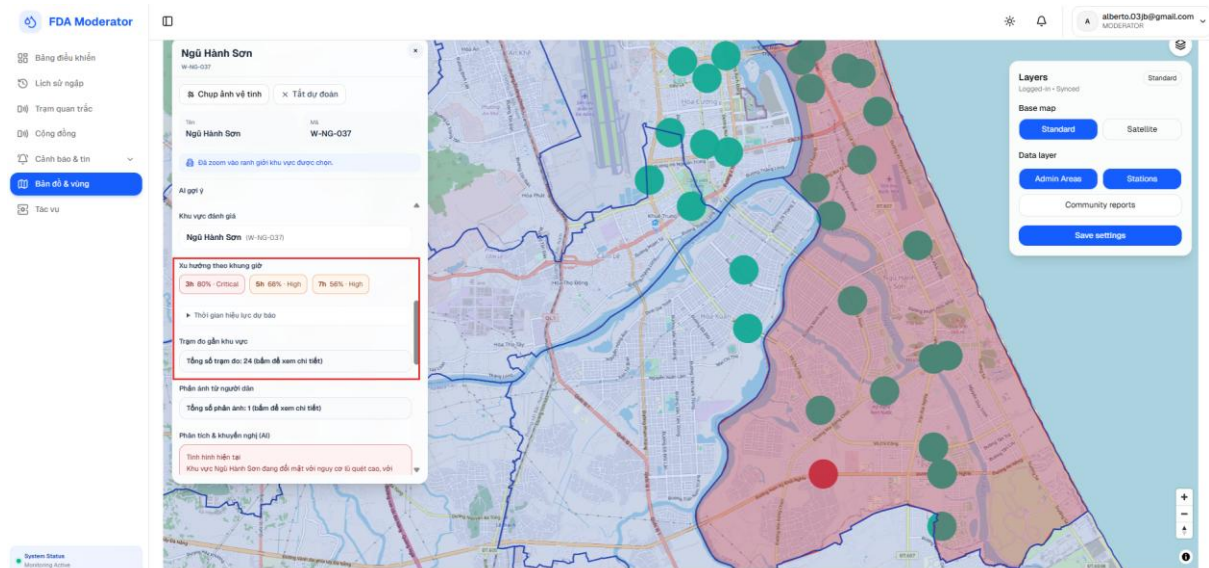


Figure 87: Result compare AI Precision

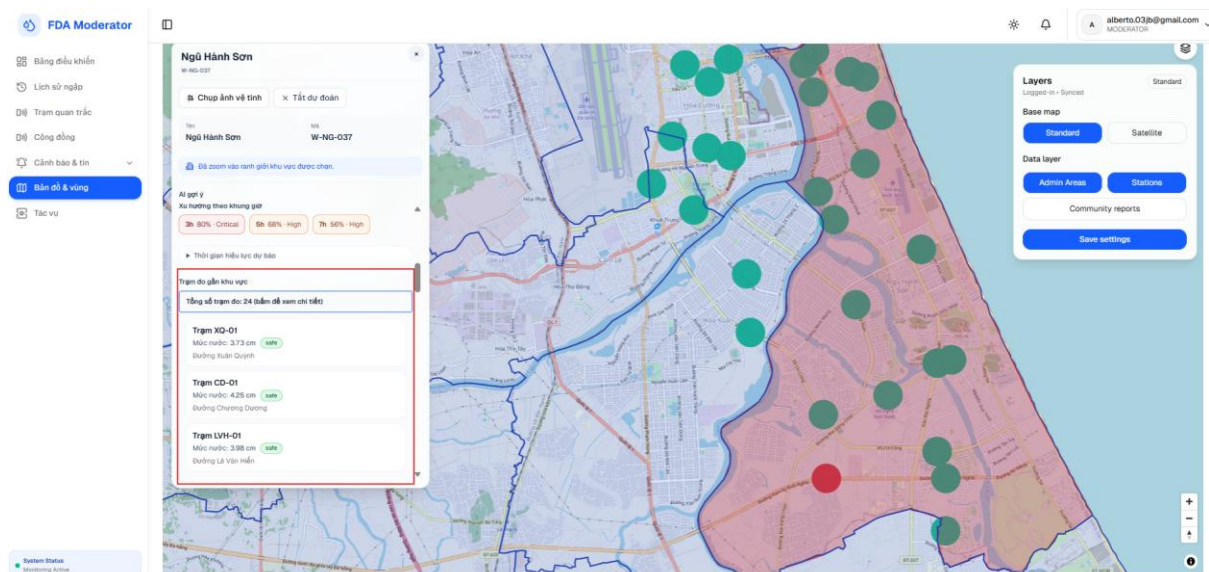


Figure 87: Result Actual Data Based on Stations

3.2.8.3 Identify High-risk Flood Zones (AI)

This feature highlights areas with the highest flood risk on the interactive map using AI-driven probability data, enabling authorities to prioritize response and mitigation efforts.

Step 1: Trigger AI Flood Prediction

After navigate to **“Area Detail Card”**, click the **“Dự đoán AI”** button (purple/violet button with a sparkles icon).

The button label changes to **“Đang gọi AI”** and becomes disabled while the request is in progress.

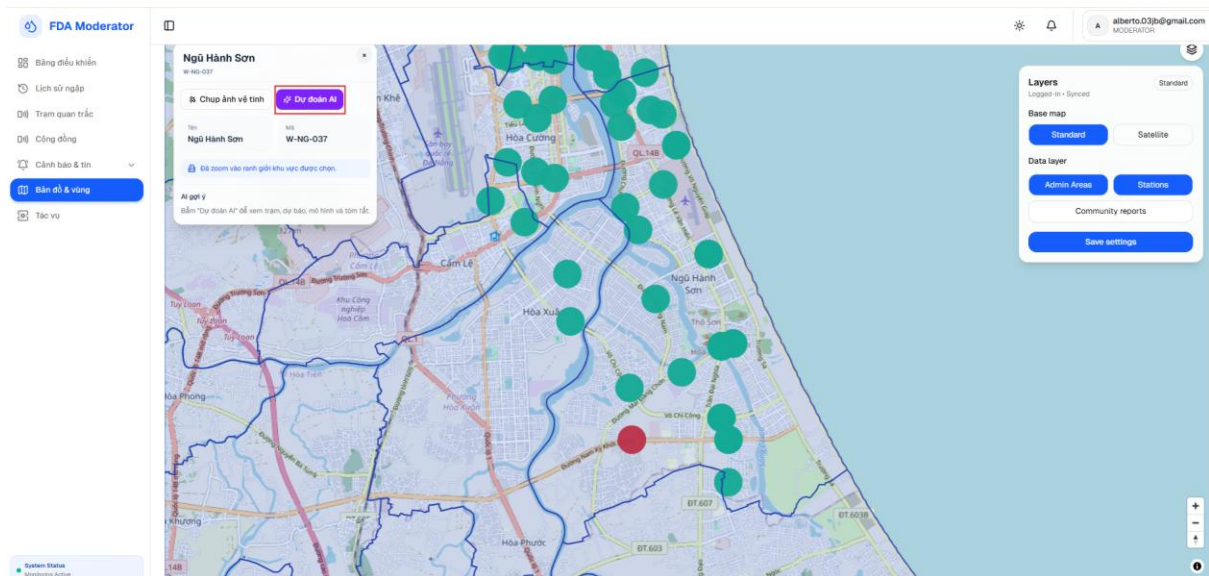


Figure 87: Filter Payment Records

Step 2: Review result of identify High-risk Flood Zones (AI)

The result of identify High-risk Flood Zones will be shown in 2 different sections:

- **“AI gợi ý”**: displays main criteria, including: Xác suất lũ, Theo xác suất, Độ tin cậy, Thời tiết, Độ bão hòa đất, Địa hình, Tương đồng lịch sử.
- **Highlighted Area**: The analyzed area is highlighted on the map with a severity color corresponding to the results shown in the **“AI gợi ý”** section.

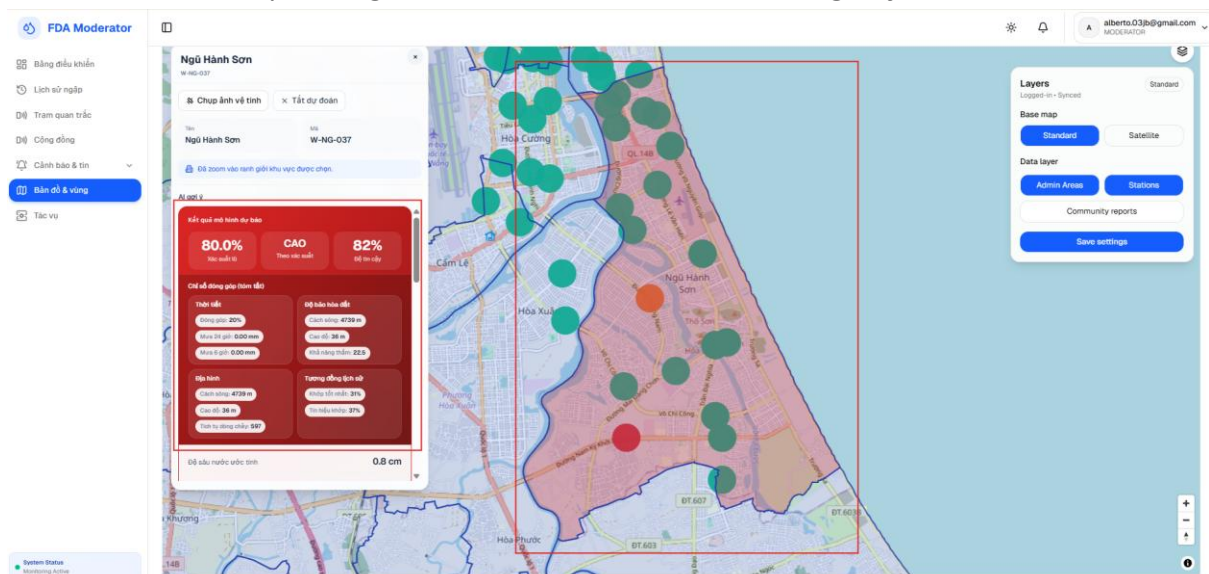


Figure 87: Results of identity High-risk Zones(AI)

3.2.9 Community Management

3.2.9.1 View Community Flood Report

Step 1: Navigate to Community Management

Log in as a **Moderator**.

From the left-hand sidebar, click "**Cộng Đồng**".

The page "**Phản ánh cộng đồng**" is displayed with:

- A quick status switch area (**Lọc nhanh theo trạng thái**) at the top
- A filter form (**Bộ lọc phản ánh cộng đồng**)
- A report list section (**Danh sách phản ánh**)
- Pagination controls at the bottom

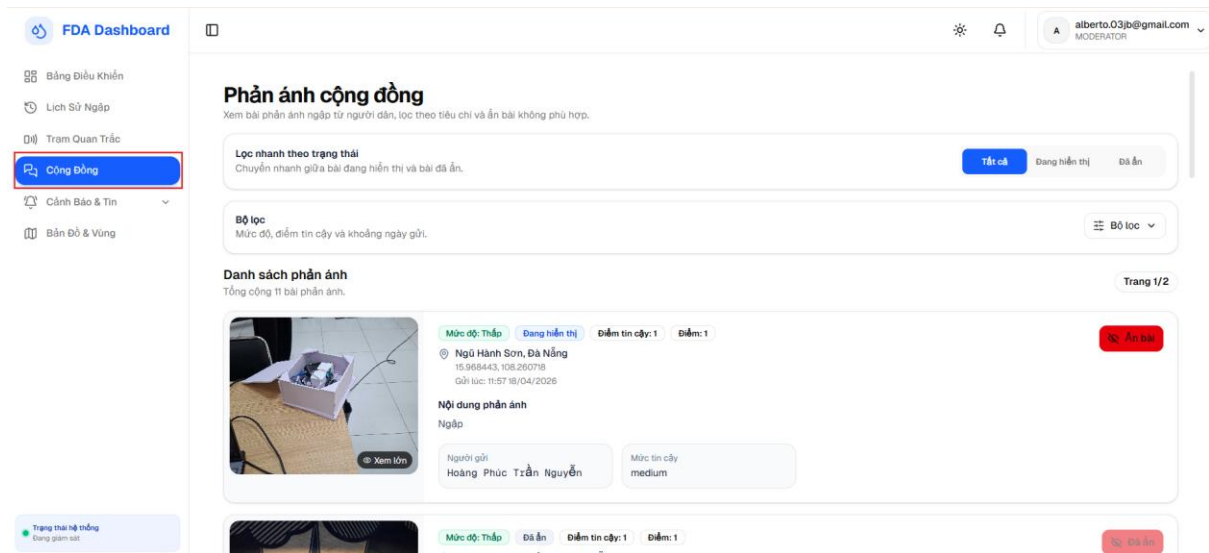


Figure 87: Community Management - Page Overview

Step 2: Use Quick Filter Tabs (Status)

At the top of the page, use the quick tabs under "Lọc nhanh theo trạng thái":

- Tất cả
- Đang hiển thị
- Đã ẩn

Selecting a tab immediately changes the status filter and refreshes the report list.

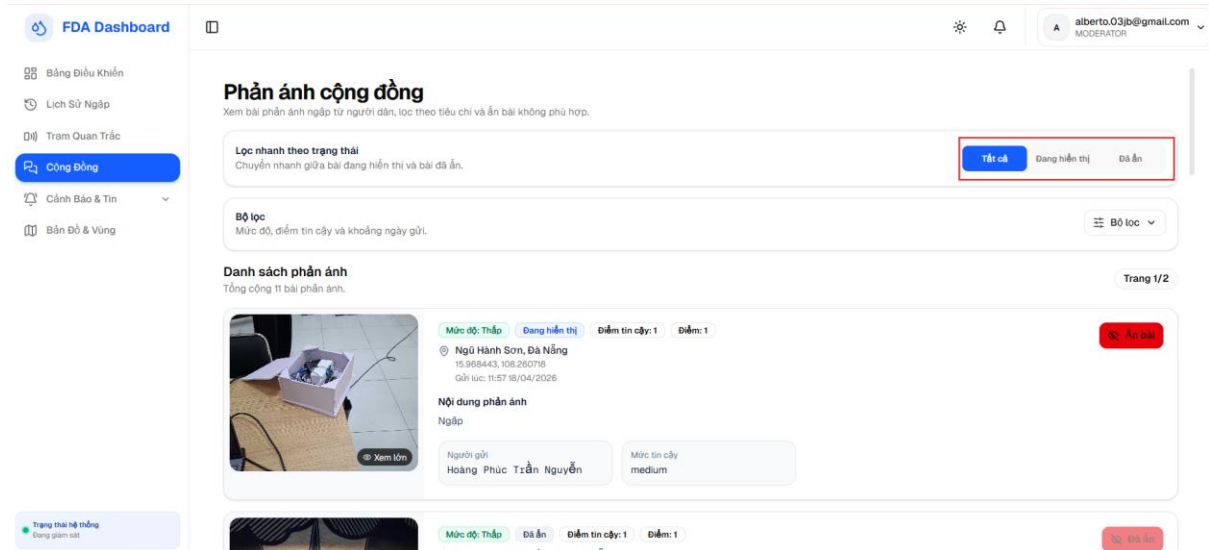


Figure 87: Community Management - Quick Status Tabs

Step 3: Configure Detailed Filters

In the "Bộ lọc" section, configure optional filters:

- **Trạng thái:** Tất cả / Đang hiển thị / Đã ẩn
- **Mức độ:** Tất cả / Thấp / Trung bình / Cao

- **Điểm tin cậy tối thiểu:** numeric value (example: 1)
- **Từ ngày:** start date (date input)
- **Đến ngày:** end date (date input)

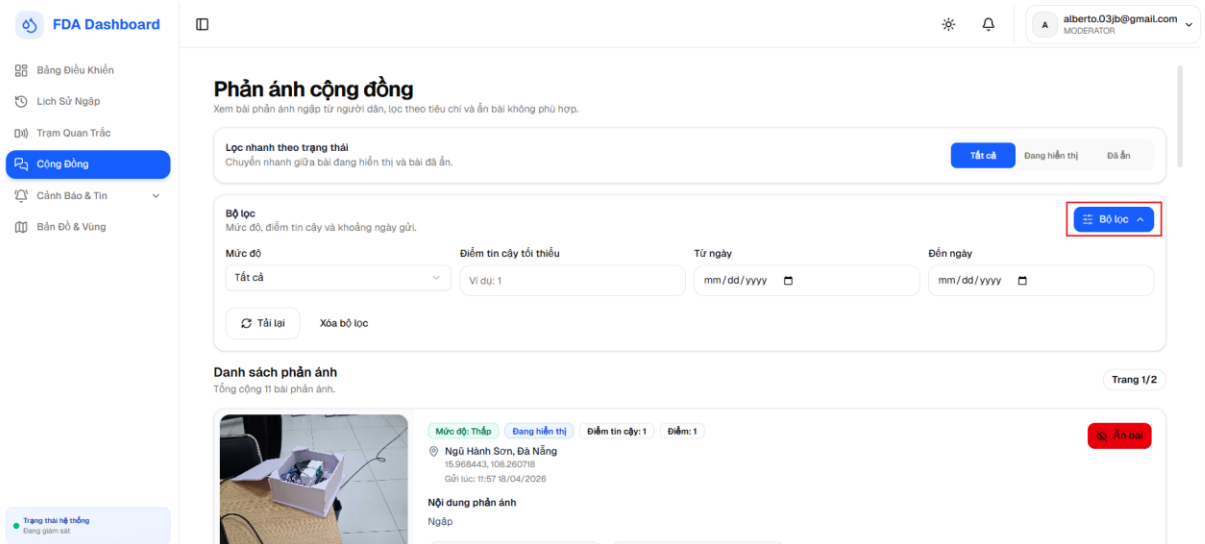


Figure 87: Community Management - Filter Panel

Step 4: Reload or Clear Filters

Below the filter inputs:

- Click **"Tải lại"** to fetch the report list using the current filters.
- Click **"Xóa bộ lọc"** to reset all filters back to default values.

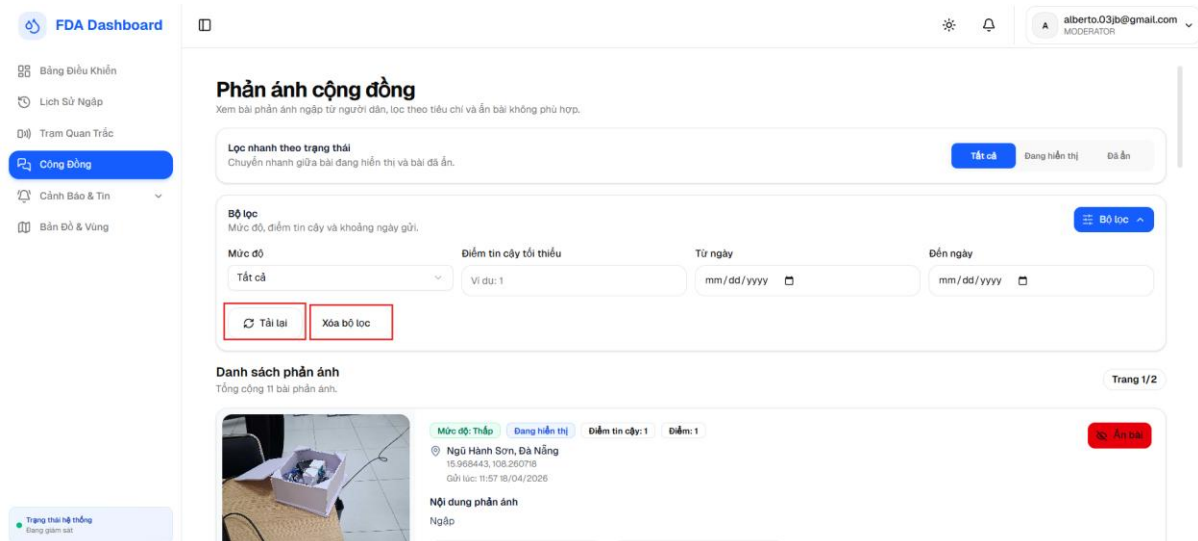


Figure 87: Community Management - Reload and Clear Filter Buttons

Step 5: Review the Report List

In the **"Danh sách phản ánh"** section:

- The header shows total report count: **"Tổng cộng [N] bài phản ánh"**

- A page badge shows the current page: "**Trang [current]/[total]**"

Each report item typically contains:

- A preview image/video area (left)
 - Button "**Xem lớn**" opens the media preview dialog
 - Thumbnails appear if multiple media files are attached
- Metadata badges (right)
 - **Mức độ:** Thấp / Trung bình / Cao
 - **Trạng thái:** Đang hiển thị / Đã ẩn
 - **Điểm tin cậy:** \ (trustScore\) (if available)
 - **Điểm:** \ (score\) (if available)
- Location block
 - Address (or “Chưa có địa chỉ”)
 - Coordinates: latitude, longitude
 - Submit time: "**Gửi lúc: [dd/MM/yyyy, HH:mm]**"
- Report content block
 - Title: "**Nội dung phản ánh**"
 - Description (or “Không có mô tả.”)
- Reporter information block
 - **Người gửi:** reporter name/email (if available), otherwise “Không rõ”
 - **Mức tin cậy:** confidence level (if available)

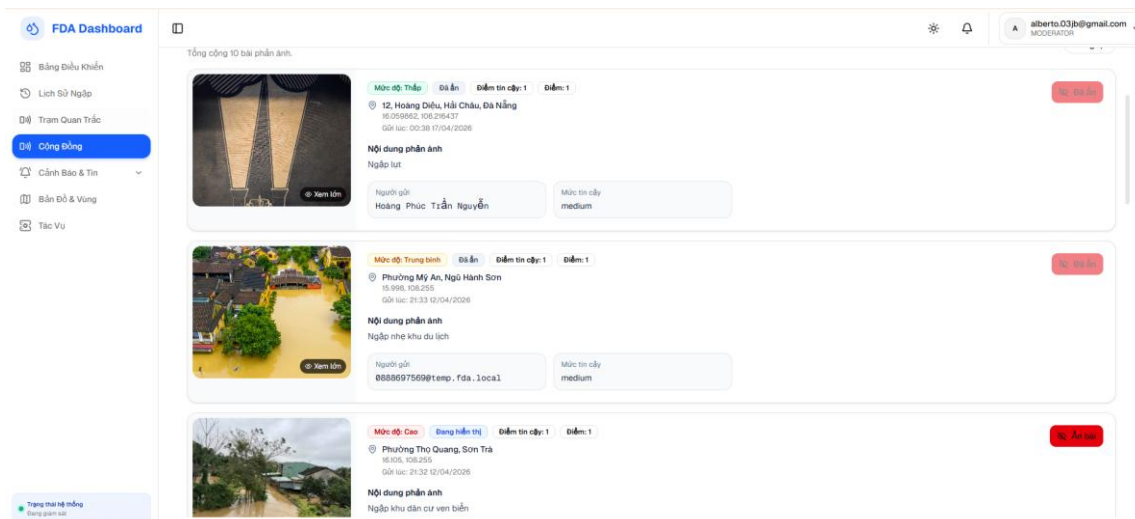


Figure 87: Community Management - Report Card Details

Step 6: Open Media Preview (Image/Video)

To inspect attachments:

- Click "**Xem lớn**" on the preview area (or click a specific thumbnail).

- A dialog opens titled "**Xem tệp đính kèm**".
- The dialog displays:
 - Video player (if mediaType is video), or
 - Full-size image viewer

Close the dialog to return to the list.

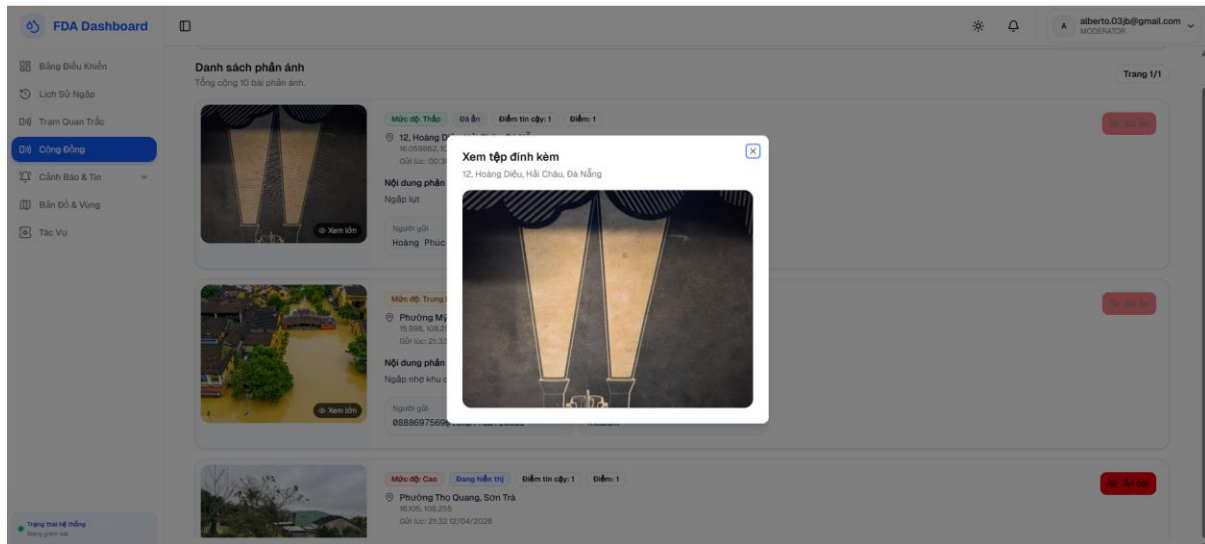


Figure 87: Community Management - Media Preview Dialog

3.2.9.2 Hide Invalid Report

This feature allows Moderator users to hide reports that are invalid, duplicated, or contain unsuitable content. Hidden reports are removed from the “**Đang hiển thị**” view and become visible under the “**Đã ẩn**” filter.

Step 1: Locate a Report to Hide

From "**Danh sách phản ánh**", identify a report that should be hidden.

Reports can be pre-filtered using:

- Quick tab "**Đang hiển thị**", or
- Filter field **Trạng thái = Đang hiển thị**

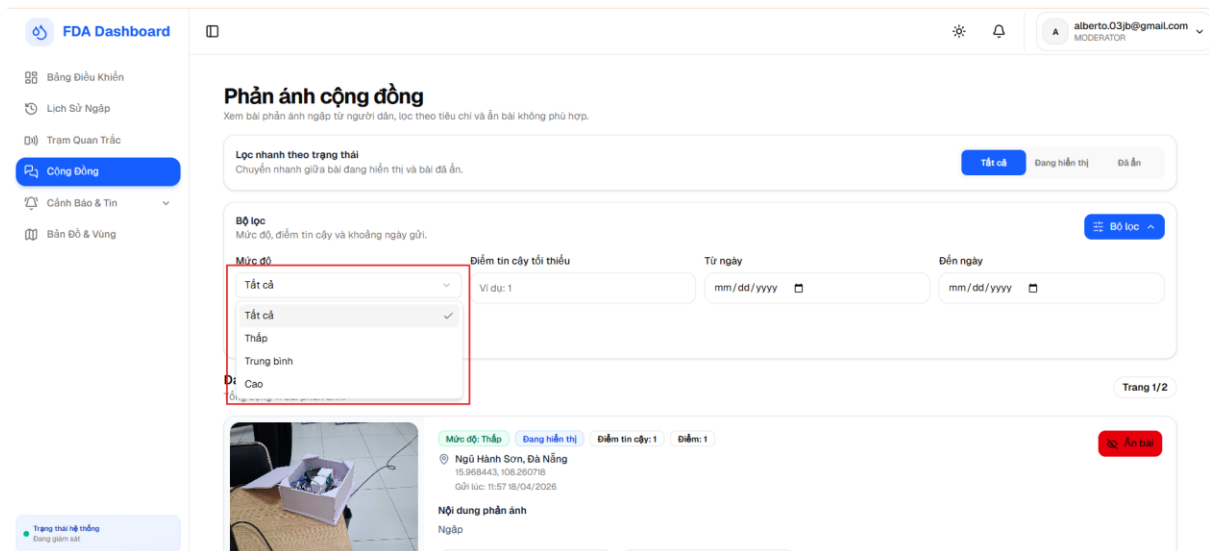


Figure 87: Hide Report - Filter for Published Reports

Step 2: Click the Hide Action

On the right side of the report card, click the "**Ẩn bài**" button.

If the report is already hidden, the button is disabled and displays "**Đã ẩn**".

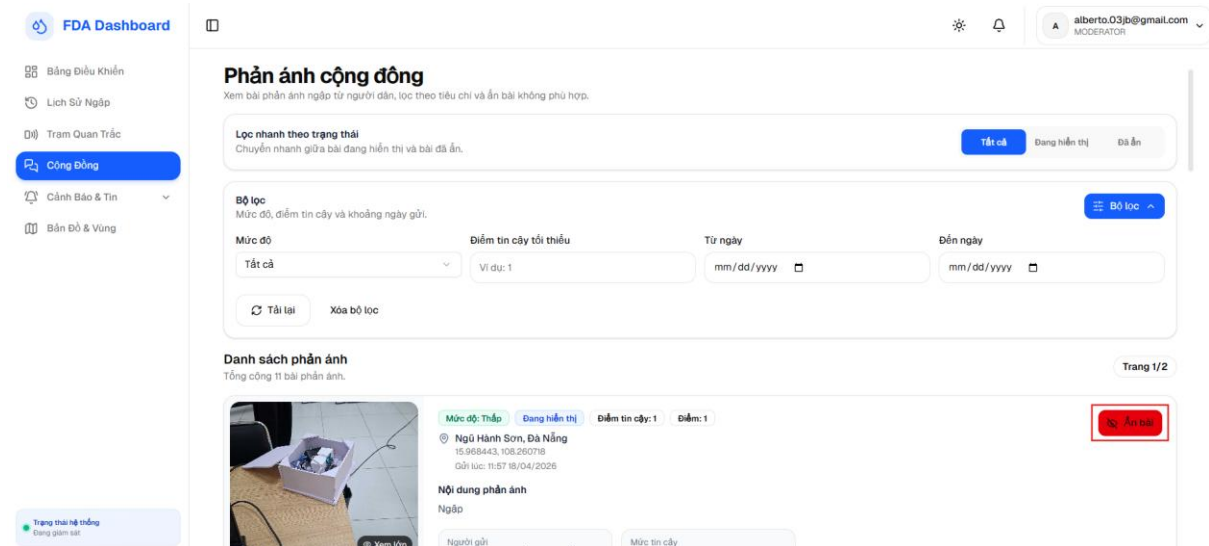


Figure 87: Hide Report - Hide Button Location

Step 3: Confirm the Hide Operation

A browser confirmation prompt appears (system dialog):

- Message: "**Bạn có chắc muốn ẩn bài phản ánh này không?**"

Click **OK** to proceed, or **Cancel** to abort.

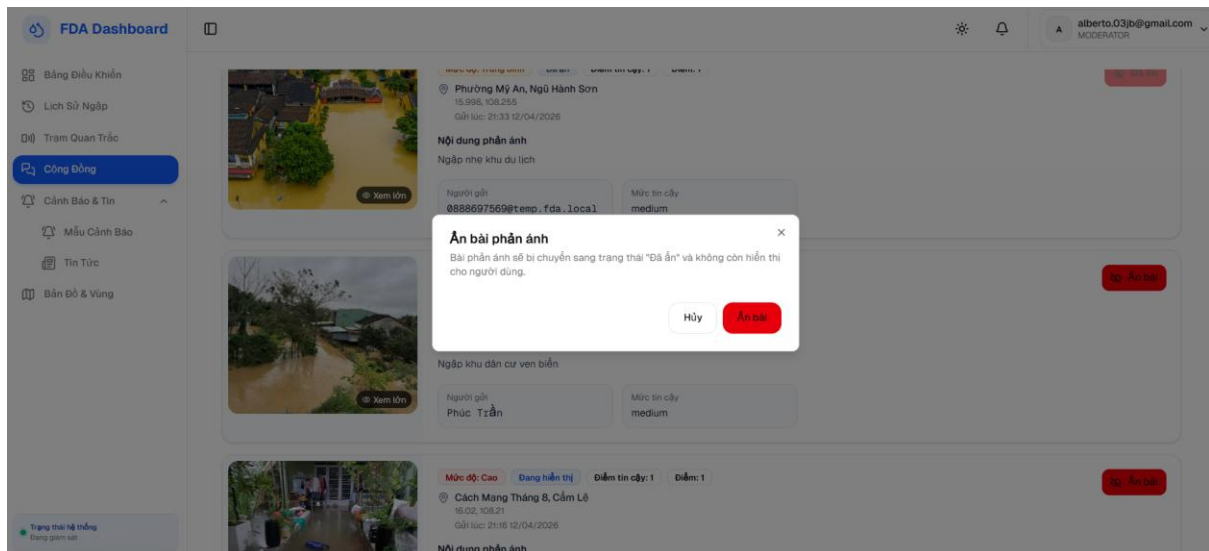


Figure 87: Hide Report - Confirmation Modal

3.2.10 Flood History and Trends

3.2.10.1 View Flood History

This feature allows authorized users (Admin, Moderator) to view historical water level readings for a selected station in short time windows (hours). The screen includes a station list panel (left), KPI summary cards (top-right), and a line chart that renders the historical readings.

Step 1: Navigate to the Flood History Module

From the Admin/Moderator Dashboard, in the left-hand sidebar, click "**Lịch Sử Ngập**".

The Flood History page is displayed with:

- A station list "**Trạm Quan Trắc**" panel on the left
- A time preset selector "**Khoảng thời gian**" on the left
- A "**Làm mới**" button at the top-right of the page header
- A KPI strip (4 cards) above the chart area
- A line chart of "**Phân tích ngập**" in the analysis panel

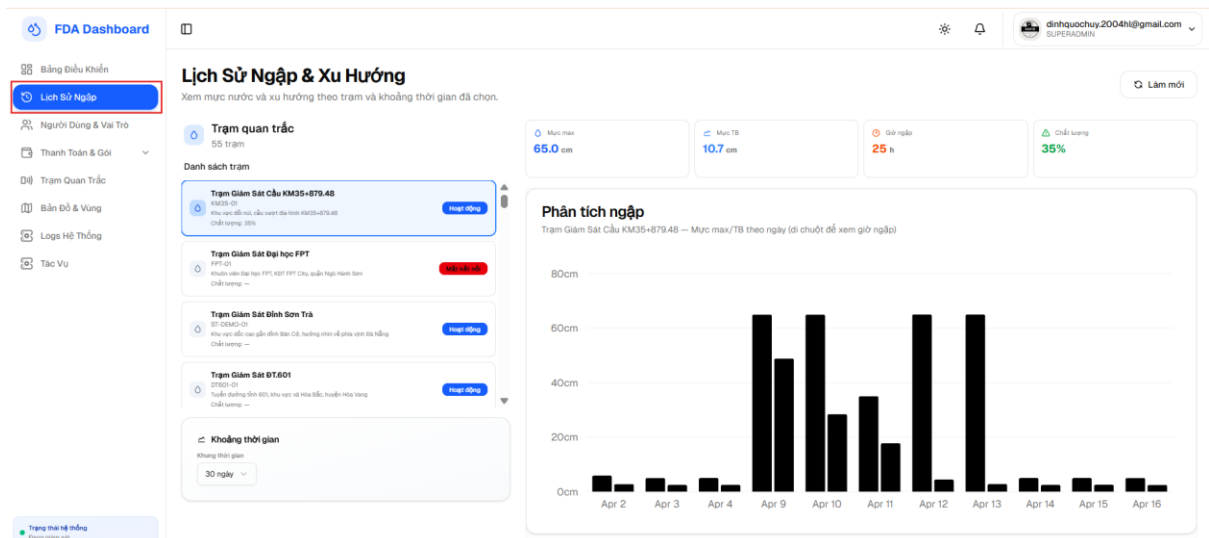


Figure 87: Flood History - Page Overview

Step 2: Select a Monitoring Station

In the left panel, under "**Danh sách trạm**", click one station item to select it.

Each station row shows:

- Station name and station code
- Optional location description
- Data quality (percentage), when available
- Status badge such as "**Hoạt động**", "**Mất kết nối**", or "**Bảo trì**"

After selection, the station row is highlighted, and the page automatically loads data for the selected station.

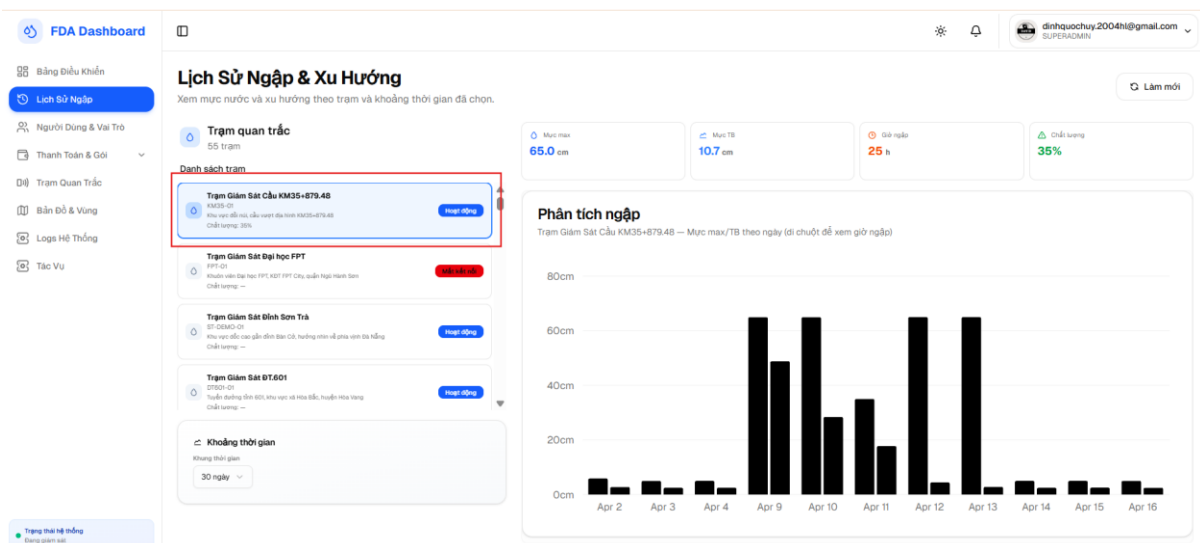


Figure 87: Flood History - Select Station from List

Step 3: Choose a Short-hour Preset

In the left panel, under "**Khoảng thời gian**", open the preset dropdown and select one of the short-hour presets:

- 6 giờ gần nhất
- 12 giờ gần nhất
- 24 giờ gần nhất

These presets correspond to the Flood History view (hourly series) and display with the line chart.

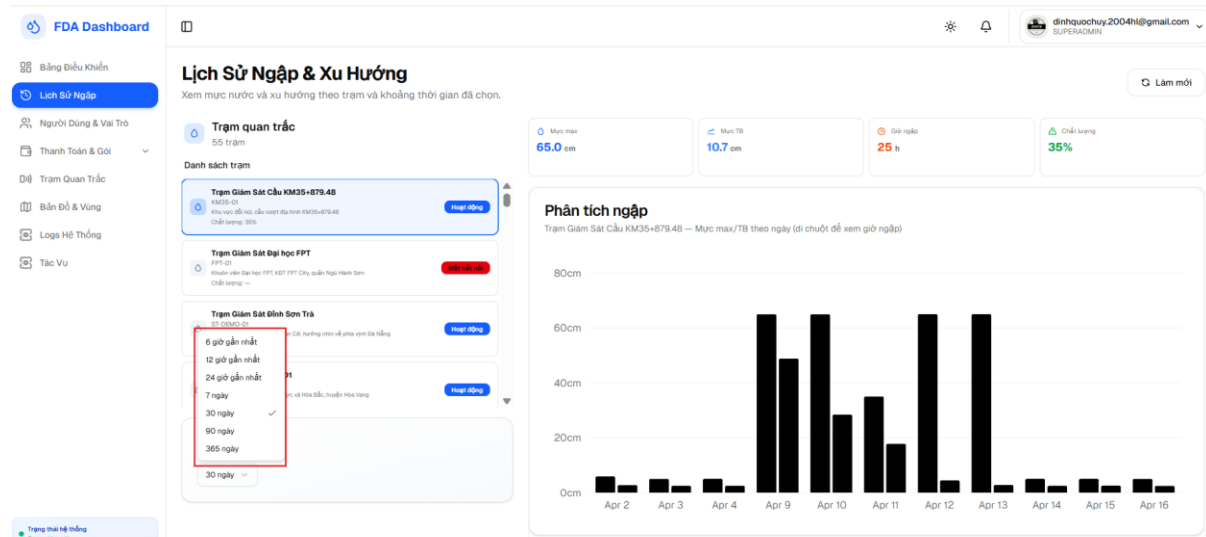


Figure 87: Flood History - Select Short-hour Preset

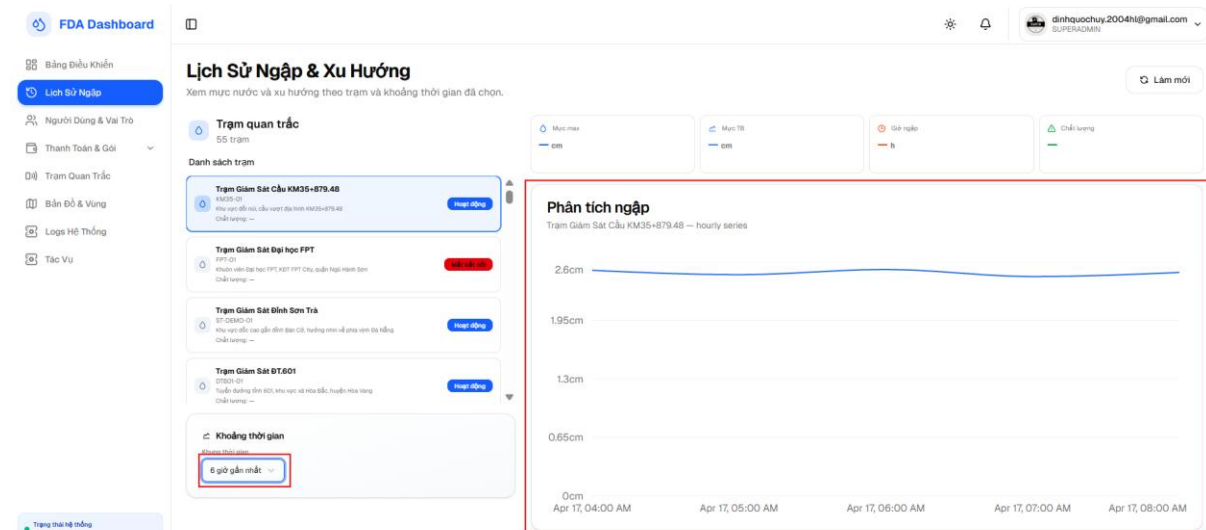


Figure 87: Flood History - Latest 6-hour Preset

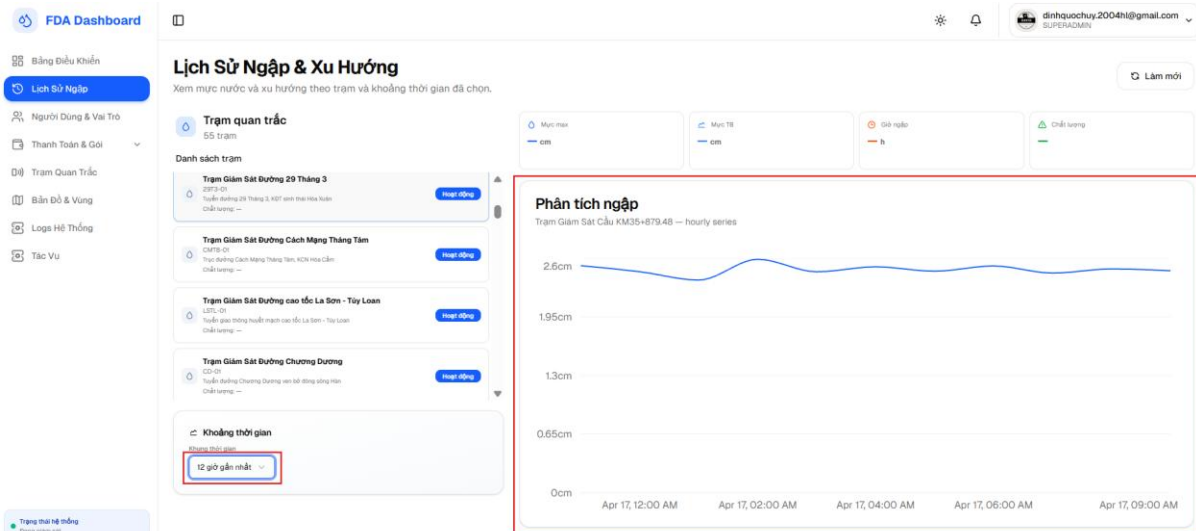


Figure 87: Flood History - Latest 12-hour Preset

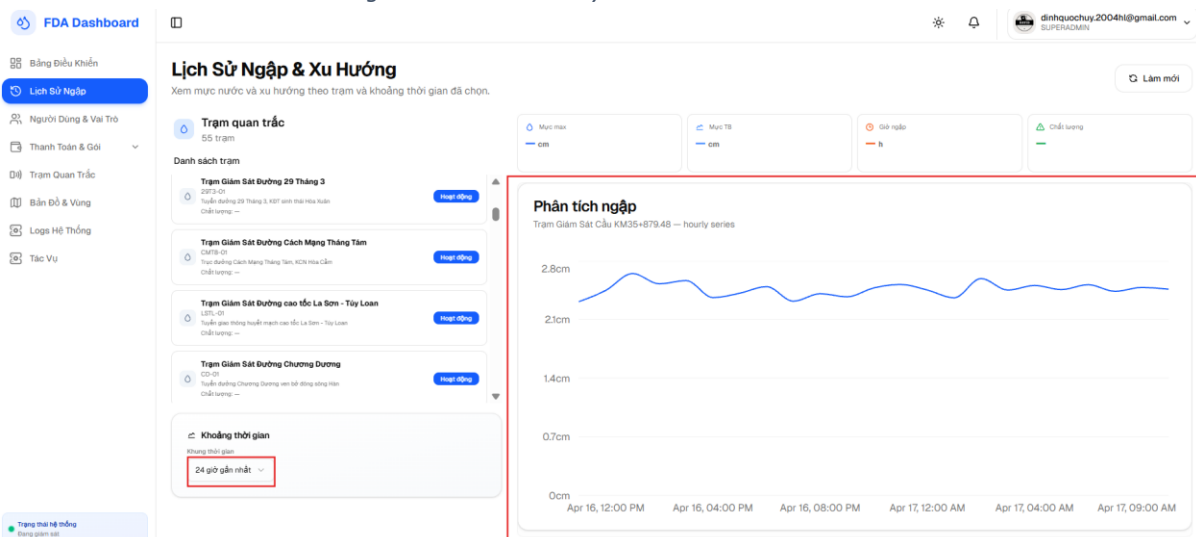


Figure 87: Flood History - Latest 24-hour Preset

Step 4: Refresh Data

Click the "Làm mới" button to reload the chart and KPI values for the current selection.

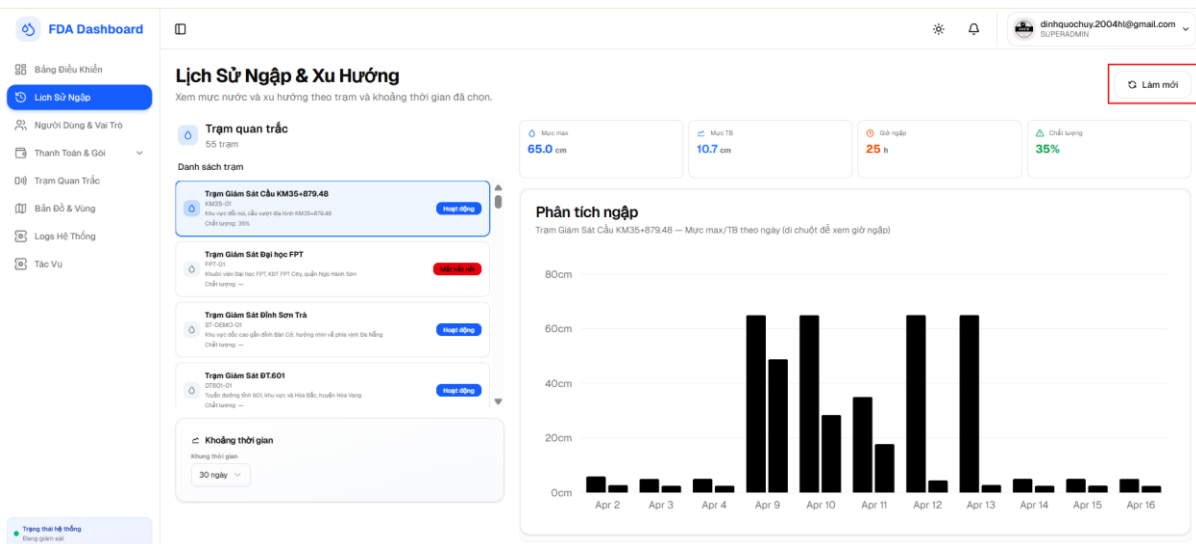


Figure 87: Flood History - Refresh Button Location

Step 5: Review KPI Summary Cards

Above the chart area, a KPI strip displays 4 summary cards:

- **“Mức max”**: Maximum water level within the selected time window (unit: cm)
- **“Mức TB”**: Average water level (unit: cm)
- **“Giờ ngập”**: Total flood hours (unit: h)
- **“Chất lượng”**: Data completeness percentage (unit: %)

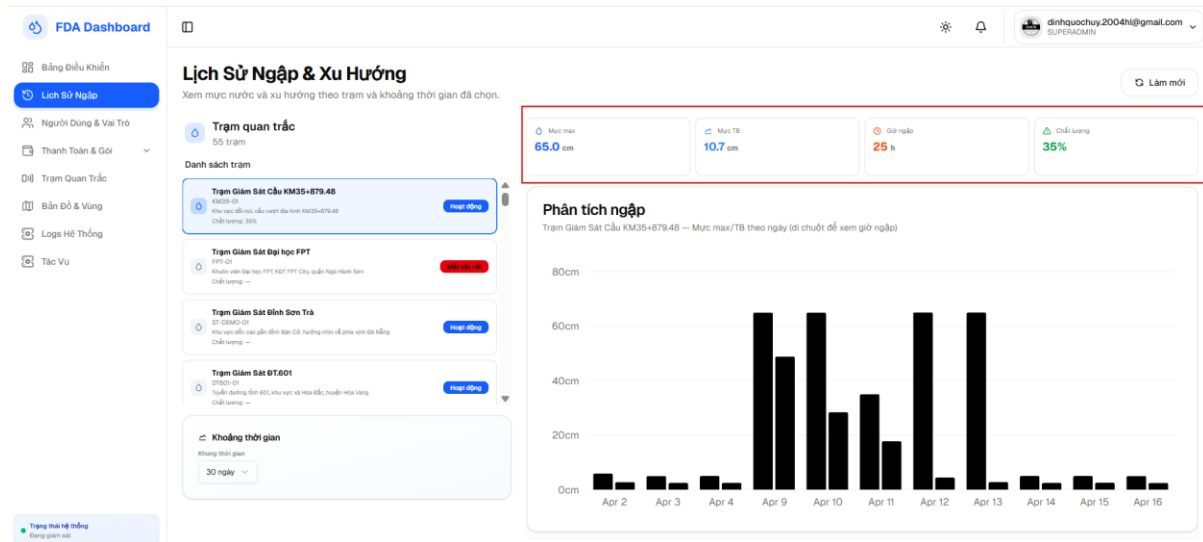


Figure 87: Flood History - KPI Strip

Step 6: View the Historical Readings Line Chart

In the analysis panel, a line chart titled "Phân tích ngập" displays historical readings for the selected station:

- **X-axis**: timestamps formatted by date and time
- **Y-axis**: water level in centimeters (cm)
- **Tooltip**: hover to see the exact reading value and time; when available, the tooltip also shows severity and quality flag details

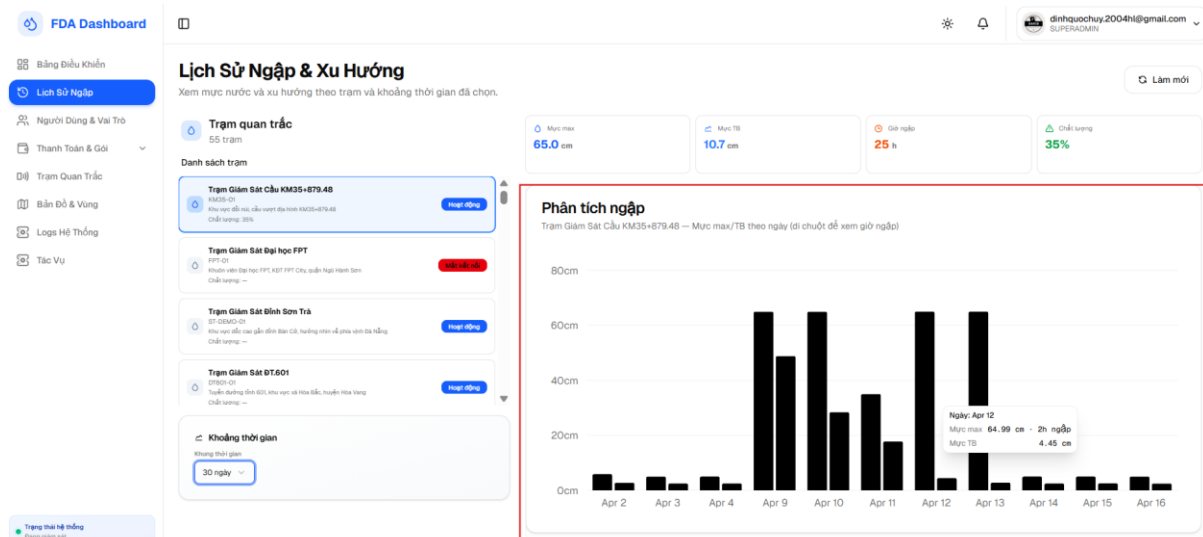


Figure 87: Flood History - Line Chart (Hourly History)

3.2.10.2 Flood Trends

This feature provides aggregated trend analysis over longer periods (days/months/year). The same page is used; changing the time preset automatically switches the chart style and data aggregation level. It supports:

- Recognizing recurring flood patterns
- Comparing periods at a high level
- Summarizing severity over longer time windows

Step 1: Choose a Trend Preset (7/30/90/365 Days)

After navigate to the **Flood History Module** and **Select a Monitoring Station**, open the "**Khoảng thời gian**" dropdown and select one of the trend presets:

- 7 ngày
- 30 ngày
- 90 ngày
- 365 ngày

Click "**Làm mới**" to ensure the latest data is loaded.

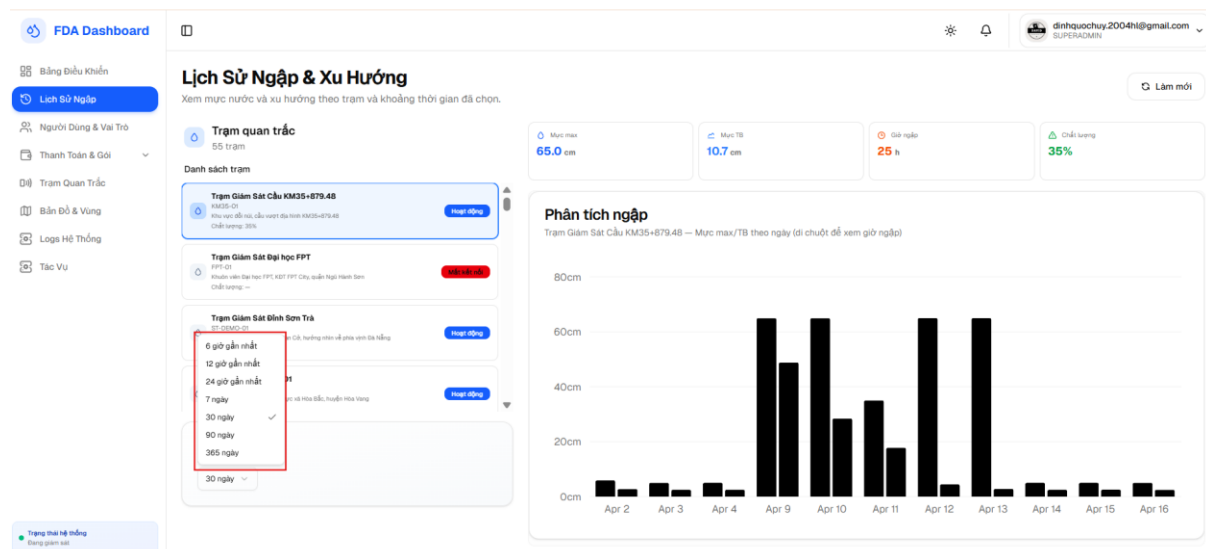


Figure 87: Flood Trends -Select Trend Preset

Step 2: Understand Chart Switching by Preset

Depending on the selected preset, the analysis panel switches to the corresponding trend visualization:

2.1 Preset: 7 Days

The page renders an **area/line trend chart** (trend analysis view) based on aggregated values per period.

Expected chart behavior:

- The chart card title remains "**Phân tích ngập**"
- X-axis shows period labels by date

- The plotted series represents aggregated water-level trends (e.g., average levels by period)

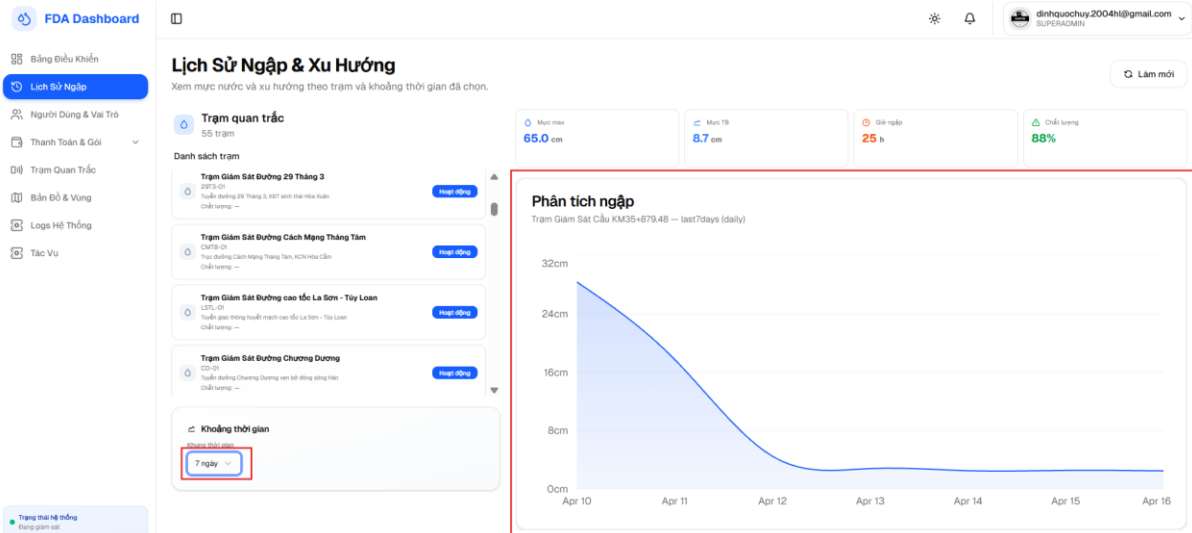


Figure 87: Flood Trends - 7 Days (Area Trend Chart)

2.2 Preset: 30 Days

The page renders a **bar chart** that summarizes water level by day.

Expected chart behavior:

- Bars represent aggregated metrics such as max and average water level per day
- Hovering shows the value in cm and includes flood-hours detail when present in the dataset

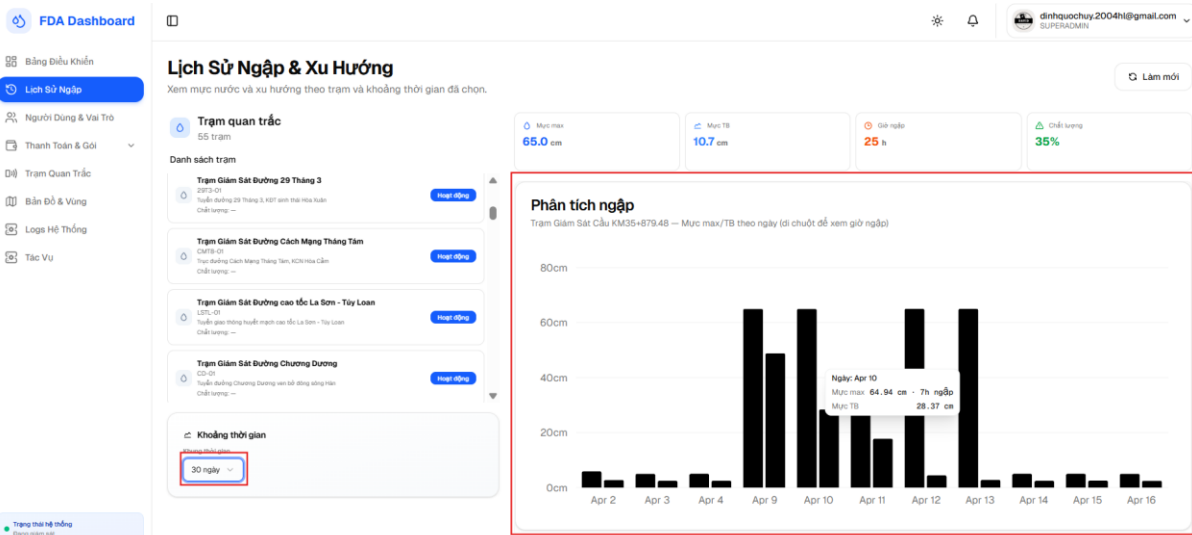


Figure 87: Flood Trends - 30 Days (Bar Chart)

2.3 Preset: 90 Days

The page renders an **area/line trend chart** (trend analysis view) over a longer period window.

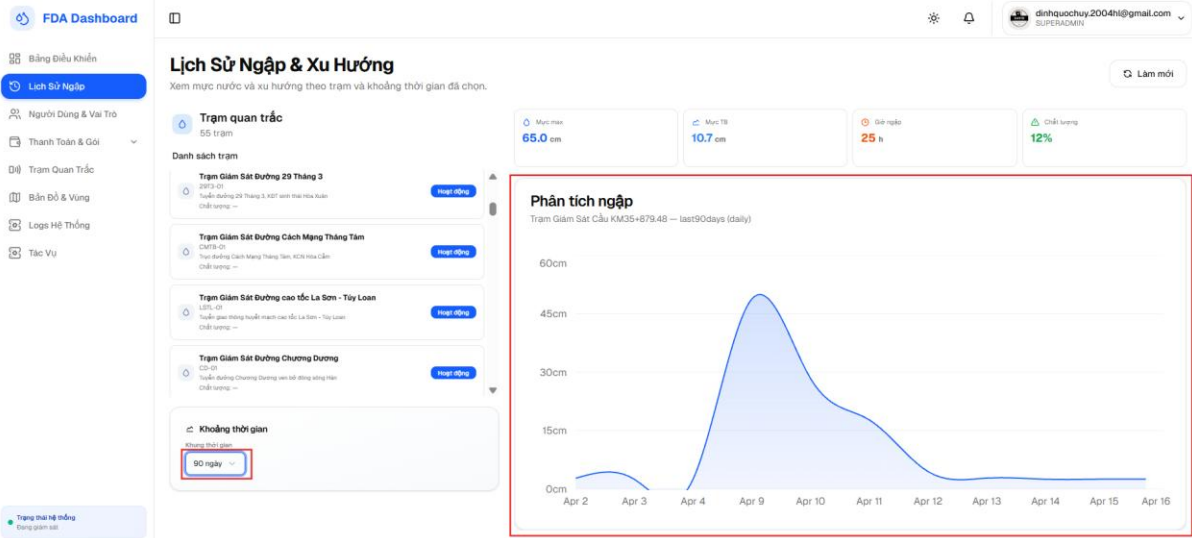


Figure 87: Flood Trends - 90 Days (Area Trend Chart)

2.4 Preset: 365 Days

The page renders a **heatmap** labeled "**Mẫu ngập theo năm**" to visualize yearly patterns.

Expected heatmap behavior:

- Each day is represented as a cell in a week-by-week grid (T2...CN)
- Cell intensity reflects flood intensity (based on flood hours for that day)
- Hovering a cell shows a detail panel with date, flood duration (hours), and max level (cm)

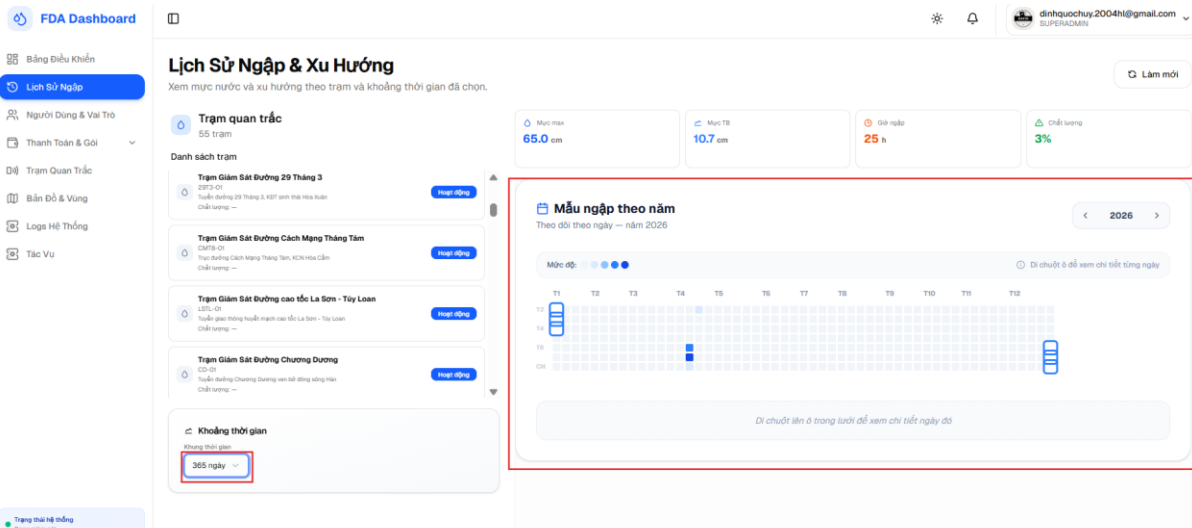


Figure 87: Flood Trends - 365 Days (Annual Heatmap)